



AUSTRALASIA

EIGHT
LECTURES
PREPARED FOR
THE VISUAL INSTRUCTION
COMMITTEE
OF THE
COLONIAL OFFICE
BY
A. J. SARGENT M.A.

V/- NET

VISUAL INSTRUCTION COMMITTEE HANDBOOKS, No. 2.



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AUSTRALASIA

EIGHT LECTURES

Prepared for
The Visual Instruction Committee of the Colonial Office
by
A. J. SARGENT, M.A.

LONDON

GEORGE PHILIP & SON, LTD., 32, FLEET STREET

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1913

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THE VISUAL INSTRUCTION COMMITTEE

APPOINTED BY THE SECRETARY OF STATE FOR THE COLONIES

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PREFACE

These Lectures, like the last series, have been written for the Visual Instruction Committee of the Colonial Office by Mr. A. J. Sargent, and have been revised at the offices of the High Commissioners for the Commonwealth of Australia and New Zealand. The lecture on the Pacific has been revised by Sir Everard im Thurn. The slides are derived from pictures painted and photographs taken by Mr. A. Hugh Fisher on behalf of the Committee, supplemented by photographs supplied from various sources. The Committee wish to acknowledge the great help which they have received from the High Commissioners and Agents-General and their staffs in the matter of slides, and in addition their acknowledgments are due to, among others, the Honourable Victor Nelson Hood, the Agricultural Departments of Queensland and Western Australia, the Secretary to the Commissioner for Queensland Railways, and the Government Geologist of Western Australia.

MEATH.

November, 1912.

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Six Lectures on the Sea Road to the East.

By Mr. A. J. SARGENT.

Eight Lectures on Australasia.

By Mr. A. J. SARGENT.

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[In these two books the ordinary edition and the lecturer's edition are combined.]

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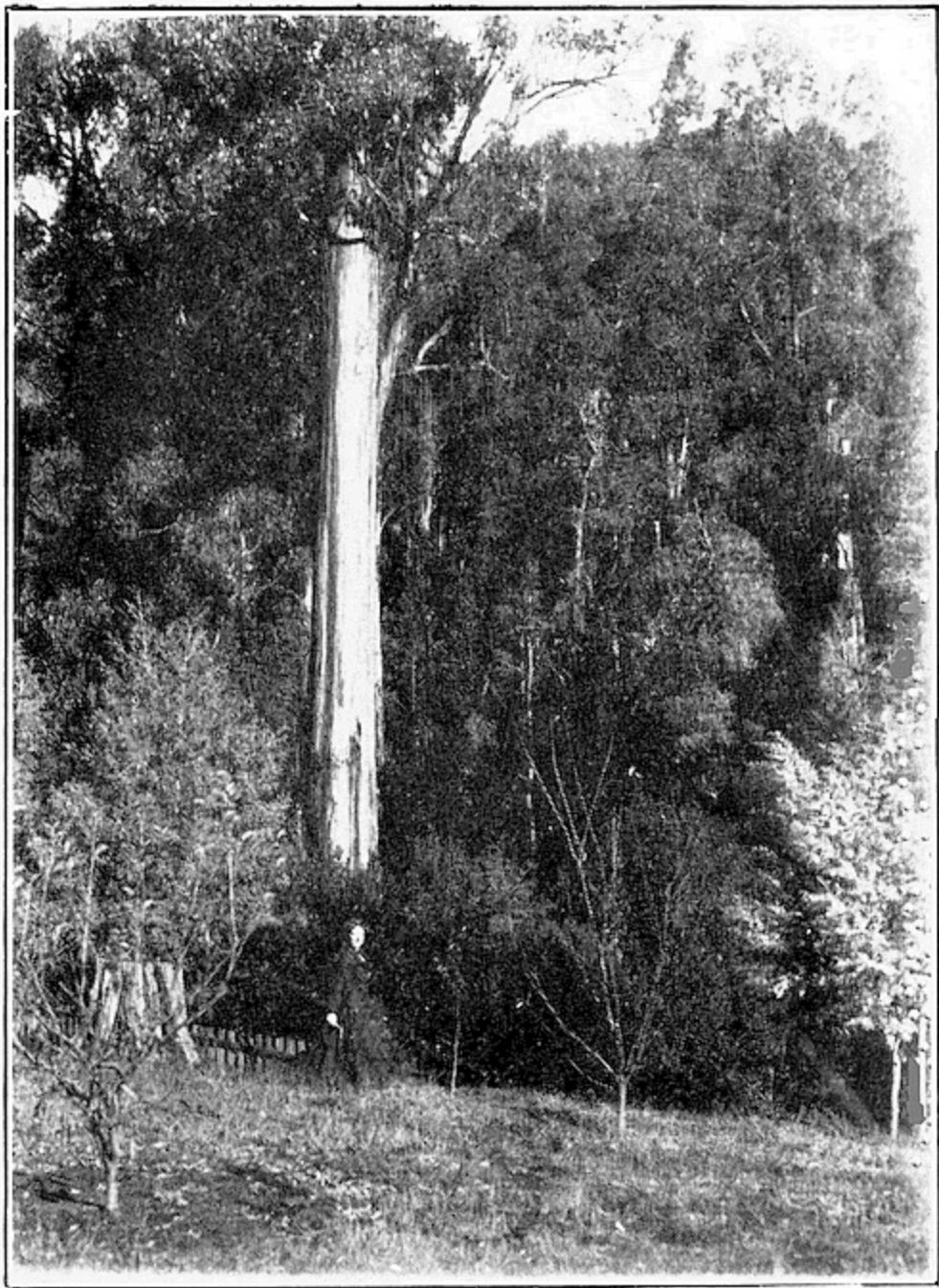
NOTE.—The reader is asked to bear in mind the fact that these lectures are illustrated with lantern slides. The numbers in the margin of the text are the numbers of the slides, of which a complete list will be found on pp. [143–152](#).



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NATIVE BEAR.

[See [page 7](#).



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GUM TREE.

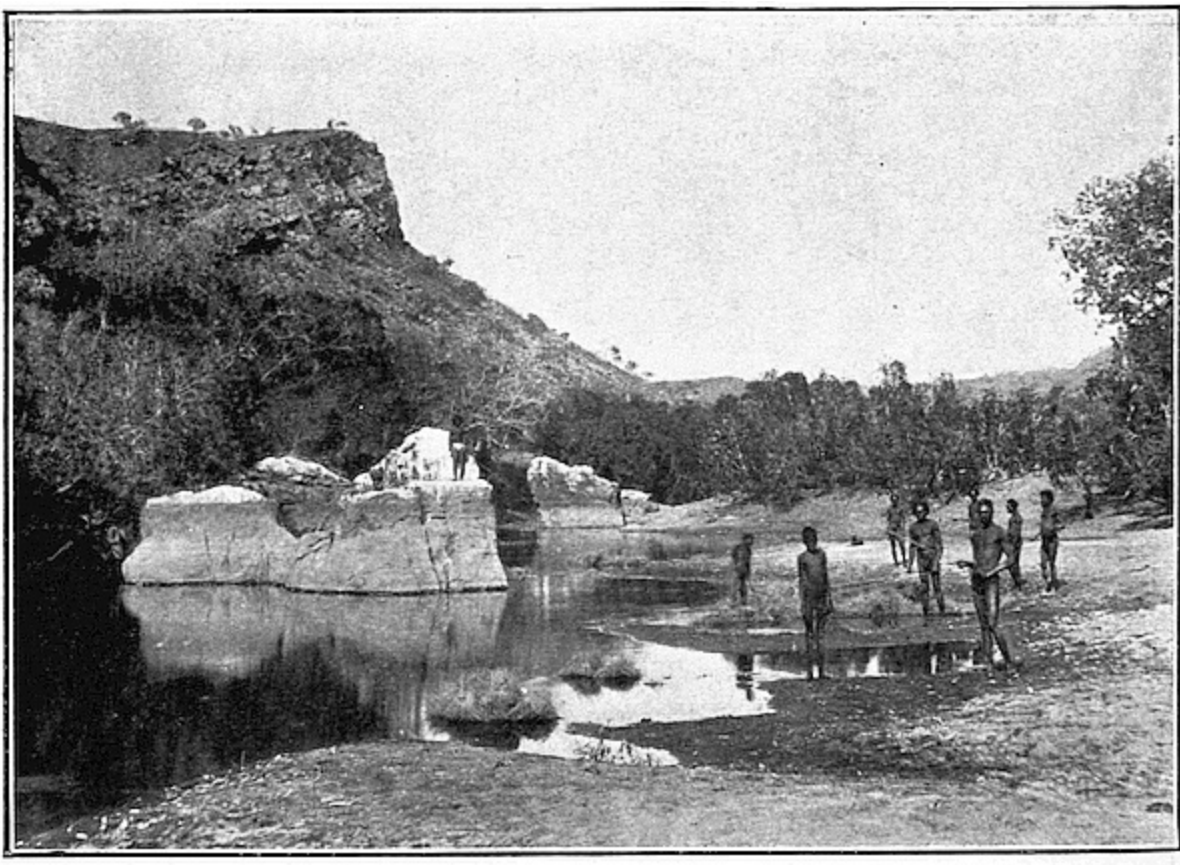
[See [page 10.](#)



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[See [page 9.](#)

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NATIVES FISHING.

[See [page 12](#).

LECTURE I

AUSTRALASIA

For nearly two thousand years the existence of a great Southland, in the ocean of the southern hemisphere, corresponding to the land mass of the Old World in the northern, was a matter of doubt and dispute among geographers. In the sixteenth century, this land begins to appear vaguely on globes and charts; possibly the information was due to the Malays and Arabs, who were skilful sailors and made long voyages in the Pacific and Indian Oceans. It has been thought that the Portuguese, approaching the Malay region from the west, while the Spaniards came from the east, may have been acquainted with the northern coast of the new continent; since they certainly had some knowledge of the northern coast of New Guinea early in the sixteenth century. But our first definite information may be said to date from the beginning of the seventeenth century, when a Spanish exploring expedition, under de Quiros and de Torres, sailed across the Pacific and discovered the New Hebrides group. One of the islands they named la Australia del Espiritu Santo, under the idea that it formed part of the great southern mainland. De Quiros then sailed back to Mexico, but Torres continued his voyage northwestward through the straits which still bear his name, and so to the Spanish Possessions in the Philippines.

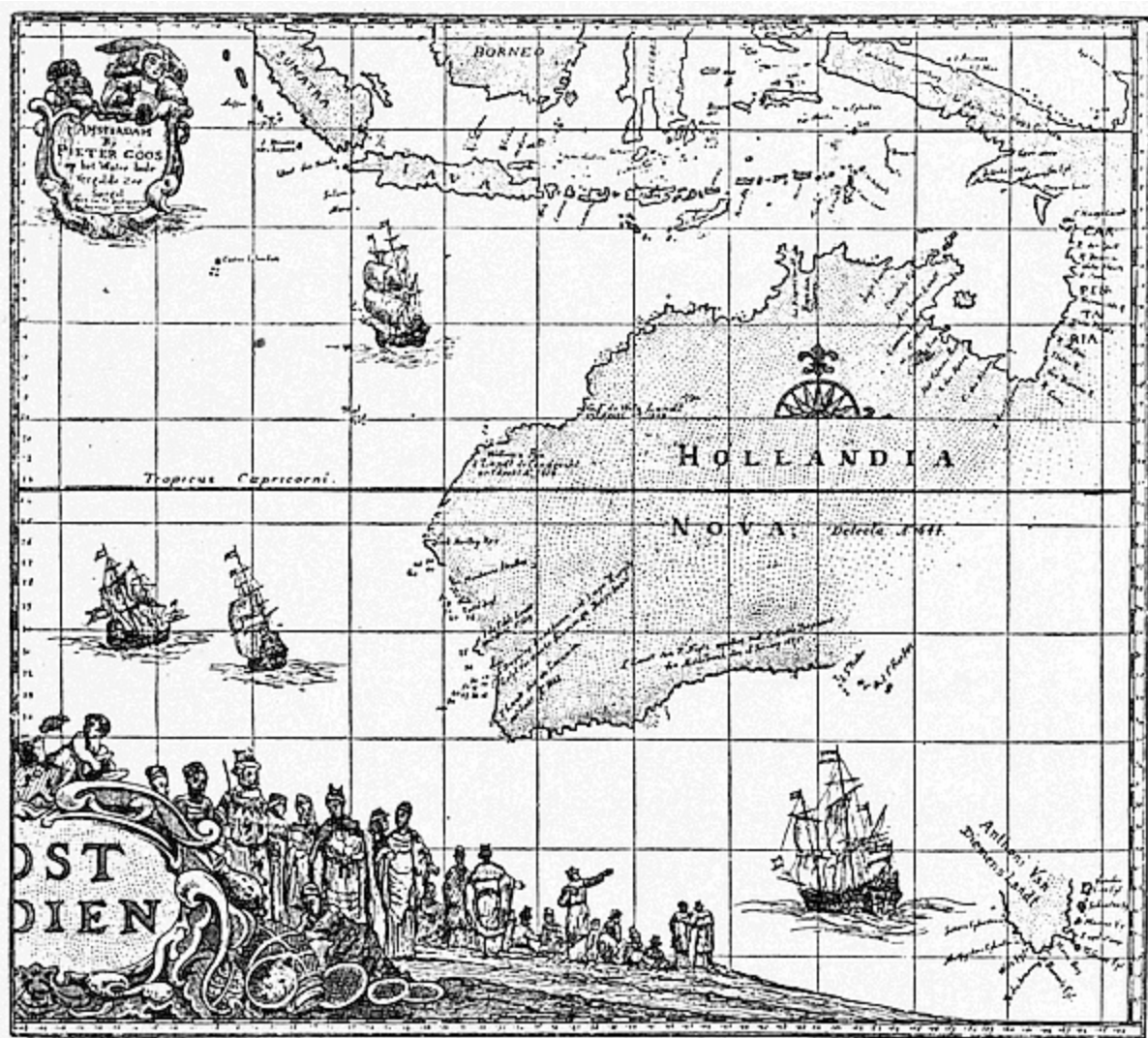
But the day of Spain as a sea power was passing away, and it was to the Dutch traders that the western world owed its first real acquaintance with Australia. The early discoveries were often accidental and did not lead to settlement or regular intercourse with the natives. This was due partly to the backward state of the science of navigation, partly to the fact that the voyagers on long expeditions usually lost from half to three-quarters of their crews from disease. The Dutch had the advantage of a local base, as they were already firmly established in the East Indian Islands; so that Australia was really discovered from the Indies.

Trade, not exploration, was the main motive of the Dutch. So, in the early part of the seventeenth century, trading ships were dispatched from Batavia by the Dutch East-India Company to explore the north and west coasts of Australia.

We find all along these coasts the names of the ships still surviving, as in Arnhem Land and Cape Leuwin; those of the captains, as in Dirk Hartog Island, Houtmans Abrolhos, Edels Land and Nuyts Land; while the Gulf of Carpentaria is a memorial to Peter Carpenter, the then Governor of the great trading company.

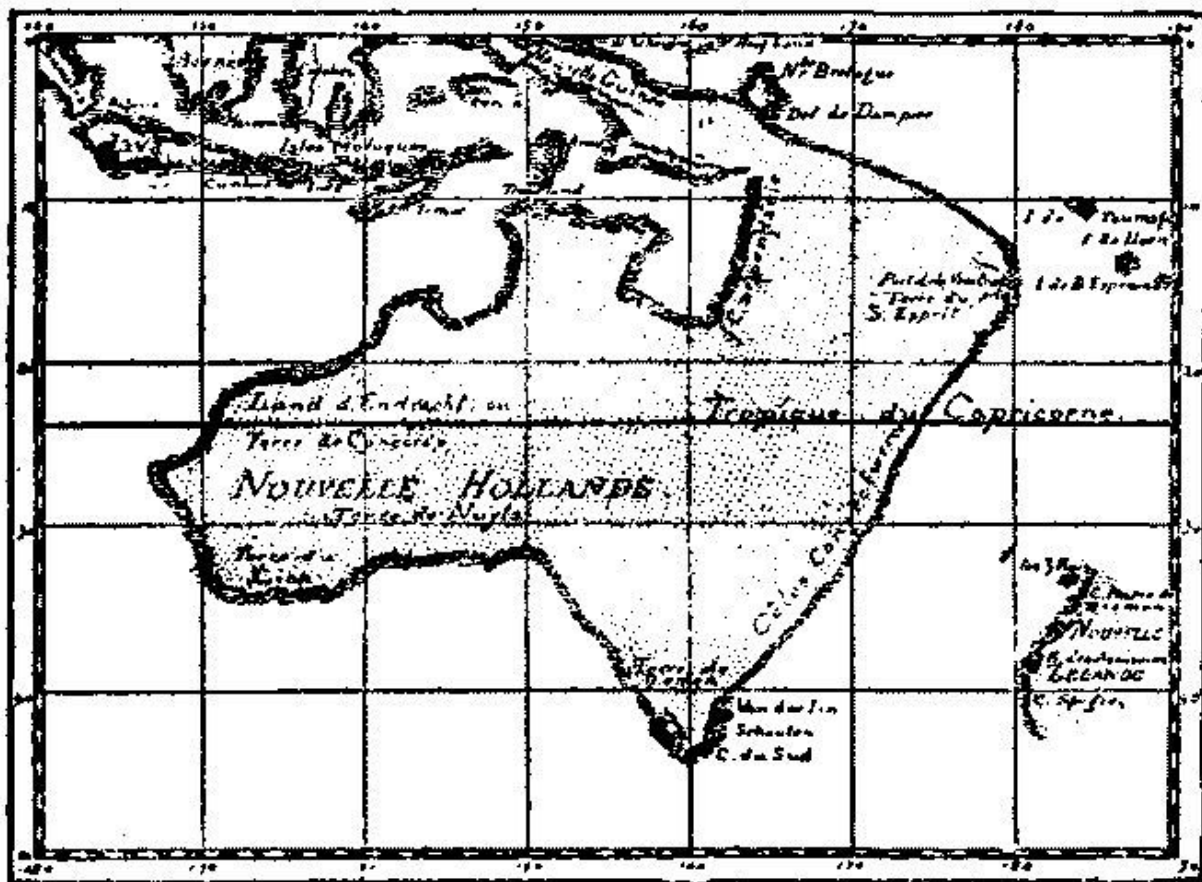
The most important voyage of all was that of Abel Tasman, in 1642. He started from Mauritius, to discover a passage south of the Australian continent; and after landing in Van Diemen's Land, now Tasmania, he sailed across the Tasman Sea and up the west coast of New Zealand, and so back to Batavia by way of Tonga and the Fiji islands and the north coast of New Guinea. He left behind him another name, New Holland, for the whole continent.

These names throughout Australasia bear witness to the skill and energy of the Dutch navigators; but they came only as traders, and the west coast of Australia, which they knew best, had little to attract them to permanent settlement. Sand and grass, hostile natives, and dangerous reefs marked by a series of shipwrecks sum up their impression of the new land. Here are two views of the coast; it does not seem attractive for sailors. We can quite understand why they made no effort to open up a trade like that of the East Indies, especially as they missed the east coast, the most promising region for European settlement. Here are two charts showing our knowledge of Australia, the first soon after Tasman's voyage, the second nearly a century later. 4,5



MAP OF PIETER GOOS, 1660.

The discovery of the east coast was to come more than a century later; it was made by an Englishman, at a time when England was looking for new outlets, both for trade and settlement, as an offset to her losses in the continent of North America. It is true that Dampier visited the west coast towards the end of the seventeenth century, and wrote an account of the animals, natives, and plants; but he does not seem to have been much more favourably impressed than the Dutch, and nothing came of his visit.



MAP OF R. DE VAGONDY, 1752.

But in the eighteenth century England and France were the great rivals in colonisation, and voyages of discovery and plans of annexation became the order of the day. It was a revival, in a less forcible though more scientific form, of the old rivalry with Spain and Portugal in the sixteenth century. State influence was behind the explorers, as it had been in the days of Drake and his freebooters. Captain Cook sailed in a ship belonging to the Royal Navy, and not in a trader. In our own country his memory is still kept green in the little port of Whitby where he served his apprenticeship to the sea, and where the ships were built in which he made his great voyages of discovery; in the Antipodes it is for ever associated with the beginnings of a great and growing Empire. Here we see his statue in Sydney, and here is a picture of his ship, the *Endeavour*, approaching New Zealand.

Cook came westward, across the Pacific, with an expedition in 1769, to observe the Transit of Venus at Tahiti, and then sailed south-west to New Zealand. ⁸ He sighted land at Poverty Bay and sailed south as far as Cape

Turnagain; then he went right round the North Island and through Cook Strait, until the Cape was again sighted, scattering English names and charting the coast as he passed. Next he circumnavigated the South Island, steering outside Stewart Island, which he imagined to be a peninsula; and so to Cape Farewell, in the north-west corner of South Island. He had obtained a fairly accurate idea of the nature of the coast, and had proved that New Zealand was not connected with the supposed Antarctic continent.

From Cape Farewell he struck westward, and sighted the mainland of Australia at Point Hicks, on April 19th, 1770, but failed to discover Bass Strait. From Point Hicks Cook sailed along the whole of the east coast to Cape York, giving to the bays and capes as he passed them the names of his crew, of British admirals, officials and politicians of all kinds at home. At various landing places he came into collision with the aborigines, and at Endeavour River he stayed for two months to repair his vessel which had been damaged through striking on the rocks. Finally, before sailing westward through Torres Strait, he landed on Possession Island and formally claimed the whole region discovered for the British Crown.

When, in 1798, Flinders and Bass proved that Van Diemen's Land was an island, the rough work of discovery was complete. Australia, New Zealand and many of the smaller islands were known and charted, and the discoveries of the Spaniards and Dutch were linked up, by the aid of a great deal of miscellaneous exploration carried out by the French navigators who followed us in the South of Australia, in New Zealand, and in the islands of the Pacific.

The navigators, French and English, were in the habit of formally annexing the country wherever they landed; but annexation of this kind was of little value without effective occupation. This was to come later, during the course of the next half century. From 1800 onwards, we may say that the different parts of this vast region begin to have a separate and individual history and development, though behind all are the conditions common to most of the area, conditions which distinguish it sharply from the rest of the world even to-day, in spite of minor differences between its separate parts. There is ample room for variation, since the continent of Australia is about three-quarters the size of Europe, or seven times the area of Germany and France together; and it includes nearly every type of temperate and tropical climate.

Whether we look at the animals, plants, or aborigines of Australia, we are at once struck with the fact that they belong to an entirely different order of life from that which we find in the other great continents. The whole region seems, from a very early age, to have been cut off effectively from the rest of the world, and to have developed along lines peculiar to itself; though since the advent of the white man we have the artificial introduction of European and other plants and animals, which bid fair in many cases to oust the native products, just as the white man has displaced the original inhabitants.

The first animal which we naturally think of in connexion with Australia is the kangaroo. His family ⁹ is large and varied, from the giant standing six feet high, and dangerous to attack, to the dwarf measured ¹⁰ by a few inches. He is a creature of the wide open plains, living on grass, though in Queensland and New Guinea are to be found some which climb the tall gum trees and feed on their shoots. An ancestor of the present kangaroo, whose remains have been discovered, was a formidable monster standing twelve feet high.

The kangaroo is merely the best known representative of a very large group, the marsupials or pouched animals. Closely allied to the kangaroo is the wombat, a clumsy badger-like animal which feeds on leaves and burrows in the ground; he is quite harmless, though, like the kangaroo, he had a huge ancestor who seems to have been as large as a rhinoceros. The bandicoots, ratlike burrowing animals, are his cousins many times removed.

Next we have a group which looks very different but is really closely allied to the kangaroo: the *phalangers*, or opossums, as they are commonly called. The ¹¹ name is adopted from America but is applied in Australia to the wrong group, as we shall see later. Nothing is more misleading than the names given by white settlers, ignorant of botany or biology, to native plants or animals, generally on the ground of some fancied and superficial resemblance. The so-called opossums are found all over Australia and New Guinea and in many of the islands of the Pacific; but owing to the value of their fur they are becoming scarce in many districts. They live in trees, and some of the family have their legs connected by a membrane, so that they can glide from one branch to another. Hence the name flying squirrel; though they do not fly and have no connexion with the squirrel which we know. Here is another of the same group, the *koala*, or native bear; though

he again has nothing to do with bears. He is a sleepy-looking ¹² animal, with no tail, and is not given to leaping or gliding; though his claws are useful for ¹³ climbing the gum trees in which he lives. He is quite harmless, and a child can play with him, as we see here.

Other members of the same great family are far from harmless: among them are the *dasyures*, or native cats, with dark bodies mostly spotted with white. Some smaller members of this family are called weasels and mice. To this family belongs also the Tasmanian Devil, ¹⁴ whose portrait we have here; he is very fierce though small. A larger animal, the size of a retriever, is the so-called Tasmanian wolf; he is carnivorous, while most ¹⁵ of the marsupials are vegetarian. It is rather doubtful whether he should be classed with the rest or go by himself.

The only existing allies of the marsupials of Australia are to be found in the opossums and some other less known animals of South America; but the opossum of South America resembles less his Australian namesake than the other group, the *dasyures*. In Europe and Asia the marsupials existed, but only in very remote geological ages, as their remains prove. It has been argued from the existence of the opossum family in America that at some time there must have been a land connexion between Australia and South America, either by way of the islands of the Pacific or by an Antarctic continent. But the isolation of Australia must have been very ancient, since it has given time for the development of the enormous differences which we have seen among the individuals of the same family.

Even these strange animals, old as they are, are not the most primitive to be found in Australia. The *ornithorhynchus* or duck-billed platypus lives in a ¹⁶ burrow by the river; it has teeth when young, but seems to lose these as it grows up. It is said to lay eggs like a reptile, though it is a true mammal. We are not surprised to learn that when the first stuffed specimen of this strange beast reached Europe it was thought to be a fraud, put together to deceive the ignorant and unwary. Another of these egg-laying animals is the *echidna*, or spiny ant-eater, which has a kind of beak for burrowing and a long sticky tongue to capture its prey. It has sharp spines and rolls itself into a ball like a hedgehog when attacked. In addition to these curious animals there are rats and mice and the dingo, or native dog, of a type more familiar ¹⁷ to us. It is thought possible by some that the dingo, being so different

from the other native animals, was introduced by man at some very early date.

The typical Australian animals disappear as we travel away through the islands to the north, and the boundary line between Australia and Asia is usually drawn through the strait between Bali and Lombok, and then on between Celebes and Borneo; though it is much disputed whether Celebes belongs to the one region or the other. The division is only a narrow water strait, and we can understand that Asiatic birds would have no difficulty in crossing. On the other hand, as there is no great land mass to the south, there are no crowds of seasonal migrants from cooler regions, such as we find in the warmer countries of the northern hemisphere; so that the existing Australian birds are for the most part allied to the bright-coloured inhabitants of the tropics to the north. Some are peculiar to the region, such as the lyre birds with their wonderful tails, the emu, and the brush birds which bury their eggs in a mound of earth, leaving them to be hatched by the heat of the sun. Other birds, as the honey-eaters, though common in Australia, are found also in the neighbouring islands of the Pacific. Unfortunately, not content with the flowers of the gum trees, these birds now attack the fruit in the orchards and are becoming a nuisance in some districts. Finally we have the *kukuburra* or laughing-jackass, one of the most popular and best known of all Australian birds. It belongs to the kingfisher tribe and is said to be a great destroyer of small snakes. Nor must we forget the black swan, one of the most striking and beautiful of all, and still to be found in large numbers in the lakes of Gippsland and in West Australia.

Then we have lizards, snakes, and strange fish of all kinds, some peculiar to Australia, others, like the animals, allied to species found in Asia or South America. All the evidence afforded by the animal life of the continent points to a very remote connexion with the other continents, followed by a very long period of isolation during which the families of animals developed wide differences among themselves, together with special peculiarities suited to the conditions in which they lived.

Now let us turn for a moment from the animal to the plant life of Australia. In the tropical parts of Queensland there are many plants which belong to the Malayan region as a whole; this we might expect. We may see palms growing even at Brisbane; while further north they form a dense

forest, thoroughly tropical in appearance, with its creepers and undergrowth, as we see in these two pictures. But further 24 south we meet more species peculiar to Australia, in the vegetable no less than in the animal world. 25 The typical Australian plants are those specially adapted to resist hot sunshine with drought, and often sudden changes of temperature. All over the continent the gum trees give a special mark to the landscape 26 of the forest. The leaves hang vertically, so as not to give a large exposure of surface to the hot sun, and are collected in dense clusters for the same purpose of protection; so that we miss much of the spreading shade of our English trees. The leaves, too, are alike on both sides, and the colour is more uniform than in our own trees, nor do they change colour and fall in autumn, but decay gradually on the stalk. The bark of many of the gums hangs in strips, thus spoiling the appearance of the huge trunk, for huge it is, rising to four hundred feet in some species. Owing to the character of the foliage the forests are more open than our own, except in the wetter parts where there is a dense undergrowth of fern and creeper. These districts are always near the coast. As we travel away from the coast the forest thins out, except along the watercourses, and becomes 27 smaller and more stunted. In place of trees with leaves we find thorny scrub of various kinds, such as the 28 mulga, whose leaves are replaced by woody spikes still more resistant to drought. Finally we reach the salt-bush 29 and spinifex and wiry grasses of the desert interior. Often there is little but bare rock as in the 30 pictures before us. The watercourses themselves may dry up in the hot season, leaving a desolate expanse 31 of sand and stones, but the line of trees still marks the presence of moisture below. 32

The huge gum trees are, as a rule, the main feature in the forest landscape, but this is varied by curiosities like the grass tree and the bottle tree; while in the moister regions, as in the forests of Gippsland, we find 33 beautiful tree ferns of every kind. Here are specimens of grass trees and gums. Everywhere we 34 find the acacia and the wattle with its sweet scent, the banksia shooting up into great trees, and the 35 beautiful waratah; everywhere, too, where the rainfall is sufficient, flowers of the same order as those with which we are familiar in Europe grow in great profusion.

So we see that there is great variety of vegetation in Australia, though the only native plants of much use to man are the gum trees, which we value for their dense hard woods. All else that is of value, both among plants and

animals, has been introduced by the settlers, and many of the native types seem doomed to extinction.

We may now be able to understand why this country of strange plants and animals had no attraction for the Dutch trader, especially in the drier west; in fact, it could have no trade until it had been settled by Europeans bringing with them European crops and animals. In India and Malaya there was already a basis for trade in some of the natural products of the region; while the natives were sufficiently advanced in civilisation to make commercial intercourse possible. It was far otherwise in Australia. The aborigines, as we found them, were as primitive as the plants and animals. They are not black, but a dark-brown people, and their hair is waved and silky, not curly like that of the negro. They are different also from the negro in the shape and build of the head and face. There are various theories as to their origin, but the nearest correspondence seems to be in some of the ancient hill tribes of India and the Veddas of Ceylon. At any rate they are quite different from the Malays, and equally also from the now extinct Tasmanians. The Tasmanians had woolly hair, and perhaps represented the remnants of an earlier and even less developed race than the invaders from the north.

The native Australian, as the first European discoverers found him, was not an attractive being. He was looked on as little better than a wild beast and treated as such. This was partly due to ignorance of his language and customs. His mode of life was fitted by long adaptation to the peculiar conditions of his surroundings. He had developed no agriculture in his new home; not without reason, if we think of the agricultural possibilities of the country in the hands of a rude and backward people. He was equally without any of the useful animals, and had no means of procuring them. So he was reduced to the nomad life and to the utilisation of the wild roots and plants of the country and such small game as he could kill with his rude weapons. He was necessarily a hunter, with a temporary shelter for his home, often an overhanging rock as in this picture; and civilisation does not grow up in such conditions. Here we see some natives ³⁷ fishing, and here again are some armed men scouting in the bush. ³⁸

The hostility of the native to the European colonists often arose from their interference with his natural food supply, or to their careless ignorance of his semi-religious ideas or customs, such as the tabu.

His only possible clothes were the skins of animals, and his weapons and tools all belonged to the Stone Age. The axe, knife, and hammer were in universal use, and long journeys were made to obtain the right kind of stone. This involved a certain amount of intercourse among the tribes. For weapons of offence the native had the club, and the spear tipped with bone or stone; while some tribes used a special spear-thrower. The boomerang was a curved bar of heavy wood, often five or six feet long, which was used for killing or stunning at a short distance. The smaller boomerang, which returns to the thrower, was merely a toy and used in sport; here we see it in use. The stone implements are 39 all similar to those which are dug up in Europe, the relics of the Neolithic Age of man. One of the chief uses of the axe was to cut notches for climbing trees, in search of honey among other things; though they had 40 another method which we see here.

The natives were divided into tribes and sub-tribes, and had some form of tribal government, under headmen. In the south-east of the continent these chiefs had considerable authority and were sometimes treated by us as representing the tribe. Here is one of them, though he does not look imposing in his European dress. 41 They had an elaborate social system and curious marriage customs about which the learned still dispute. They had a strong belief in spirits of various kinds, though it could hardly be called a religion, and a whole series of tales and legends handed down orally, some of them showing considerable power of imagination. 42 They even had the beginnings of some ideas of art and ornament, as we can judge from the crude paintings 43 shown here.

The most interesting of their social customs was the corroboree, a great gathering for feasting and dancing, 44 often combined with some religious or social ceremony. Such meetings represented the only real social intercourse of the people and tribes, except messages by ambassadors who were sacred everywhere.

On the whole, then, they were not so low in the scale of civilisation as the early observers imagined. Even the language, with its many dialects, due to the absence of writing and the nomad life of the people, is elaborate and inflected like those of Europe.

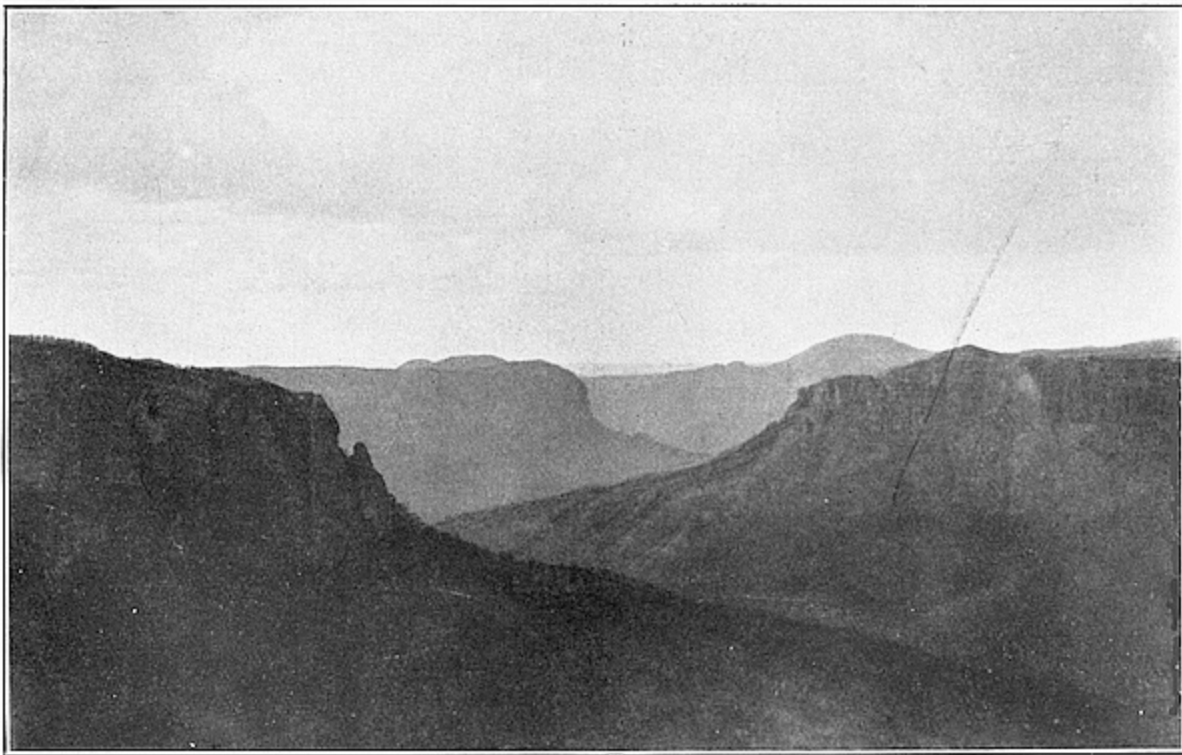
The native life in its original form is decaying, and survives chiefly in the interior and the west. Wherever white occupation has extended, the native is

dying out; in fact, in some parts he survives only on the Government 45 Reservations. Here are some of these survivors in Victoria. Here again, in Queensland, we see the 46 native converted to European clothes, though he does not seem very comfortable in them. In this district, 47 as in South and Western Australia, and the Northern Territory, they still exist in considerable number; but it is probable that there are less than 100,000 in all in the Commonwealth. In the census of 1911 an attempt was made to count them, and some 20,000 were found to be living in or near white settlements; only a vague estimate was possible in the case of the tribes of the interior, who still live their nomadic life in the more inaccessible parts of the country. But the area untouched by the white man grows smaller every year, and unless the native can change his character greatly, he is likely to die out in the north and west as in the south-east. In 1911 there were only about two thousand in New South Wales, hardly any in Victoria, and none at all in Tasmania. It was inevitable that the Australian native should be displaced from his hunting grounds. An area about equal to that of the United States could not be left in the sole occupation of a few thousand savages. Now, instead of the savage with his primitive tribal system, we have a white race, purely British in origin, with industry and agriculture of the most advanced type, and an elaborate political constitution of federated States. It is the utilisation, by the white man, of the resources of this vast area which we must study.

We have seen something of the coast of this area; let us now try for a moment to picture it as a whole. The map shows us an oblong block, which lies east and 48 west, on either side of the Tropic of Capricorn. On the north, two peninsulas, Arnhem Land and Cape York, project towards the Equator; between them is the Gulf of Carpentaria, the only deep indentation in the whole continent. On the south, Victoria, continued by Tasmania, stretches into the cool waters of the Southern Ocean. A coastal plain, broad in the north, narrow elsewhere, fringes a plateau which occupies the greater part of the oblong. The plain is narrowest in the east, and the plateau edge most marked, so that it resembles a series of mountain ranges. Rivers, short for the most part, plunge down the seaward edge of the plateau. The only large area of lowland is in the south-east, drained by a group of long rivers and shared by four of the Australian States. Some of these rivers lose themselves in the salt lakes and marshes of the inland basin of Lake Eyre; the rest, gathered up by the Murray, reach the sea at the one point where the

plateau rim disappears. The tropical lowlands, the temperate coast-plain, the plateau and the long inward slopes of the Murray system are the main features which we shall find recurring in the geography of the various States.

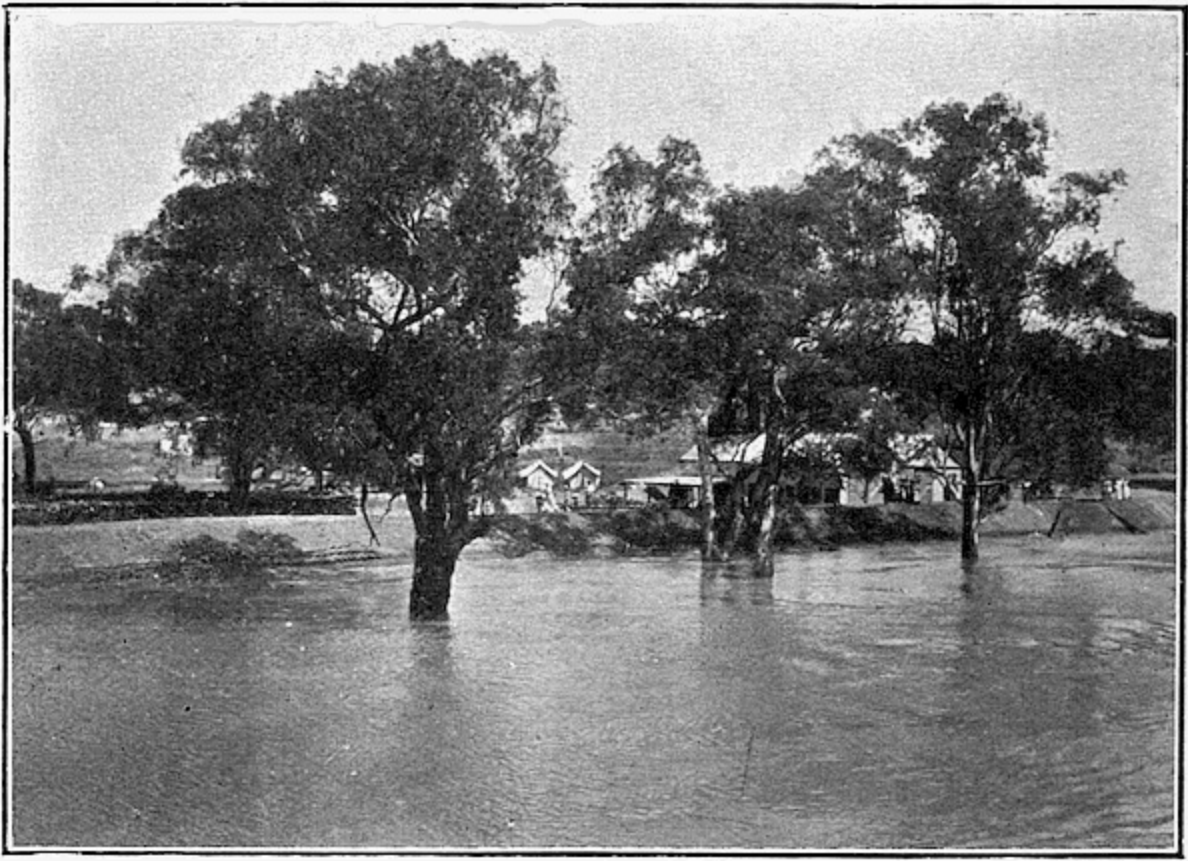
So far we have dealt only with Australia. New Zealand shows to some extent the peculiarities which mark Australia off from the rest of the world, but on the whole the differences between these two sections of Australasia are more noticeable than the likeness. We might anticipate this, since the dominant fact in the development of the peculiar plant life of Australia is drought; in the case of New Zealand it is moisture. 49 Again, New Zealand has no snakes, and lacks the peculiar animals which we have seen in Australia. In fact, its only native animals are a bat and a rat, the latter perhaps introduced by the Maoris. To compensate for this it has a large group of birds peculiar to itself, including the wingless birds, which may have lost their wings since they had no need for flight from enemies on the ground. The best known of these birds is the *kiwi*; here we see him in his natural haunts. 50 Again, there is the *takahe*, which was at one time thought to be extinct; but one or two specimens have been found, and there may be others still existing. Here is one of these stuffed. Finally there is the 51 gigantic *moa*, which probably existed at the time of Captain Cook's voyage. Now we can only see its skeleton, 52 but naturalists have attempted to reconstruct the whole bird, as we see here. One link New Zealand 53 possesses with a very remote geological past; this is the curious *tuatara* or three-eyed lizard. He belongs 54 to an extinct group of reptiles which lived in Europe many ages ago and is now only represented in the fossils which we dig up. So we see that in its animal life New Zealand differs from Australia and from the rest of the world; we shall find also a strong contrast between the native races of the two countries in their character and origin, their relations with the white settlers, and their influence on the history and development of the land which they possessed before our arrival.



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GOVETT'S LEAP: BLUE MOUNTAINS.

[See [page 23](#).



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THE MURRAY, IN FLOOD.

[See [page 27](#).



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YASS.

[See [page 31](#).

LECTURE II

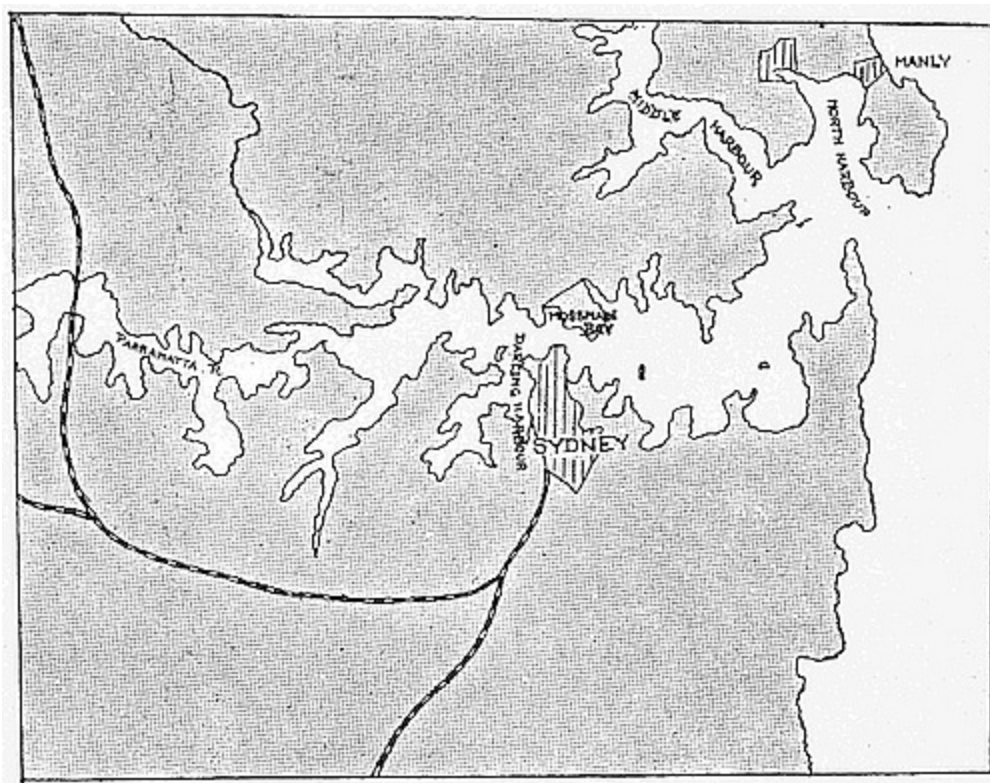
NEW SOUTH WALES

When, in 1770, Captain Cook dropped anchor in Botany Bay, he just missed discovering the finest harbour in the world. Voyaging northwards he sighted Port Jackson in the distance but did not examine it more closely, and the discovery was left to the first party of colonists, a few years later. The harbour which we are going to explore was the scene of the first real settlement, and is still the vital centre for the whole of New South Wales.

We steam through a broad channel, nearly a mile wide between the rugged points which guard it. There is no lack of room here for ships to pass one another, and our large vessel seems quite insignificant beside the towering cliffs. On our right is a broad bay, the North Harbour, with the village of Manly at its head; on the other side of Manly, across a narrow isthmus, is the open ocean, with its long rollers thundering always in surf upon the beach. But inside the Heads the water is calm as a lake. In front of us, and beyond the North Harbour, is a narrow, winding inlet, running for miles into the hills; this is Middle Harbour. There is plenty of good anchorage here, but it is mainly given up to pleasure boats; we are a long way yet from the commercial harbour. To reach this our vessel turns sharply southwards, behind the South Head with its lighthouses, and steams on for about five miles up the Main Harbour. All along, on either hand, are jutting headlands, and in the bays between, especially on the south side, are seaside villages. But we shall not see swarms of bathers on the beaches as in our own country, for there are sharks in Sydney Harbour; the only safe bathing is in the surf outside. As we approach our landing-place the houses are more closely packed together, and islands are dotted here and there in the channel. We may be reminded to some degree of parts of the Clyde, or of one of the larger inlets on the west coast of Scotland; though here we

find not only the most beautiful scenery but a great seaport and busy city in the very midst of it.

We turn at last into Sydney Cove, on the south side of the harbour, and here we are moored at Circular Quay in the very heart of the city. Further on to the west, just round the next point, we see Darling Harbour, crowded with shipping, and its busy wharves piled with merchandise. The waterway extends some miles further inland, but here in Sydney Cove is the centre of commercial activity and the landing-place of the original settlers in 1788.



PORT JACKSON.

Before we land let us look at a chart which shows us the long passage by which we have entered, the windings of the harbour and the city spreading over the surrounding hills. This will give us our bearings and help us to understand the views.

We will now land, climb the hill, and look down on the Cove. There, on the further side, is our vessel, lying close to the tall warehouses. Beyond it are the trees of Government Domain, with the tower and roof of Government House showing above them. The little bay on our left is Farm

Cove, where the warships lie 7 at anchor; and beyond it again, on the next point, we see the trees of the Botanical Gardens. Then we have Woolloomooloo Bay, running up to a new and crowded suburb, and in the distance many more points and bays, as we look along the south side of the harbour back towards the sea. Or let us climb up behind our vessel, in another direction, to the library, and look down over Farm Cove. Below us, on the little island, is 8 Fort Denison; and across the water on the north side is Mossman Bay, where a new Sydney is growing up. It is all very different from the crowded ugliness of most of our own commercial cities.

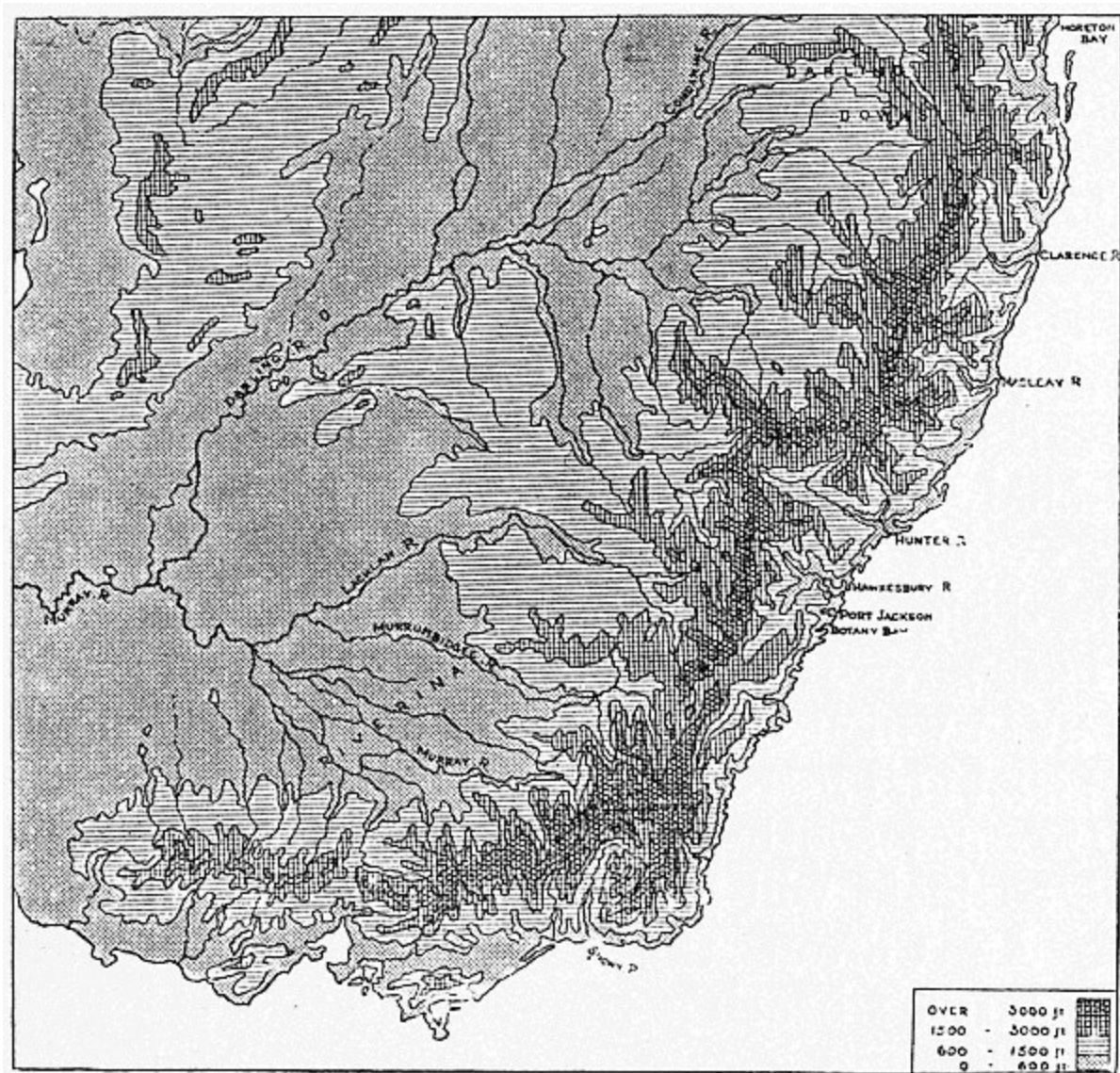
To see something of the inside of the city we walk up George Street from the quayside. On our right is old Sydney, irregular and picturesque, built on the rocky peninsula between Sydney Cove and Darling Harbour. Here is a view of one of the old streets. 9 George Street itself is very modern in appearance, with its broad roadway, electric cars, and handsome stone 10 buildings. Here is the Post Office at one of the corners; it is built of sandstone and granite which are to be 11 found in abundance in the local rocks. Across George Street runs Bridge Street, one of the oldest in the city. It takes its name from the old bridge across the little Tank stream, which has now been absorbed into the underground drainage of the city, like so many of our old streams in London. There are many Government buildings in this older part as we might expect, and at the top is the entrance to Government House.

On the west side of Darling Harbour is the suburb of Pyrmont, on another peninsula; and at the base of these peninsulas Sydney is spreading and broadening out beyond the railway station. But even here, in the new suburb, are many parks and open spaces reserved for public use; while nothing can destroy the beauty of the older portion of the city, divided up by inlets, and with glimpses down many of its sloping streets of the blue water and the hills beyond. It is not surprising that the early settlers found this spot far more attractive than the open beach of Botany Bay.

As we look down on the Cove and its neighbourhood, we must remember that we have in front of us only a small part of the great expanse of landlocked anchorage available in the harbour; there is still room for unlimited growth, though Sydney has already over a third of the total population of New South Wales collected in and around it.

We must now look beyond the actual harbour, and try to place ourselves in the position of the early settlers. We have great distances to cover, since New South Wales is just half as large again as France; we must therefore keep fairly closely to the railway; but we shall not lose much by this as the railway will carry us through all the important districts of the State.

We may travel north, south, or west, and the map can give us some idea of the character of the country ¹² through which we shall pass. Sydney lies near the middle of a long strip of coastland, shut in on its western side by the steep edge of a great plateau. In the neighbourhood of Sydney this edge goes by the name of the Blue Mountains. Here the barrier is about forty miles inland; further north, in the valleys of the Hunter and Hawkesbury rivers, the lowland widens out to nearly a hundred miles; while in some parts of the south the highlands come right down to the sea. This narrow strip was the original New South Wales.



NEW SOUTH WALES AND VICTORIA: OROGRAPHICAL.
By permission of the Diagram Co.

We can travel now by railway along the coast strip 13 to Newcastle, then up the valley of the Hunter, and finally climb the Liverpool Range on to the plateau beyond. But the journey was far from easy for the early settlers. In fact, until 1820, when a stock route was discovered from Sydney to Newcastle, the only intercourse with the north was by sea, and Newcastle grew up almost as a separate colony in consequence. The valleys of the Hunter and other rivers gave a natural direction to early settlement, since in their lower courses they flow through wide alluvial flats which are very fertile and easy to cultivate. But they are subject sometimes to devastating

floods, as the settlers found to their cost, while the heavy summer rainfall is not well suited to certain of our crops, such as wheat. So in the early days the colony was often in difficulty as to its main food supply.

The name Newcastle at once suggests coal; and coal is everywhere in this district. The surface of the country round Sydney is largely a barren sandstone; but underlying the whole of the area, from Newcastle on the north to Bulli on the south, and extending westward to the other side of the plateau edge, is a vast coalfield. Its chief development at present is around Newcastle. Here is a view of Hetton colliery, ¹⁴ Newcastle; both the name and the picture remind us strongly of the North of England. We see the coal being wound up from the shaft as in our own mines, and in the distance vessels lying at the wharves in the fine harbour. Here again is a general view of the ¹⁵ harbour in which we can clearly distinguish the loading of the coal and merchandise.

A journey southward from Sydney to the other end of the coalfield will bring us to a less familiar type of mine. At Clifton the early explorers found coal strewn on the beach; the actual seam is in the face of the cliff, and shows as a broad black band, while ¹⁶ the coal is mined by means of adits, and then run on to the little pier to be shipped. The coal is found under Sydney itself, and mining is now in progress on the south side of the harbour; but the shafts are much deeper here than at Newcastle, since the coal measures lie in the shape of a saucer, and Sydney is near the middle. We may notice here that the southern railway line ends at Nowra on the Shoalhaven river, and beyond are only a few small coast towns; so we need not at present explore further in this direction.

We will now leave the coast district for a time and climb the plateau edge to survey the country beyond. First let us consider the nature of the obstacle by which the early settlers were faced. The Blue Mountains are merely a part of the eastern rim of a great tilted tableland of sandstone, with a steep face towards the sea and a long and more gentle slope towards the west. Down this face a series of comparatively short streams come tumbling to the sea; while on the other side of the ridge, almost within sight of the sea, are the sources of the slow westward-flowing rivers, whose courses are measured by thousands of miles.

In this sandstone block the torrents have carved out deep gorges, which often widen out up-stream into broad valleys; but these valleys are deceptive and do not provide a road to the interior, since they ¹⁷ end in steep cliffs over which the streams plunge in waterfalls. Here is a view of the country at Govett's ¹⁸ Leap; we may notice the flat tops of the ridges, all about the same level, which suggest the old surface of a plateau. It was a long time before the early settlers found a path over this edge, and the available roads are still very few all along its length, as we may see by tracing them on the map; our train must twist and tunnel up one of the ridges between two of the valleys by a most difficult route, with steep inclines, instead of following the bank of the stream below. We realise that climbing a plateau is a far more serious matter for the engineer than piercing through a narrow ridge of mountains.

At Victoria, on our way up, we leave the train for a coach drive, to Jenolan. Here the scenery changes; the rock is no longer sandstone, but limestone, and the streams have burrowed out many curious gorges and underground channels as in our own Derbyshire. Here we have one of these in the form of a huge rock ¹⁹ archway through which we catch a glimpse of the country beyond; while far down below us flows the stream which bored out the arch. A little further on we find the stream running at the bottom of a lofty ²⁰ cavern, and out into a deep and narrow gorge. Here again is a view of the interior of one of these caverns, with its huge pillars hanging from the roof and rising up from the floor. These limestone tunnels and ²¹ gorges, and the sandstone valleys with their steep surrounding cliffs and narrow outlets, are a fine subject for the artist and tourist in search of beauty, but do not suggest opportunities for settlement or farming; at the same time they are evidently a serious obstacle to movement. The bare surface of the plateau is little better; in fact, the highlands in this district are still among the most thinly populated areas in New South Wales, in spite of their nearness to the capital and the oldest settlements. So we pass through quickly, and coming out by a ²² long tunnel drop down to Lithgow, where we enter an entirely different kind of country. Lithgow is a manufacturing town, with coal mines, ironworks, smoke and dirt. It really belongs to the coast region, and is here, on the inside of the ridge, only because a small piece of the Newcastle coalfield, which underlies all the country which we have been crossing, crops out in this district from under the sandstone. On

our journey inland we shall not meet with any other town of the same type, as we are now entering the great wheat-belt of Eastern Australia.

Here is a typical farm on the eastern edge of the 23 wheat-belt. Beyond the hills, which we see in the background, is the steep descent to the Hunter valley and the coastal plain. The hills are wooded, but the trees thin out and the ground becomes more open as we go westward down the long slope. We must not forget that here at the back of the plateau edge, though we are on the "Plains," yet we are still more than a thousand feet above the level of the sea. We shall realise the importance of this height later. Our next picture shows a wide expanse of level ground, under 24 grain, with the reapers at work. We are at Tamworth on the Liverpool Plains, not far from the northern end of the wheat-belt; but this belt can be traced from Queensland right round to South Australia, and from end to end the scenery is the same. There are the same open sunny plains, dotted with homesteads and small agricultural towns, and covered with the waving grain. Everywhere is the hum of machinery, reaping, binding, and threshing; for labour here is costly and as little as possible is done by hand. We may find it hard to tell, from the appearance of the country, where we are within a thousand miles, and we may be struck by the monotony of the view as we rush through it. None the less this great field of grain is impressive in its own fashion, if we use our imagination and follow the heavily loaded waggons to the station, and on to Sydney, and so across the ocean to London or Liverpool, until the grain appears as bread in the baker's shop. We are watching here the beginning of the process by which the crowded millions of Industrial Europe are fed; and the wide spaces under crop may give us some idea of the greatness of the business; for we have in the wheat-belt of Australia, in spite of its great extent, only a small fraction of the wheatfields of the world.

A rainfall map is the best aid to the understanding of the position and extent of the wheat area. The 25 map is arranged in a series of parallel zones, which show the annual fall decreasing rapidly on the west side of the Divide, as we move further away from the coast. We are crossing one of these zones where the fall is from twenty to twenty-five inches, or not far removed from that of the eastern counties of England. This is the wheat-belt of Eastern Australia, which follows the rainfall right round the inner slope of the plateau as far as South Australia.

This zone gradually shades off into another rather broader where the rainfall is much lower; from ten to twenty inches at most. So the scenery changes as we travel westward until we are lost in the country of the western plains: a great dry lowland not far above sea-level, and drained by slow-moving rivers, the Lachlan, Darling, Murray and their tributaries. The railway runs north-westward, through interminable miles of grass and scrub, until it ends at Bourke, the head of navigation on the Darling. It is the land of the sheep and nothing else. We may gain some idea of the enormous extent of land in this part of Australia, available for pasture or agriculture in some form, by placing upon it for comparison the eastern part of the United States, as in the map which we see here. 26

New South Wales possesses nearly as many sheep as the rest of Australia together, and most of these are to be found on the inland slope of the plateau and far out into the plains, more especially in the Riverina district, between the Murray and the Murrumbidgee. We have left behind us the wheatfield and the reapers; the loaded waggons which we pass, drawn by long teams 27 of horses, are carrying great bales of wool to the railway. We may follow the wool back to the shearing sheds 28 where again all the work is done by machinery; then we go on to the sorting shed, and so to the railway and the showrooms at Sydney, where thousands of samples 29 are displayed for the benefit of the buyers for the markets of Europe. We can see the great flocks of sheep before and after the shearing at the homestead 30 or follow them as they are driven to pasture; and everywhere in this great river plain we find the same thing repeated. The rainfall is not sufficient for agriculture; but in ordinary years it will provide good grass for the sheep; and there is also the drought-resisting salt-bush to eke it out. Sometimes the rain fails, and then there is neither food on the ground nor water in the creeks and pools, and millions of sheep die, as in the great drought of 1901–2. The dry climate gives the best wool in the world, but it is not without its drawbacks; though the large profits made by the farmer in ordinary years more than compensate for an occasional period of drought.

The uncertainty of the rainfall shows itself in another way, in the peculiarities of the rivers. Of all the great rivers in this basin, the Murray alone, fed by the melting snows of the Australian Alps, has a good supply of water at all seasons; the rest are variable. The Darling, Lachlan and Murrumbidgee are navigable for long distances in favourable seasons, and

sometimes are flooded and overflow their banks, turning the surrounding country into a huge shallow lake; but at other times they become, in places, little more than 31 strings of detached pools. Here is a lagoon on the Murrumbidgee, and here is the Murray evidently in 32 flood, to judge from the trees growing out of the water; another view shows us the river in its ordinary 33 state. By way of contrast here is a small creek in the Riverina district; the road crosses it by 34 a ford, so that it is evidently not very deep, and would soon dry up. But after heavy rains, further up-country, the creek may become for a short time a roaring torrent. Settlers new to the country have often made the mistake of camping in the evening on the near side of a creek of this kind, only to find in the morning that the ford has vanished and that they must stay where they are until the water subsides. One of the most remarkable features in Australian weather is this sudden change from drought to flood, which not only transforms the rivers but in a few days gives a covering of rich green pasture where before was a parched desert supporting only the hardy salt-bush.

When the rivers are full we can see the shallow draught steamers collecting wool and other products; but the 35 want of water is not the only drawback. The rivers wind greatly in their courses over the level plain, so much so that at one place it is said that the steamer takes a whole day to pass a particular house, owing to the river bending right back upon itself. The river banks are marked by lines of gum trees, by which the eye can trace them for many miles across the level. Except for this, the whole area crossed by these rivers in their lower courses is one vast treeless plain, covered with grass and scrub in the rains, but at other times dry, dusty, and monotonous. It extends into Queensland and Victoria, but its greatest development is in New South Wales: for though the other colonies have large flocks of sheep, it is here that sheep-raising is the one industry above all others; in fact, under ordinary conditions, no others are profitable or even possible. In this country, next to the sheep, water is the most valuable commodity.

The greater part of the Murray-Darling basin is filled up by recent rock sediment and river alluvium; but the narrow belt of country with a moderate rainfall, lying between the plateau edge and the western plains, has not depended for its development solely on agriculture. All along it the older

rocks crop out, and in the older rocks we find the valuable minerals in which Australia abounds.

Gold, in its alluvial form, occurs all along the agricultural belt; and since the time of the first discovery near Bathurst, in 1851, the search for gold has often caused an inrush of people who have abandoned mining for the more secure and pleasant business of growing wheat or rearing sheep. Though much gold is still produced, New South Wales is not by any means the chief of the States to-day in this respect; but gold has been woven deeply into her history. One of the most usual methods of obtaining gold is still by dredging alluvium; but in place of the shovel and washing-pan we have the ugly machine dredger scooping out the creeks and flats where the gold is to be found. We must look elsewhere for gold-mining from the rock on a large scale; though this is increasing in New South Wales in connexion with the development of mining for other minerals, especially copper.

Well out in the plains, and south of Bourke, at the end of a branch line of railway, is the town of Cobar; it stands just where the old rocks are disappearing underneath the recent deposits. Here is one of the chief centres of copper mining; and, once the work was started, mining for other ores naturally followed. It is a desolate country, rendered more so by the nature of the industry. The furnaces for the rough smelting of the ores need fuel, but coal is far away; so that the country round has been stripped of its small supply of timber, and has nothing left to relieve its ugly monotony. The ore, partly worked, is sent by rail all the way to Lithgow, on the coalfield, to undergo the further process of refining. The importance of these mining fields to the State lies not so much in the money value of the products as in the fact that they give rise to railways and traffic and so to a further spread of the settled agricultural population. The minerals, and especially gold, have played a great part in the settlement of the less accessible or less attractive regions of Australia.

The old rocks, which disappear at Cobar, under the alluvium of the Murray basin, crop out again at the surface in the far west, and give us one of the chief silver-producing areas of the Continent. The natural outlet of the district is by Spencer Gulf, as Sydney is more than twice as far away; and the development of these mines has been largely due to the people and capital of South Australia. Here is one of the most famous mines ³⁶ at Broken Hill; and here we have the camel team, the only means of transport

away from the railway. We ³⁷ are in the semi-desert area, and the existence of the mining population is only made possible by collecting the water from the neighbouring hills in great reservoirs, ³⁸ such as we see in the picture before us.

We have still to see the south-east corner of the State, where we shall find some of the most picturesque scenery and a country rather different from that which we have so far visited. We take a line running south from Sydney, not along the coast, but following the river valleys and so up again on to the plateau at Goulburn. Here we branch off from the main line southwards through the Monaro Plains; this is a high pasture land, thinly populated, though there is a growing agricultural industry in some of the more favoured spots. To the east of the plain are the Coast Ranges, to the west the Snowy Mountains; both extending over the border into Victoria. Cooma, the terminus of the railway, about fifty miles from the State boundary, lies nearly three thousand feet above sea-level. North-west of Cooma is the town of Kiandra, in the Alps, where we find snow and winter sports as in Switzerland. South-west of Cooma is Mount Kosciusko, rising over seven thousand feet, the highest mountain in ³⁹ Australia; here the snow lies even in summer. We reach it by a road following the valley of the Snowy ⁴⁰ River, and can ride or even motor up the track almost to the summit. Here are two views of the river and ⁴¹ its tributaries. Kosciusko is not an imposing peak as we see from these pictures, but merely a flattened ridge ⁴² lying on the top of a great tableland, so perhaps we may be somewhat disappointed in the outcome of our visit. ⁴³

From Goulburn we begin the long descent to the level of the Murray. We are again crossing the agricultural belt, and forty miles west of Goulburn we break our journey at Yass. Here, on the banks of a small stream, the site has been fixed for the ideal city, the future capital of Federal Australia. Notice that we can have here no great industrial and commercial centre, but merely a town like Bathurst, a centre of farming and country life. Perhaps in this it will be more representative of the real Australia than are the larger cities. In position Yass is nearer Sydney than Melbourne; but it is roughly halfway between Brisbane and Adelaide; so that it is fairly central for the long belt which contains most of the population of Eastern Australia. The city is not to be allowed ⁴⁴ to grow haphazard; here we see the surveyors' camp and the surveyors at work, mapping out the ground. ⁴⁵ In the distance is Black

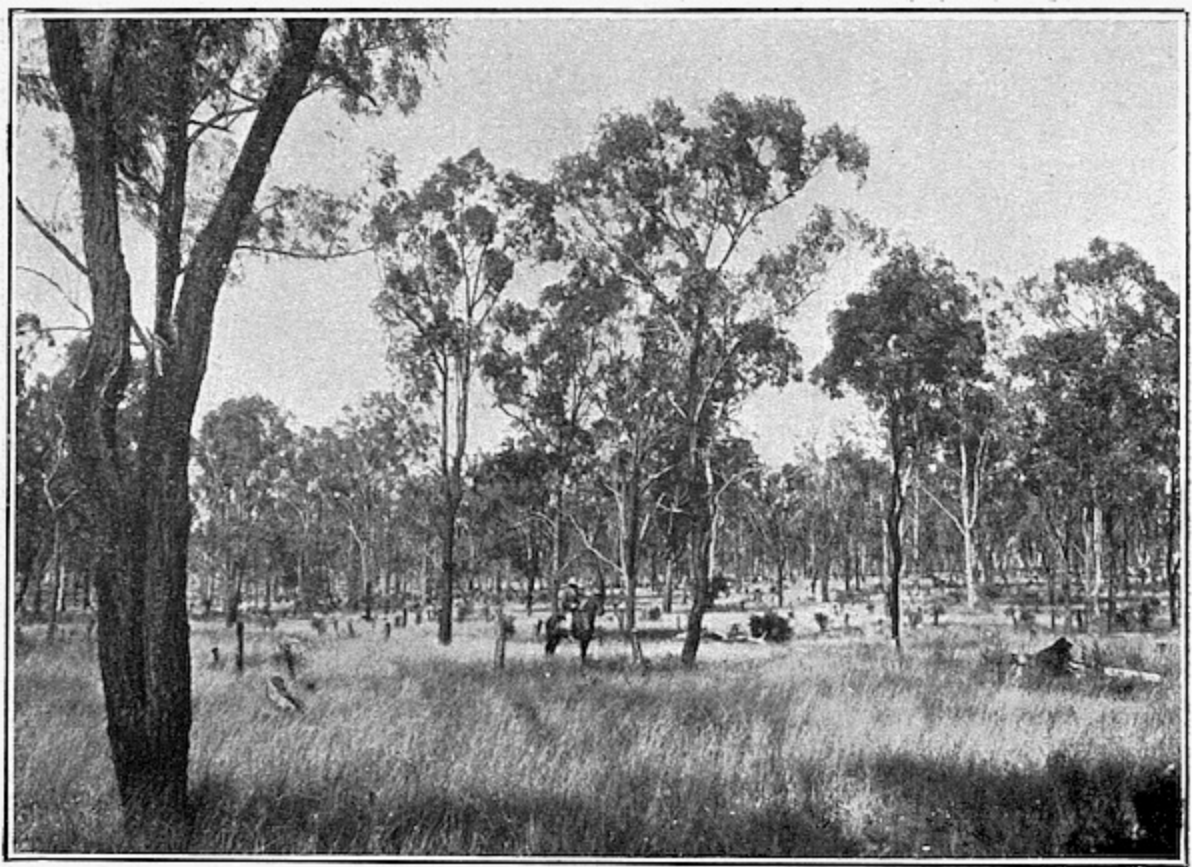
Mountain. The whole scene is quiet and rural, but it will be very different in a few 46 years' time. This deliberate choosing of a site for a new city is common in Australian history; we may contrast with this the way in which centres of population have grown up in the course of ages in old countries almost of their own accord.

We continue our journey down the slope, and crossing the Murrumbidgee at Wagga Wagga reach Albury, the border town. Here, it is necessary to change trains to continue the journey to Melbourne, for unfortunately the different States of Australia did not plan their railways on the same scale. In New South Wales the gauge of the lines is the same as that in England; in Victoria and in part of South Australia there is a broad gauge; while all the other railways in the continent are on a narrow gauge of three feet six inches. This has been adopted as the most economical for opening up a new country; but the differences have led to great inconvenience and loss, where through connexion is made between the main railway systems of the various States. We may remember how, in our own country, the Great Western Railway was forced to abandon the old and comfortable broad gauge, so as to be able to work in connexion with all the other lines which had adopted a narrower gauge; Australia has still to face the problem of unifying her railways in this respect.

We have travelled for many miles over the railways, (13) and now perhaps we may begin to notice certain peculiarities in their arrangement. First there is the main-line system connecting up the capitals. This runs north-east and south-west from Sydney, roughly parallel to the coast. Only a short stretch of this is on the low coastal plain; the rest is inside the plateau edge. The line descends through the Victorian Mountains to the sea at Melbourne; but goes inland again on its way to Adelaide. Branching from this system, or starting independently from the coast, is a whole series of lines running inland, roughly at right-angles to the coast. Some are very short, some very long; and they commonly end at a small town on one of the rivers. We can trace them right round from the line between Normanton and Croydon, in North-West Queensland, to that ending at Oodnadatta in the desert region of South Australia. Except round Bathurst, and in the country at the back of Melbourne, we shall not find many branches or cross connexions. This curious arrangement can only be understood in the light of the resources

and historical development of the country; we have already seen something of its meaning in New South Wales.

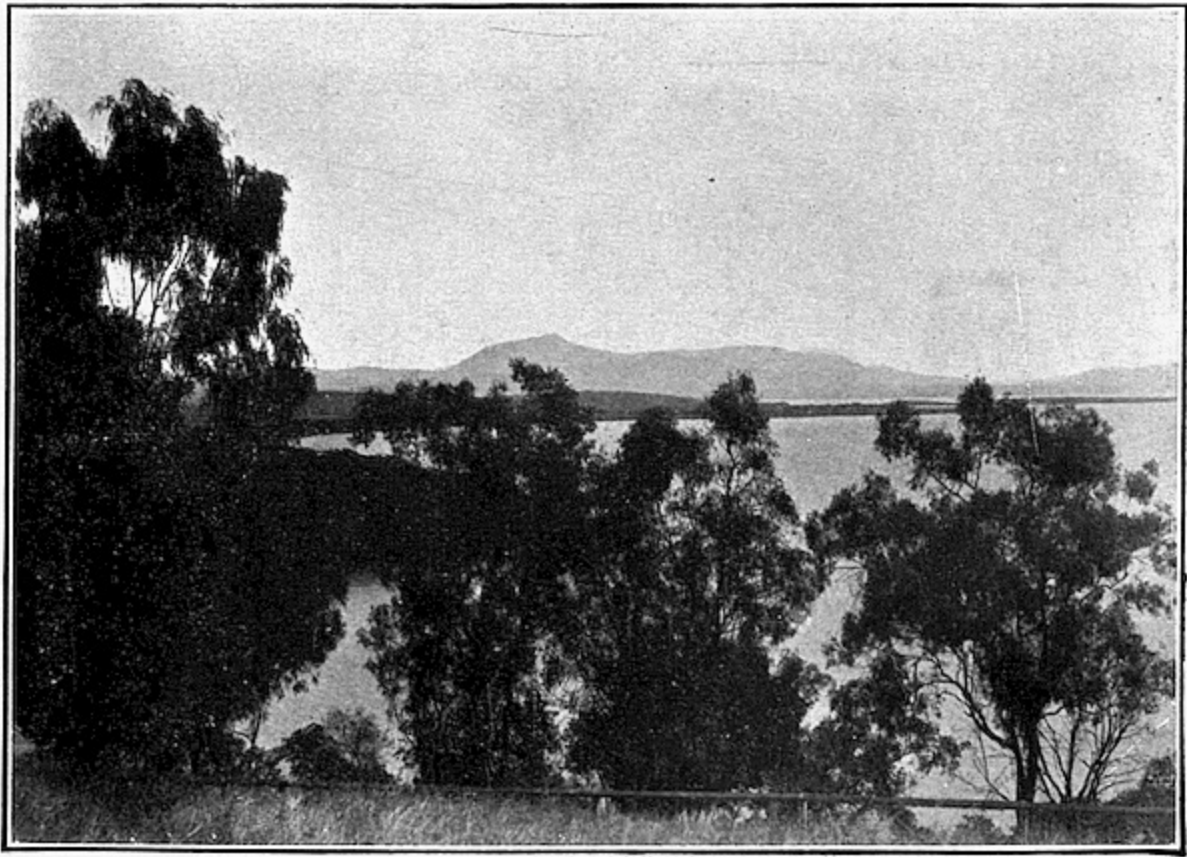
We noticed that the line from Sydney left the sea at Newcastle to follow the valley of the Hunter and scale the edge of the Liverpool Downs. For two hundred miles north of Newcastle the coast district lacks a railway; but in Clarence county, in the extreme north of the State, there is a detached piece of line running for a hundred miles not far from the coast, and touching it at one or two points. This line has a meaning.



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[See [page 40](#).

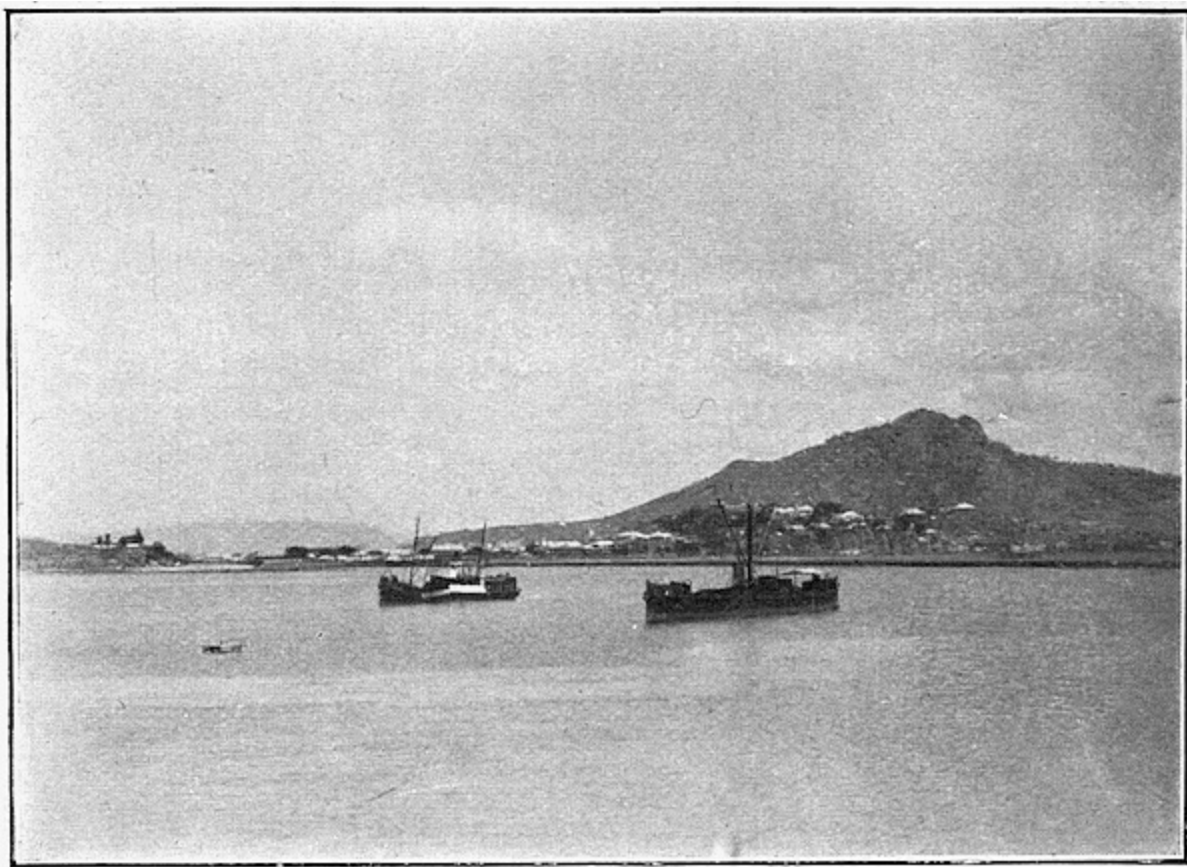
ON THE DARLING DOWNS.



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[See [page 41](#).

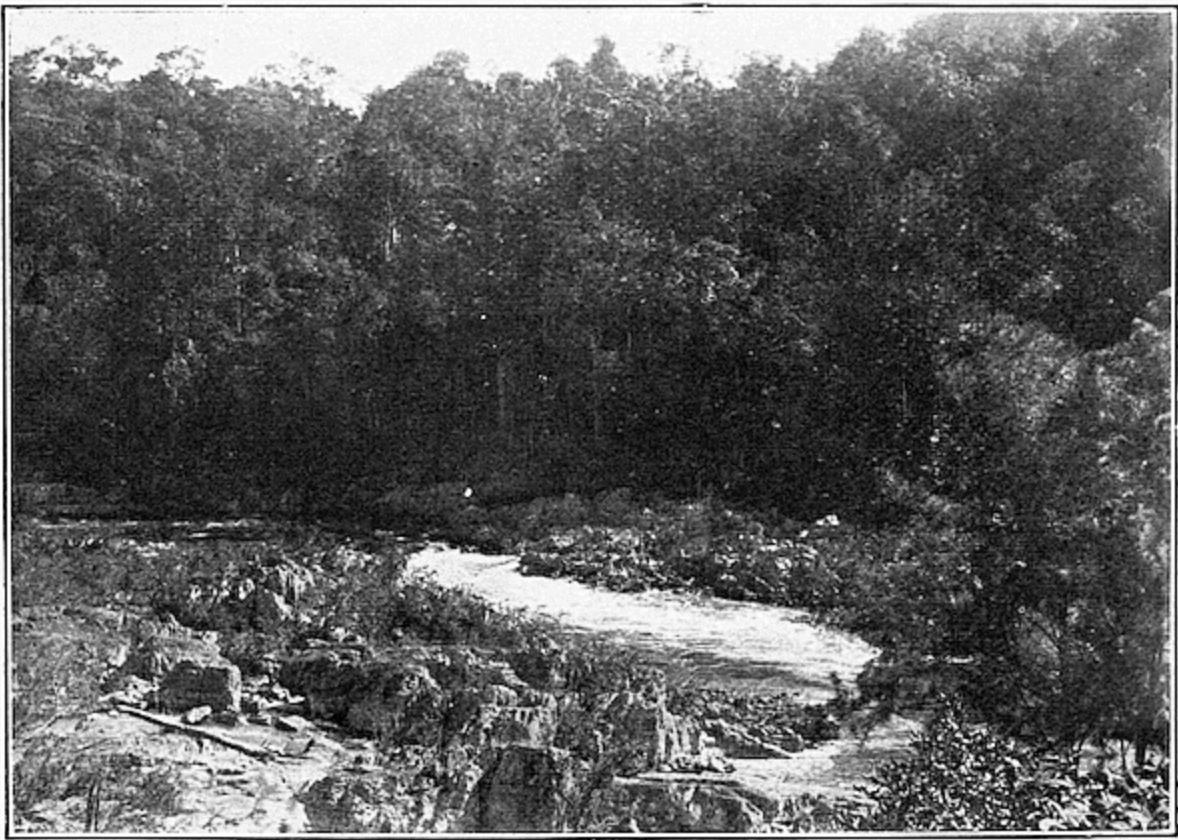
GLADSTONE.



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TOWNSVILLE.

[See [page 43](#).



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[See [page 46](#).

ABOVE THE BARRON FALLS.

The district is warm, as it is low-lying, and not very far from the tropic; while the south-east winds bring abundant rains. It has been found to be well fitted for dairy cattle, and is displacing to some extent the coast district south of Sydney where the industry first started. Sydney still provides a good local market for the dairy products of this northern region, but there is also a growing trade with more distant places. We can understand now the need of a railway to open up the country and connect it with the sea.

Here is a typical dairy farm with the cattle clustered ⁴⁷ in the shade on the banks of the creek. We notice the abundance of trees, and the curious dead bare look of some of them. These have been ringbarked, that is, a strip of bark has been cut away right round the trunk; this process kills the tree quickly, and the dead wood can then be burnt off. It is a rough and extravagant method of clearing, and some of the forest, which grows luxuriantly in this rainy district, is put to better use. In the wetter parts are to be found the cedar and other soft ornamental woods; in the drier, are the

various hard woods of the eucalyptus family, especially the ironbark and the blackbutt. It was the timber which attracted the first 48 settlers to this district, though the difficulty of transport has prevented them from making much use of it up to the present. Here we see the felled timber lying ready for removal; it must be dragged over rough 49 tracks, often for long distances, by teams of horses and bullocks. We can gain some idea, from these pictures, of the huge size of the trees and the difficulty of forest development. Evidently the forest further inland can only be attacked by the aid of the railway. We shall find similar conditions in Queensland; in fact, we may look on this coast strip as giving us geographically the beginning of the Queensland coast conditions; for round Grafton, at the southern end of the railway, we find the cultivation of the sugar-cane.

We have seen the beginning of the new capital of Federated Australia; we will now, before visiting Queensland, cross Torres Strait, with its innumerable islands and reefs, for a glimpse of Australia's new tropical colony. British New Guinea, or Papua as it is now officially styled, was annexed in 1888, owing to pressure from the Australian colonies, and more particularly Queensland. From the first, Queensland, New South Wales and Victoria contributed to the cost of administration; and in 1906 the new Commonwealth Government took over the entire control.

British Papua is a curiously shaped corner, carved 50 out of the eastern end of the great island of New Guinea. The western end of the island is entirely Dutch; the eastern we share with Germany. We may think of British Papua as two separate blocks, as the Gulf of Papua almost divides the territory into two. In the west is a rectangular area, with a low marshy coast, fringed with mangroves and split up by river deltas, especially that of the Fish River. The dividing line in this district between British and Dutch territory is merely a line of longitude. The country is mostly unexplored, except along the Fish and Strickland Rivers, and the natives are still fighters and cannibals. On the north of this block of country, and continuing south-east through a long narrow peninsula, is a high mountain backbone, on the other side of which is German territory. The eastern peninsula is mountainous everywhere; while the whole country is wet, densely forested and difficult to penetrate. The peninsula ends in a string of islands, mostly volcanic.

The colony is in the first stage of organisation, when the main problem is to reduce the native to some kind of order. Let us see what he is like and how he lives. 51 Here are two inhabitants of the coast district; they seem very different from the aborigines of Australia. Notice their frizzy hair, standing out in a great mop, and their bracelets and necklaces. The Papuans are fond of personal adornment. Here is a girl from the 52 same district; she wears an elaborate girdle of grass. Behind her we see the end of a curious canoe, with an outrigger. The canoe is important to the Papuan, since he commonly plants his village at the water's edge. Here is a village, and here is a nearer view of 53 some of the houses; they are merely covered platforms, built on piles. Fighting and headhunting are 54 still the amusements of the tribes which are not yet brought under our control, but conditions are changing rapidly for the better. Here is one of the instruments 55 of the change, the native village constable, who seems quite proud of his office. Behind him, law and order 56 are represented by the visiting magistrate with a small force of armed constabulary. The chief difficulty in opening up the country is that of movement. Everywhere we find forest, mountain, and unbridged streams. 57 Here is the kind of track through which the explorer must force his way, and here we see two methods of 58 crossing a stream. The native bridge hardly seems calculated for heavy traffic. We may realise from 59 these pictures the nature of the task of controlling the natives of the interior, such as the formidable pair in 60 front of us. Even when reduced to order the Papuan is not anxious to develop the country by work on plantation or mine.

Port Moresby, the capital, stands on a fine bay in a relatively dry district. Here a few score white people represent the influence of civilisation. The climate forbids effective settlement. Here we see a 61 European house with its staff of servants, and here is the steamer which links Papua with the mainland. It 62 will be interesting to see how Australia solves the various problems of her new tropical Dependency. In Queensland we shall find similar problems, though in a modified shape.

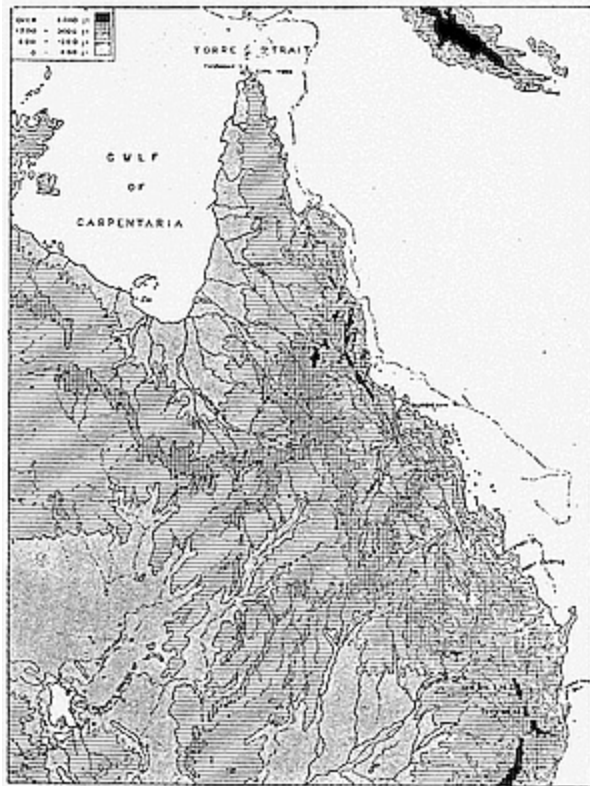
LECTURE III

QUEENSLAND

The land route from New South Wales to Queensland does not at present follow the sea-coast. The railway at Newcastle turns up the valley of the Hunter River, climbs the steep edge of the plateau, to run along the eastern rim of the Liverpool Plains and the Darling Downs, and then descends again by a steep pass to the sea-level at Brisbane. At the little frontier station of Wallangarra we must change trains, since the Queensland railways, as we have already noticed, are on a narrower gauge than those of New South Wales. This would be a very serious matter but for the alternative route by sea to Sydney; this is the natural route for heavy goods, since nearly all the important towns of Queensland are on or near the sea-coast. Before the advent of the railway, the sea was the sole means of intercourse for all the towns on the eastern rim of Australia; even in our own country, where the railway system is highly developed, the coasting trade is still of very great importance.

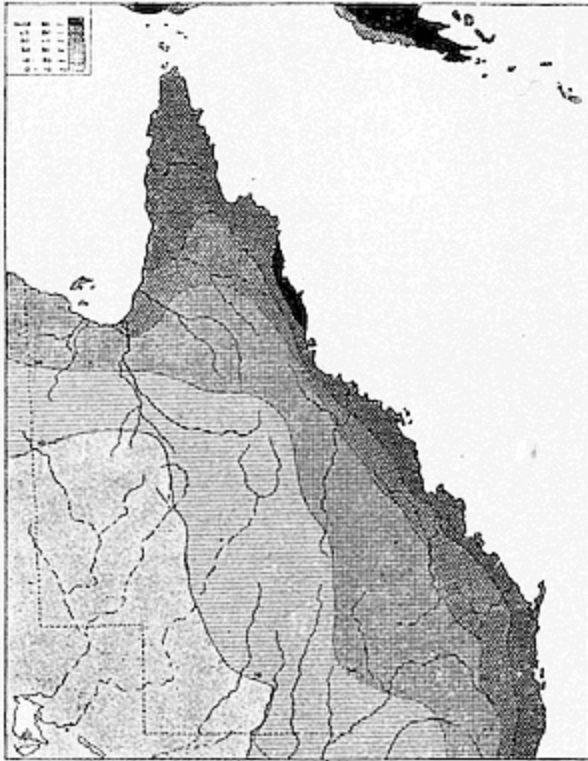
The course of the railway suggests that the structure of the country is not unlike that which we have seen in New South Wales. This is true of the Darling Downs area, but further north the map shows us a somewhat different type of country. The eastern part of the State consists in the main of a broken and irregular highland mass; the west of rolling plains, sloping gently towards the interior or the Gulf of Carpentaria; but we look in vain for a long, well-marked escarpment, such as we find further south. The mountain ridges seem to run in every direction, sometimes, as near Cairns, forming a definite coast-range, at others striking inland or running down in spurs to the sea; so that the country is split up into a number of distinct basins, each with its own group of rivers flowing in the most diverse courses. Thus, from Cairns to Brisbane a great stretch of country, broadest in the middle, narrow at both ends, drains into the Pacific; but the Burdekin

and the headwaters of the Fitzroy flow for long distances parallel to the coast, before turning and breaking through the ranges to the sea. Another group in the south joins the Murray-Darling system and belongs physically to New South Wales; the rivers of the north-west and of the Cape York peninsula drain for the most part into the Gulf of Carpentaria, while a large block of country in the west and south-west is occupied by intermittent streams, which in time of flood find their way into the inland basin of Lake Eyre, in South Australia. The country has not that simplicity of relief which we found over the greater part of New South Wales, and, as we might expect, the rainfall does not show such clear and symmetrical divisions. The fall from the south-east winds is more irregular and more widely distributed inland than in New South Wales; while in the north there is an area with tropical rains of a monsoon type.



QUEENSLAND: OROGRAPHICAL.
By permission of the Diagram Co.

On our railway journey we have crossed from one State to another, but we must notice that, except in the south-east corner, the boundaries of Queensland have no relation to the natural features of the country; they are merely mathematical lines. The reason for this is to be found in the method by which the settlement was carried out. Moreton Bay was one of Cook's landing-places in 1770; but the Brisbane River, flowing into it, was not discovered until 1823; the first settlement was formed in 1824, and the Province remained part of the mother State until 1859, when after much agitation it became an independent settlement. The interior was not explored at the time, so that the only way of determining a boundary was to follow a line of latitude or longitude. A similar method has been used in more recent times in parcelling out unexplored regions of Africa among the European Powers. The western boundary of the new State was the line of



QUEENSLAND: RAINFALL.

longitude 138° E., and what is now the Northern Territory remained nominally part of New South Wales, which thus consisted for a short time of two areas widely separated.

The very name of the city of Brisbane recalls the connexion with the older colony, since Sir Thomas Brisbane was Governor of New South Wales in the years 1821–5, at the time of the first settlement. The city stands, not on the shores of Moreton Bay, as we might expect, but twenty miles up the river, on both banks, which are connected by the Victoria bridge. Here is a view over the bridge from the north

bank; 5,6 and here is a wide view of the river beyond the city. There is plenty of space in Brisbane, with its suburbs, for the population of 100,000; there are parks and gardens everywhere, and a large number of fine public buildings. Here are the library and the Executive Buildings in a beautiful garden, with a statue of Queen Victoria; here again the Parliament House, and here one of the main streets of the city. We have nothing like this in any town of the same size in England, but we must remember that Brisbane has been built for the future, and is the capital of a State more than three times the size of France.

Brisbane lies in the extreme corner of Queensland, not, it may seem, a very good position for the capital; but the south has the more temperate climate, while behind Brisbane is some of the most fertile land in the State. Westward a railway runs for nearly five hundred miles, at right-angles to the coast-line, to Charleville on the Warrego; we are about to make a rapid survey of the country from it. Twenty-five miles out, we pass through Ipswich, at the head of the river navigation; it is a busy town with valuable coal mines and the main railway works of the State. Then we climb again the steep plateau edge, which we descended on our journey from Sydney,

and come out at Toowoomba, on the Darling Downs, fifteen hundred to two thousand feet above the level of the sea.

The Darling Downs country was reached in 1827, by Allan Cunningham, botanist and explorer, who travelled by way of the Hunter River and the Liverpool Plains; but the journey was difficult, and the first occupation was not until 1840, when Patrick Leslie brought over a few sheep and settled in the neighbourhood of Warwick. Others soon followed; the direct road to the coast was discovered, and the basin of the Condamine River became a great pastoral country where fortunes were made by the early squatters. There is less rain here than on the eastern side of the plateau edge, but it is sufficient for agriculture, and there is plenty of water in the streams. Much of the soil is volcanic in origin, and of great fertility, so that the land is wasted on 10 sheep. Here we see the natural grass in this fertile region, and here is a great sheep run. The rancher's 11 home, which we have next, suggests comfort and success. At the present time, with the aid of the Government, 12 the great pastoral properties are being broken up gradually, and sold or let to farmers; so that a district which started as a sheep run bids fair to become one of the most important agricultural areas in Australia. Toowoomba, the chief centre of this fertile district, has 13 already the air of a busy and prosperous market town, as we may judge from the picture here.

As we travel further west the country becomes drier and rather less fertile, so that agriculture gives way more and more to sheep. The conditions are not unlike those which we found to the west of Sydney; and we have seen that the Condamine and other rivers of this area all belong to the New South Wales river system. In fact we are crossing the northern end of the long agricultural belt which lies behind the coast ranges. Though we are close to the Tropic, the climate is not very different from that further south, owing to the height of the plateau above the sea. There are even slight frosts in the winter. Further north, we shall find a marked change; wheat disappears and sheep give way more and more to cattle. The cause of this lies in the different conditions of rainfall and temperature. To visit this country to the north we must return to Brisbane and resume our journey along the coast.

Our first port of call is Gladstone, nearly three hundred miles north of Brisbane, on the landlocked inlet of Port Curtis, one of the finest natural harbours on the whole coast. Here is a view of the bay and the 14 jetty. As

we see, there is no very great trade at present, no line of wharves and warehouses; the importance of 15 Gladstone is in the future; its chief business at present is the shipment of meat and cattle. A short railway journey takes us to Rockhampton, which lies some 16 distance up a river, the Fitzroy; in this it resembles Brisbane. Near Rockhampton we find a steamer 17 loading frozen meat from the factory. From Rockhampton the central railway runs nearly due west for over four hundred miles, to the Thompson River on the Bowen Downs. There is also a railway along the 18 coast to Brisbane, linking up the various small seaports; but this line is a late construction. The typical railway of Queensland starts from the coast and runs directly inland; and the development of the country follows the same course.

As we follow this inland line, in the wetter districts near the coast we find cattle everywhere. Further west, where the rainfall grows less, there are more sheep; but the area of considerable rainfall is much greater than in the districts further south, owing to the broadening out and irregularity of the eastern highland mass. The whole of this moist area is particularly suited to cattle.

Before the settler can begin either cultivating the soil or raising cattle, there is much preliminary work to be done. A large part of Eastern Queensland is covered with forest or scrub, which must be cleared by cutting down and burning. Here we have a settler starting 19 operations in a rough camp: he seems to have a difficult task before him. Here he has reached the stage of a 20 permanent hut, with a small area of cultivated land round it, and beyond, roughly fenced pasture with the remnants of the forest showing in it. Here again are 21 the cattle feeding in pasture where the trees have been ringbarked and so partly destroyed. To destroy, cut down, and burn trees which may yield valuable timber may seem an extravagant method, but the settler here has even less choice than in the Grafton district of New South Wales, where we saw the same methods employed. The timber is certainly valuable. Here are specimens of the hardwoods which we use for street pavements and railway sleepers, and for other purposes where great strength and endurance are needed; these are the blackbutt 22 and Queensland Karri pine, and there are many other varieties. The difficulty is to get them to the markets where they are wanted. Here we see one method; a trainload of sleepers is leaving the sawmills, 23 bound for

India; but the railways are few and in many districts the logs must be dragged at a slow rate by bullock teams to the banks of the nearest creek, where they lie perhaps for months until there is enough water to float them down to the larger rivers and so to a seaport. This method is commonly used for the cedar which grows near the coast, and we have already seen it in New South Wales. Except in a few places easily accessible, the natural forest wealth of Queensland is as yet hardly touched.

The railway which we have been following from Rockhampton runs almost along the Tropic. Townsville, our next landing-place, over eight hundred miles north-west of Brisbane, is well within the Tropic, yet in many ways it resembles Rockhampton and Gladstone. From this point another long line runs inland to end on a tributary of the Diamantina, about a hundred miles north-west of the terminus of the line from Rockhampton. We may notice that both these lines, after crossing the wide area of coastal drainage, reach the streams flowing into the Lake Eyre basin. They thus form real links with the interior. After Brisbane, Rockhampton and Townsville are the most 24 important ports in the State. Townsville lies on a broad open bay, and the harbour has been made at 25 great cost by building out long jetties. This has been done because here is the sole outlet for a very large area of country to the west. Here we find more meat and 26 cattle; and again at Bowen, a small port on a fine natural harbour which we passed further south, they are landing cattle for the local stockbreeders. In short, the whole of this north and north-eastern district is 27 the home of cattle; cattle running on the cleared scrub lands and pastures of the interior, or fattening on the rich alluvial lands round the creeks near the coast, 28 where the surroundings, as we see from this picture, are very different.

The coast region itself has something of far greater interest than cattle. There is here a higher uniform temperature than in the uplands, and in some parts a very heavy rainfall; the result is the growth of tropical and sub-tropical plants, and of these plants one of the most interesting to us is sugar. We find sugar-cultivation in patches, all along the coast from Brisbane northwards; and a little even in the extreme north of New South Wales, since the change in climate is very gradual in the coast region. But the most important districts for sugar lie near or inside the Tropic, where the cane is grown on the rich alluvial lands of the coast plains and river valleys.

On our voyage to Townsville we passed Mackay, one of the chief centres of sugar-production; outside the Tropic is Bundaberg, where the cane is grown by the aid of irrigation; while in the far north is the important district of Cairns. The fertile land is covered with dense scrub and must be cleared; here is a clearing in progress; ²⁹ notice that the scrub is different from that which we saw up-country: it is palm. Within two years of planting the shoots the cane is ready for cutting, and in the northern districts we can go on for many years cutting the new canes as they spring up, without the need of re-planting. Here is the cane growing by irrigation in ³⁰ Bundaberg, one of the drier districts, and here is the reaping of the harvest. We notice that the cutters are ³¹ white men. In the early days of sugar, the cultivation was on large estates, each manufacturing its sugar in its own mill. The heavy work was done by Kanakas, imported from the islands of the neighbouring Pacific. This is no longer allowed, and the cane is grown more and more on small farms, by white labour. The growers then sell it to a central mill, where the cane is crushed and the juice extracted. Some of the mills are owned by the small farmers on the co-operative principle, but more often the miller has nothing to do with the growing of the crop. Here we see a trainload ³² of canes bound for the mill. There is now a Government bounty given for sugar produced entirely by white labour, and it seems to have been proved that, on the small farming system, it can be grown thus and show a good profit. But there are other and more attractive occupations for white people in Queensland, and though the whole coast, right round to the Gulf of Carpentaria, is suitable for sugar, it does not seem likely in the near future to become a large industry in a White Australia.

Sugar is one of the most interesting and valuable of the tropical products of the coast region; but many others are grown, some for the market, others hardly beyond the experimental stage. Let us look ³³ at some of these. Here at Woombye are pineapples and bananas growing; and on the Johnstone River ³⁴ huge crates of bananas are being shipped for the markets in the towns further south. At Kuranda, on ³⁵ the Barron River, is a large coffee plantation where we may see the bushes growing and follow the berry as it is dried and husked. Here we notice the ³⁶ pruning of the coffee bush, and here is a fine specimen with the pruning completed. Not far away, in the ³⁷ State Nursery at Kamerunga, we find all kinds of tropical plants growing side by side. The nursery is an official experiment ground, since the Government is anxious to test the possibilities of the region for all kinds of

economic plants. We walk down a fine 38 avenue of palms and visit the quarter where they are experimenting with rubber trees. Here is a large 39 plantation of the trees, and here we see the method of tapping. In a corner we come on a curious African 40 rubber tree, in which the juice exudes, not from the bark, but from the fruit. More prosaic, but none the 41 less useful, are the fibre plants, such as sisal hemp, of which we have a fine specimen here. Cotton-growing 42 has also been attempted on a small scale, and the Commonwealth Government provides a bounty for its encouragement. The main difficulty is the high cost of labour for its cultivation. Here we see the picking 43 of the cotton. We may gain some idea from these gardens of the great variety of tropical and sub-tropical plants, all of which will grow well on some part of this northern coast. The only need is labour to clear the scrub and make full use of the fertile alluvial soil and the warm rains.

Clearing is very necessary in this region since the bush grows with great luxuriance. By travelling inland a few miles from Cairns we may find a picture 44 of the bush in its natural state. The coastal plain south of Cairns is very narrow, for in the background, a short distance inland, are various mountain ranges such as the Bellenden Ker range, said to be the highest 45 in Queensland. Only twenty miles from Cairns, in the foothills of one of these ranges, the little Barron River comes tumbling down in rapids and falls, amid 46 some of the finest scenery in Australia. Here is the river a short distance above the falls; and here are the 47 falls themselves from above and below. The railway, following the course of the river, brings us to the 48 little township of Atherton. Here is the main street. We notice here a Chinese joss-house; this may serve 49 to remind us that there are other immigrants besides the Kanakas in this northern region of whom account 50 must be taken by the advocates of a White Australia. The Chinese have already spread all over the Malay 51 region to the north, and might equally occupy tropical Australia if special measures were not adopted to check their immigration.

In the country round Atherton we have all kinds of typical bush scenery. At one spot we pass giant fig-trees overhanging the road; next we enter the 52 denser bush. We find lakes in the bush, of which the chief is Lake Eachem, which we have seen before. Everything suggests warmth and moisture. The rivers, lakes, and natural vegetation of this region, together with the fertile soil in the low-lying strip between the mountains and the

sea, give us a picture of the conditions which prevail along the greater part of the Queensland coast. Here is a tropical garden, typical 53 of the coast. Similar conditions extend right across the northern edge of the Australian continent.

We have seen that the coast strip at Cairns is very narrow: behind the mountain edge we shall find the rivers flowing west into the Gulf of Carpentaria. As we travel westward we enter rather a different type of country. There is less rain than on the coast, but more than in the interior further south. We must remember that in this part of Queensland we are getting near the zone where a heavy monsoon rain sweeps in from the Pacific in summer. All this country is well fitted for cattle; but it is as yet thinly peopled, since there are few railways, so that it lacks an outlet to the sea. The existing railways are intended to serve the mining districts, since the gold of this region has been the chief attraction up to the present time. There are goldfields scattered all along the eastern side of the highlands. On the Cooktown line is Palmerville; on the Townsville line are Charters Towers and the Cape River field; near Rockhampton is Mount 54 Morgan, one of the most famous mines in the past; it has given rise to a large town which we see here. Here, too, we see the crushing of the ore, preliminary 55 to the chemical extraction of the gold. Near the coast, north of Brisbane, is the Gympie field, and in the far north, near the Gulf of Carpentaria, is Croydon. There are many smaller fields, and we can easily see the connexion between the railways and the gold. There are other minerals, too, in this area, 56 so we see a whole township based on tin, and another growing up round a copper mine. 57

Some of the railways have done more than develop the mining areas; they have been pushed westward into the pastoral country on the long inland slopes of Queensland. We have already followed the southern lines for some distance, and we must now carry our exploration beyond the railway zone, for a very large area in the State is as yet quite untouched by railways. This area will be greatly reduced by the execution of the scheme contemplated by the State Government for linking up all the railways both along the coast and inland. First there is the country in the basin of the Flinders River, round the southern end of the Gulf of Carpentaria; this is being opened up slowly from the sea. It contains much good pasture land, and already feeds a large number of cattle. Further south is the drier country

where the streams drain towards the Lake Eyre basin; this is part of the Rolling Plains which occupy West and South-West Queensland and the north-west corner of New South Wales. This area is most interesting owing to the possibility of obtaining water from beneath the ground.

The rainfall is somewhat scanty, and in the dry weather the creeks lose themselves in the desert sands or become mere strings of waterholes. There may be enough food for the cattle or sheep, but they must have water also, and this can only be found underground. In the higher country to the east, where the rains are relatively heavy, are large areas where the water sinks readily into the ground, as the rock is porous. A long way below the surface it is held up by impervious strata of clay or rock and begins to creep away downhill towards the west and the Gulf of Carpentaria. To the west the surface also becomes impermeable, so that the water is confined between the two layers, and can be reached anywhere by boring through the top covering. Sometimes in these boreholes the water does not rise as far as the surface of the ground, and so must be pumped up; but more often it spouts out with great force, as we see it here. Some of these artesian wells give over a million gallons a day, and penetrate for three or four thousand feet into the earth. The water is often warm and charged with salts; but this does not seem to harm the cattle. The water from the bore is run off into trenches extending perhaps for miles, so that there is plenty of room for the cattle to drink. By the aid of these wells cattle can live in the country all the year round; and even the desert stock-route from South Australia to Queensland can be kept open in the dry season. The utilisation of a simple geological fact is changing the whole face of the country.



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[See [page 55](#).

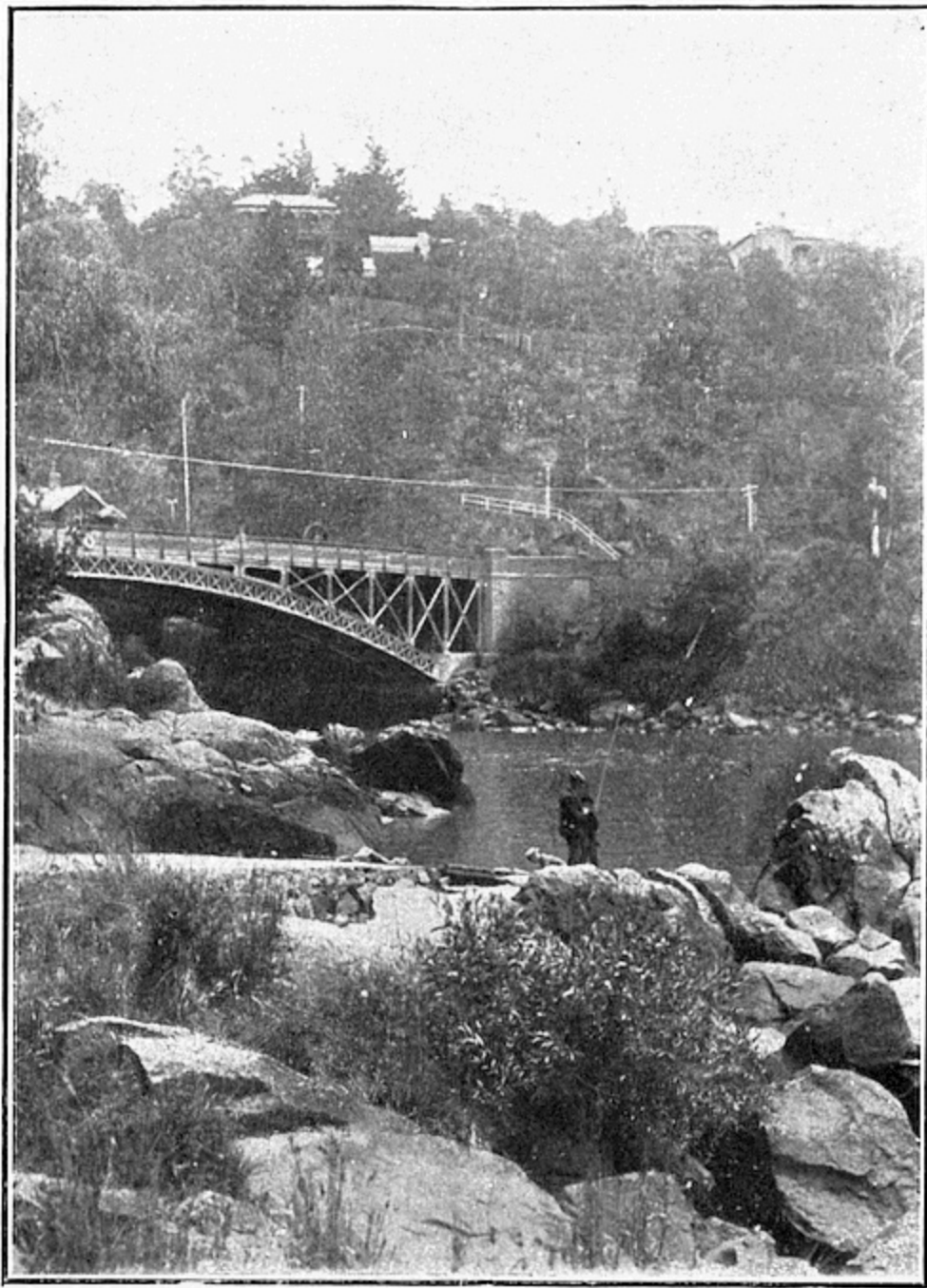
ON THE BANKS OF THE YARRA.



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WASHING GOLD DUST.

[See [page 56](#).



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KING'S BRIDGE: LAUNCESTON.

[See [page 60](#).



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[See [page 65](#).

SCAFFOLDING A TREE.

We have already seen that most of the chief towns of Queensland lie on or near the coast, and that there are many harbours, often protected by islands. The towns are there because the country was developed for the most part from the sea, and in fact settlement inland is still confined to the

neighbourhood of the railways running up-country from these coast towns. The harbours are there because the land has sunk and the coast is partly drowned, giving deep sea-inlets, often where the sea has flooded some old river valley. As the coast gradually sank, the coral builders were at work, piling up their reefs in the warm shallows. So we get the Great Barrier Reef, stretching for fifteen 59 hundred miles along the coast and leaving a calm though rather dangerous channel between itself and the mainland. The reef is really a series of reefs, resting on a platform of older rock, and pierced with numerous openings, especially where the larger rivers enter the sea; for the coral will not grow in fresh water. In the neighbourhood of this reef, especially in the north, towards Thursday Island, fishing for pearl oysters is largely carried on. The oysters are valuable for the mother of pearl, rather than for the pearls themselves, which are very small. Here, too, is found 60 in great quantity the trepang, or sea slug as it is sometimes called, looking like millions of brown cucumbers 61 crawling over the reef. Here is a portrait of one kind of the trepang; it does not seem very appetising, but the Chinese consider it a great delicacy. Nearly the whole of the trepang gathered in this region is exported to China, after being first cooked and dried. The rest is eaten by the natives, as it does not appeal to European tastes.

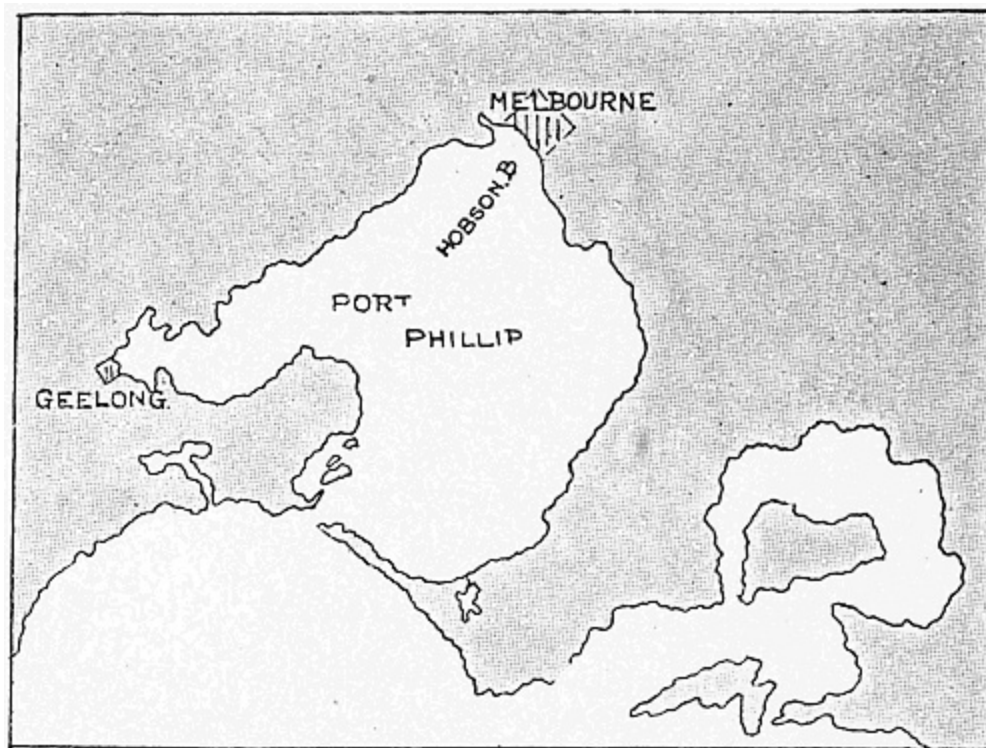
In New South Wales we did not encounter the aboriginal Australian, since there he has practically disappeared. But he still survives in considerable numbers in parts of Queensland, where the country is less favourable for white occupation or has been settled for a shorter period. Government at the present day protects the aborigines as far as possible; but none the less they are steadily dying out and do not count in the future development of the country. We have already seen how, for the hard work of the plantations, the brown men from the islands of the Pacific were brought in, as there was no native labour available. The Kanaka has now been rejected, and Australia steadily refuses to admit the Chinese, who seem to be able to adapt themselves to any country and any climate. So the future of this northern part of the continent depends largely on the extent to which the natural resources of the country, as distinct from its minerals, can be exploited by purely white labour. Some parts of the coastlands are clearly not fitted for European occupation, and the policy of a White Australia is only rendered possible by the fact that the elevation of much of the country within the Tropic greatly modifies the climatic conditions. If

Northern Australia had been a great lowland, its history must have been far different.

LECTURE IV

VICTORIA AND TASMANIA

Following the example of the original settlers, we will approach Melbourne from the sea, as in this way we ¹ shall perhaps get the clearest view of the peculiarities of the State of Victoria. From Sydney to Cape Howe, we are still following the coastal plain of New South Wales, with the plateau edge in the background. But when we round Cape Howe and turn westward the coast changes: a series of mountain ridges runs down to the sea, ending in promontories with fiord-like inlets between them. The eastern end of Victoria is occupied by an irregular mountain mass, trending on the whole north and south. West of the mouth of the Snowy River, the coast scenery changes again, and we have Ninety Mile Beach. Here long banks of sand, brought by the strong currents from the west, have silted up the mouth of an old river valley. The water is thus held up and spreads out into lagoons, which communicate ² with the sea here and there through narrow channels. Further west still are isolated mountain ranges, one of which ends at Wilson Promontory, the southernmost point of the mainland of Australia. In the bight between this promontory and Cape Otway, lie Western Port and Port Phillip, wide bays with narrow entrances. Beyond Port Phillip the coast is fringed by the Otway ranges; and then follows a low plain, with few inlets or good harbours, a region of lakes and swamps. So we see three great irregular curves or bights, with great variety of coast-line; Port Phillip lies at the top of the middle curve. It ³ is a drowned valley, like the lagoons further east, and is almost blocked at the mouth by the drifting sands. It is the only good natural harbour on this part of the coast, and is still the centre of settlement and of the area of densest population.



PORT PHILLIP.

Though the harbour of Port Phillip was discovered at the beginning of the nineteenth century, it was over thirty years before a permanent settlement was planted on its shores. Something was learnt of the country in 1824, when Hume and Hovell, travelling overland from New South Wales along the route now partly followed by the railway, reached the spot where Geelong now stands. The coast at that time was unoccupied, except for a few whalers who were settled at Portland in the far west. The real occupation of Victoria was brought about by the fusion of two distinct streams of immigrants, one coming by land, the other by sea. In 1836 Sir Thomas Mitchell, one of the greatest of the New South Wales explorers, came over from the Murray basin and discovered the fertile plains at the back of Port Phillip; Australia Felix he named this country of promise. His report on the country led to a rapid movement from New South Wales over the border southward. But the journey from Sydney by land was long and arduous, and the southern part of Victoria, like the coast region of New South Wales, was most easily and naturally settled from the sea.

In 1835 a Tasmanian, John Batman, representing a syndicate of Tasmanians, surveyed the site of Melbourne and tried to buy it from the

natives; but the New South Wales Government refused to sanction the arrangement. Still other Tasmanians followed, and a body led by Fawkner actually settled on the Yarra in 1835. The two parties naturally quarrelled and the matter was complicated by the fact that New South Wales claimed the whole territory. The British Government supported this claim, and, as a result, in spite of the difficulty of communication, the people of Melbourne had to send their parliamentary representative to Sydney for some years. At length, in 1851, the Port Phillip district became independent and was re-named Victoria. The name of the State and its capital easily remind us of its history; for the official founding of Melbourne was in 1837, when Queen Victoria came to the throne and Lord Melbourne was Prime Minister.

Melbourne and Port Phillip have not the picturesque appearance of Port Jackson and Sydney; the broad, lagoon-like harbour does not lend itself to scenic effects, and we have not the deep-water inlets penetrating the heart of the city which add so much to the beauty and utility of Sydney. Melbourne is built on the level, with broad, straight streets and fine buildings, modern and handsome, a typical Australian city. Here are Collins 4,5 Street and Bourke Street. Here again are some of the Government offices; the statue in the foreground is 6 one of General Gordon.

Melbourne is a true city of the plains; we have already noticed that Port Phillip itself is merely part of a drowned plain. On either side of Melbourne, between the mountains on the north, which form a continuous wall from east to west of the State, and the broken ranges of the coast lies a great lowland, a series of plains dotted with marshes and detached hills. This is the Great Valley of Victoria. In the west it is largely covered with lavas, poured out from volcanoes now extinct. We can trace many of the old craters, especially in the district round Ballarat. Here is one 7 of them from the inside; notice the shape of the rocks. It is the lava and the river alluvium which have made the Great Valley the most fertile area of the whole State. Eastward the plain narrows for a space and then broadens out again in the valley of the Latrobe, behind the lagoons and the Ninety Mile Beach. The early settlers were quick to notice the fine grasses on these open plains; they started with sheep, but with the growth of communications dairy farming and butter making have increased greatly. Here we have a typical view on the plains, not far from the great Lake 8 Korangamite. It is open rolling country, and the building in the foreground is a large dairy. We may see the

butter being brought in from a branch factory to the central collecting station, and can watch the latest methods of working it by machinery. We shall find the same scenery and the same industry all over this area.

Here is another typical scene: a string of draught 9 horses is being brought in for sale, and we can follow them back to their feeding ground on the rich grass of 10 the open country. Victoria, with a much smaller area, values her horses at about four-fifths of the total value of those of New South Wales. They are not raised only for farm purposes as the picture before us proves. It is a race meeting, and we might imagine ourselves 11 in England but for the strange shape of some of the carriages.

We have had a glimpse of one aspect of Victoria; the port, the city, and the plain. Now let us turn to the mountains. We have seen that the east end of the 12 State is almost filled up by a mass of highland, and we may notice that the railways only touch the outer fringe of this district. It is out of the world and thinly peopled, though much of it is well fitted for cattle. Westward the highland becomes narrower and sends out spurs on either side, leaving the Great Valley on the south, and on the north and north-west a broad plain sloping down to the Murray River. This corresponds somewhat to the slope west of the Divide in New South Wales, but the climate, as we shall see, is not the same.

As we leave Melbourne and follow the Yarra up-stream 13 we soon notice a change of scenery. At Warburton, the rail head, we are well within the highlands. The river runs through forests of eucalyptus and fern, and we notice rapids below the primitive bridge. It 14 is evidently a mountain stream. This district is one of the playgrounds of Melbourne, and we stumble on 15 the Christmas camp of the Boy Scouts, who are known in Australia as well as in our own part of the Empire. 16 A few miles away to the north is Healesville. We notice here, as we drive through, that there are some 17 trees which seem to have shed their leaves. Perhaps these are some English trees of which many have been imported, and they find the weather too dry and hot; for close by we find the native forest with trees in full leaf. In spite of the presence of English trees, we may easily recognise the country as Australia by the great gum trees, with their bare trunks, and the thorny acacia growing below. The gums and the 18 tree ferns are everywhere on the hillsides. Even the house and garden, which we see here, have a slightly 19 foreign look, and do not seem adapted for the conditions

of an English winter. In fact, there is no winter, as we understand it, in this part of Victoria, though snow may lie for months on the heights of the Alps to the north-east. We must remember that Melbourne lies in nearly the same latitude as Seville. In the hills to the north of Melbourne we find the same scenery, with its abundance of streams and trees. Here is a woodland scene not far from Mount Macedon in this district, 20 where the Governor of Victoria has his summer home.

Let us now travel westward by the railway to Ballarat, which lies on the south face of the narrow ridge which forms the water-parting between the Victorian Valley and the north-western plains. Ballarat is a fine town, second only to Melbourne, and planted in far more picturesque surroundings. Here is Sturt 21 Street, named after one of Australia's greatest explorers; looking down it we can just see Mount Warrenheit in 22 the distance. We can wander in the Botanical Gardens where the aloe is in flower, or stroll by the lake and 23 admire the black swans. But we have not come here only for the scenery. Ballarat represents the second great factor in the development of Victoria: gold.

Within ten years of 1851, when gold was discovered here in paying quantities, all the chief fields of Victoria were opened up, and there was a sudden rush of settlers to the country. Many of the goldfields are so near Melbourne that it may be considered as a centre of mining as well as of pastoral industries. Thus we may account for the fact that it has to-day concentrated in its neighbourhood nearly half of the total population of the State. The gold most easily reached was in the underground leads, the channels of old streams, or fissures in the rock. In these were found nuggets and gold dust. Here we see the primitive methods of mining. A group of miners is sinking a simple shaft and raising 24 the soil in buckets, while another washes it in a pan to separate the heavier gold dust. The pick and shovel 25 and the strong arm of the miner are the chief instruments needed for this form of mining; and the fact that the goldfields are in the midst of fertile country, with farms all around them, makes the life much less hard than in some of the fields of the far interior of Australia which we shall visit later.

The modern method of mining is to attack the quartz rock by the aid of machinery; and the shafts are often carried to a great depth. Here, instead of the tents of the miners, we see what might be the top 26 of a coal mine, with elaborate machinery for winding. We shall examine this type of mine elsewhere; for though gold has made Victoria in the past, it is not now the chief gold-producing State in the Commonwealth. So we pass on, after a glance at a quartz reef 27 cropping out from the ground—a sign that has often guided the prospector in his search.

We have seen that as we travel eastward along the ridge on which we are standing, the forest grows more dense and settlement thinner, while roads and railways disappear. But the greatest change is found as we travel north and north-west from Ballarat. We have crossed to the inner slope of the highlands and are entering a very dry country. In the districts which we have visited the rainfall is not unlike that of the 28 Midlands of England, though most comes in the winter time from the westerly winds. The extreme east of the State has also, like Sydney, a good deal of rain in summer from the Pacific. But the great plain sloping to the Murray is cut off by the highlands from the moist winds of the oceans and exposed to dry hot winds sweeping down from the deserts of the interior. The rivers end in shallow lakes and marshes on the arid plain; and we may notice that the railways push out into this district and stop in similar fashion. It is a region of sandhills, heaths, and a dense scrub, called *mallee*; dreary and desolate at first sight, but not altogether without promise. The soil is very fertile, being composed of the old river silts, and with light rains at the right season, or by the aid of irrigation, it will grow fine crops. Here we see the beginning of the process of cultivation, by the rolling down of 29 the *mallee* scrub.

With large areas of fertile land lying waste for want of water, and water in abundance in the rivers, we should expect that attempts would be made to bring the two together. In New South Wales, near Yass on the Murrumbidgee, a great scheme is in progress. The Barren Jack dam, when finished next year, will hold up and make available for agriculture, a mass of water comparable to that of the Nile at Assuan. It will preserve, for the dry season, the water from the winter rains and the melting snows of spring. Other schemes are proposed for the Lachlan and Murray; up to the present, however, the chief development of irrigation has been in Victoria, on the streams flowing into the Murray and on the main river itself. Here we see

the process of impounding the water; notice the woods in the background 30 which show that we are on the upper course of the stream, near the hills and the region of heavier rainfall. Here again is a great reservoir being excavated, 31 to hold the flood water; and next we may see the water flowing out into the irrigation channel. 32 In another place the water is being pumped up into high-level tanks for distribution. The river has 33 gradually built up a flood plain at a higher level than its usual channel; so that the water must be raised before it will flow over the fields. At Mildura and Renmark, the latter in South Australia, a large fruit-growing industry has been developed on the basis of irrigation. So we find lemons and apricots, and above all the currant and the vine which give us our dried currants and raisins.

But the supply of water in the rivers is limited, for the rivers are not broad or deep, in spite of their great length. One of the chief difficulties of all irrigation schemes is to avoid damage to the interests of people living lower down the stream, or interference with the navigation. In this matter the interests of all three States of the Commonwealth must be considered, since the Murray basin is divided among them. As we have already noticed in the case of Queensland and New South Wales, the State boundaries only coincide in part with natural features of the country.

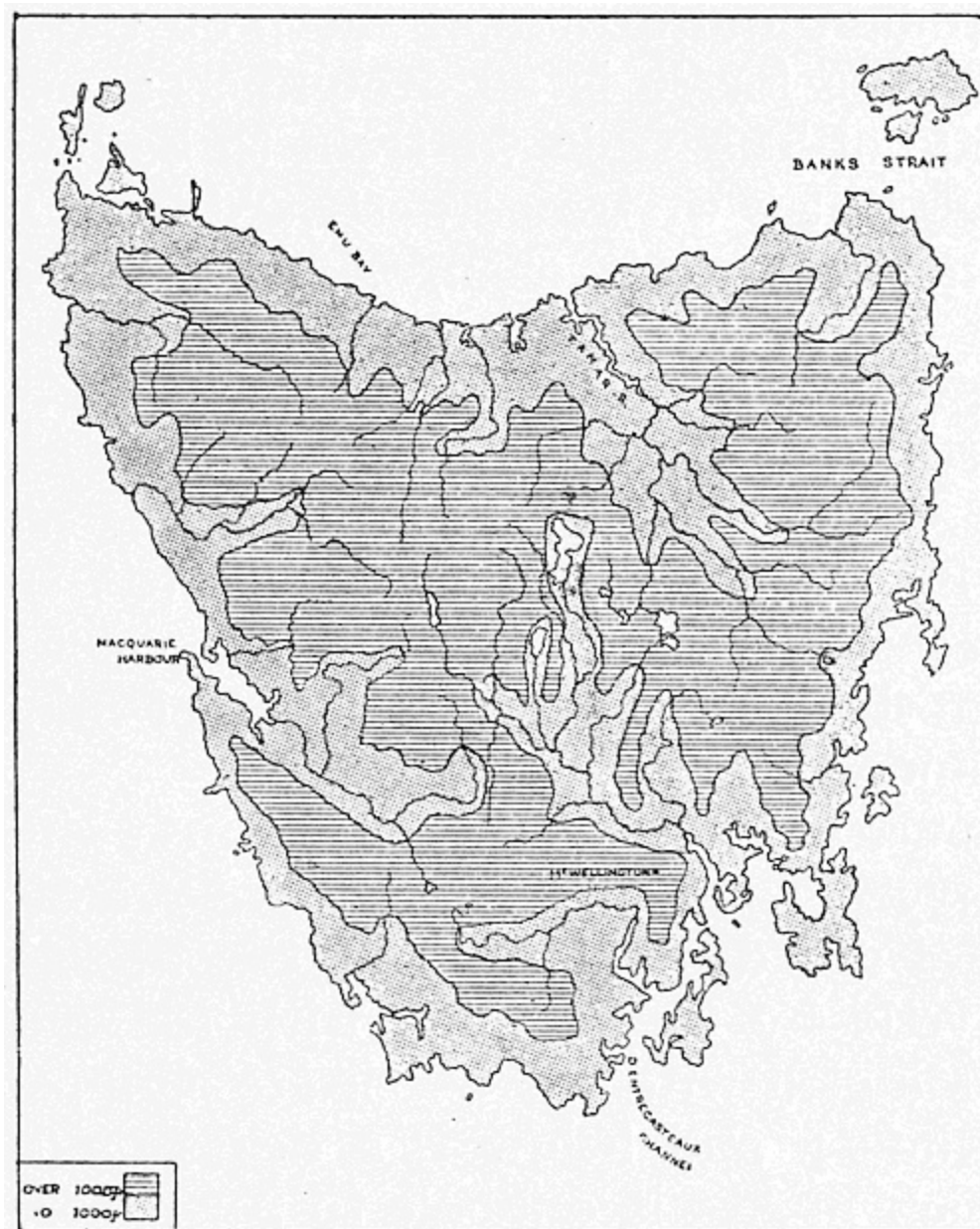
The whole character of the river basin depends on the distribution of the rainfall. In the case of all three States there is a similar arrangement: first the coast belt, which is more varied and irregular in Queensland and Victoria than in New South Wales; then the highland edge, and then the back slope with a zone of moderate rainfall which shades off gradually into desert conditions. This zone narrows as we come southwards until it almost disappears in South Australia. But before following it out into this last State we will cross Bass Strait to visit Tasmania. Victoria, though a mere corner of the Australian continent, is about the size of Great Britain, while Tasmania is not very much smaller than Ireland, and both could support a very dense population. We must bear these facts in mind during our rapid journey through the country, since the maps in our atlases, for the most part, give us utterly wrong impressions as to the area of Australia.

If we look at a chart showing the depths of the sea, 34 we may notice that Bass Strait is shallow while the surrounding seas are deep. From Wilson Promontory we can trace a connexion through Flinders and Cape Barren

Islands, to the north-east horn of the curved coast of Tasmania. A similar bridge runs through from the north-west horn, through the Hunter Islands and King Island. We may be reminded of the shallows of the English Channel and the North Sea, and can easily imagine that Tasmania, like England, at some very remote period formed part of the neighbouring continental mainland. We find, moreover, a general similarity between the plants and animals on the opposite sides of the Strait; but there are also some marked differences which suggest that Tasmania was separated very long ago and so has had a peculiar and isolated development.

We cross Bass Strait and steam up the winding estuary of the Tamar to Launceston. The names ³⁵ remind us of England, and round us are the counties of Devon, Dorset, and Cornwall; but we must not be too ready to take the names as a guide to the character of the country or the climate. Our vessel lands us at a busy wharf in the middle of the town, for Launceston, though second to Hobart in size, is the chief commercial centre for the island. Here is a general view ³⁶ of the town: it is modern and well planned and has fine houses and gardens in the suburbs, like the other Australian capitals.

A glance at the country round soon shows us why the early settlers were reminded of the south-west ³⁷ corner of England. At King's Bridge we can leave the broad estuary of the Tamar and turn up the narrow valley of the Esk. Notice the fishermen: the streams here abound with fish, mostly introduced from England. The Esk here flows through a rocky ³⁸ and wooded gorge; and we might easily imagine ourselves in Devon or Cornwall. Higher up the scenery is spoilt by an ugly building with great pipes; it is the ³⁹ power station for the electric light of the town. The water power is too valuable to waste, so the picturesque has been sacrificed to the practical needs of the people.



TASMANIA: OROGRAPHICAL.
By permission of the Diagram Co.

As we leave Launceston and travel up the valley of the Tamar, through beautiful open country, it is still easy to imagine that we are somewhere in the south of England. The apple orchards are everywhere. Here 40 is one on a hill slope, but we notice that it is quite unlike the grass-grown orchards so common in Devon. The trees are grown as low bushes, in straight rows; it is less picturesque but more profitable. They produce the fine even fruit which we can buy in the London market. The climate is cooler and moister here

than on the mainland and so is well fitted for most of our English fruits. There are the same orchards and the same English crops all along the line of the railway southward to Hobart. The line follows a narrow sheltered lowland. On the east is a broken mountainous country; on the west a great solid plateau, ⁴¹ rising up in tiers or steps, and occupying all the centre of the island. We shall not cross this central block, since it is without road or rail and almost without inhabitants, a bleak bare roof to the island, feeding a few sheep in summer, but even then subject to biting storms and snow.

From Launceston a short journey by rail will take ⁴² us to Scottsdale among the hills in the north-eastern corner of the island. Here we meet the bullock team ⁴³ hauling timber, and droves of sheep on the road, while the eucalyptus forest is all around us; on this drier ⁴⁴ eastern side of Tasmania we seem to be back again in Australia. But if we travel westward from Launceston we shall notice a great change. At first our way lies over the plains, between the northern edge of the plateau and the sea. Then, at Burnie, we leave the sea and strike southwards towards a new country. We have turned the north-west corner of the plateau, and between its steep western edge and the sea we find the plain broken by a detached range of mountains rising from the level. Here is the chief mining district in the island.

Let us use our eyes as the train runs swiftly through this country. The gaunt gum-trees have disappeared; everywhere are dense forests of the evergreen beech, ⁴⁵ called myrtle by the settlers, with its small feathery leaves. Mingled with the beech are clumps of pine, of various kinds; and below is a dense undergrowth of scrub which makes it difficult to penetrate the forest. The rivers have cut deep gorges in the surface of the plateau: here we see one of these with its slopes clad ⁴⁶ thickly with trees. It is a rugged country, largely unexplored, and would have few inhabitants but for the mines. It lies in the track of the strong west winds, the Roaring Forties, and has a rainfall three times as heavy as that of the sheltered eastern valleys, a rainfall only to be compared to that of the wettest parts of the West of Scotland and Ireland. The vegetation is naturally different from that of the neighbouring regions of Australia, with their moderate rainfall and greater warmth; in fact, to find a parallel to Western Tasmania we must look to New Zealand and to parts of South America.

The railway on which we are travelling has been built solely for the benefit of the mines, and we are drawing near to Zeehan, an important centre for the 47 production of silver, lead, and other metals. Notice the mountains in the distance rising up sharply from the level of the plains. The town looks primitive and 48 unfinished; little better than a mining camp. Here is one of the smelteries at the foot of the hills. A little 49 further on we come down to the sea again at Strahan, the only seaport of importance on the west coast. It 50 lies on a fine bay in the deep and almost landlocked inlet of Macquarie Harbour. This splendid sheet of water was discovered as early as 1816, but it was too far away from the settled portion of Tasmania, and it owes its present importance solely to the presence of the mines in the country behind it.

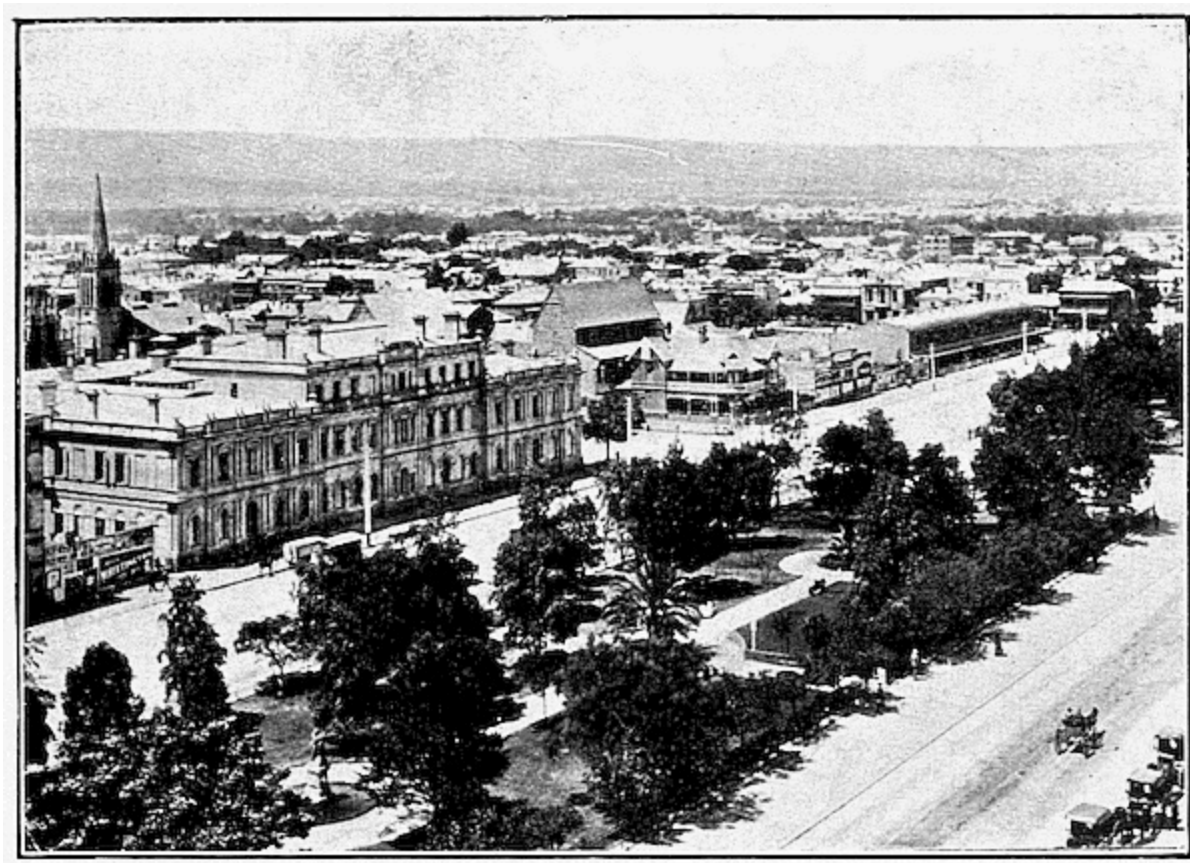
A short distance inland is Queenstown, where much 51 smelting is going on. Notice the desolate appearance of the country round, and the stumps of the trees which have been cut down for fuel. Gormanston, with its 52 background of rugged mountain, is equally desolate. Close to this town are the famous Mount Lyell copper 53 mines. Here is the “open cut” where they are quarrying into the mountain-side, and here are some of the 54 smelting works. Here again is a general view from our hotel: the whole country is grim, scarred, and waste, 55 in great contrast to the beautiful forest scenery a few miles away. But it is the source of great wealth to Tasmania.

We shall not attempt to travel further south than Macquarie Harbour, as there is little beyond but wild forest and hill country, backed by the bare plateau and full in the path of the westerly gales from the Southern Ocean. It is without roads or railways and has scarcely a human settlement. So we return to Launceston and follow the railway southward to Hobart. The coast of the south-east corner is very different from that of the rest of the island. It is a drowned coast, with deep fiords, many islands, and irregular peninsulas barely connected by narrow necks with the mainland. On one of these deep fiords stands Hobart, the second oldest city in the whole of Australasia.

Here we have a general view of Hobart, looking 56 across the water to the hills beyond. Once again we are reminded of portions of the Clyde, and only Sydney can compare with Hobart for the beauty of its position. Here is another view from the water, with 57 Mount Wellington in the background rising into the clouds. In the neighbouring lowlands, sheltered from the west by the mountains, are more apple orchards; and in a gully near the town we

find a mass of tree ferns. Here again is Government House, since Hobart is the political capital of Tasmania: notice the lake and the trees. Everything around us suggests a mild and not very dry climate.

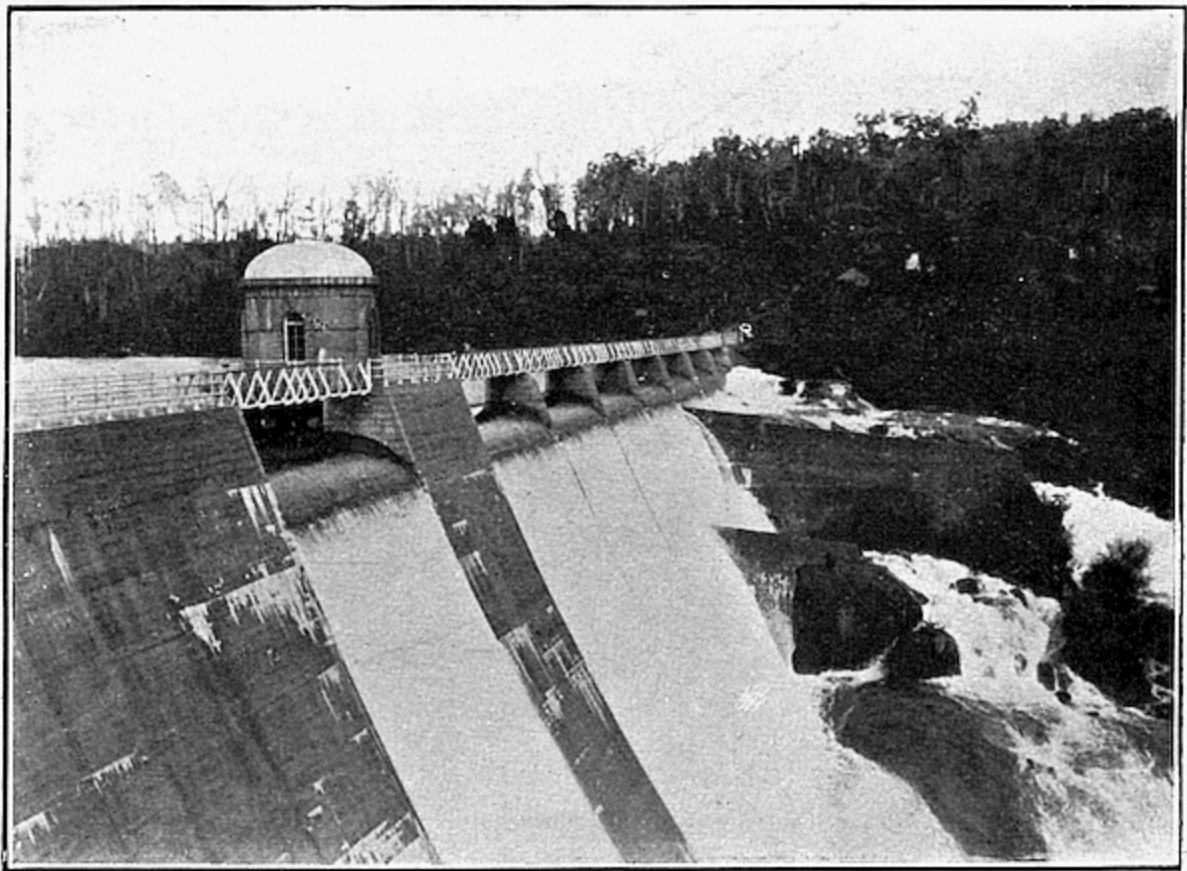
If we climb Mount Wellington, the aspect of the country soon changes. The mountain is not an isolated peak, but merely the south-eastern corner of the central plateau. From the summit is a fine view of the fertile lowland valley and the great expanse of fiords and islands. But the summit itself is a wild confusion of boulders with low scrub and heath. This is a very good guide to the nature of the whole surface of the plateau behind, and we realise that it is not a favourable country for the settler, though in some of its wilder aspects it may attract the tourist. We shall not attempt to reach the lakes lying on the surface of the plateau, but content ourselves with a short journey round its southern rim.



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ADELAIDE: LOOKING SOUTH-EAST.

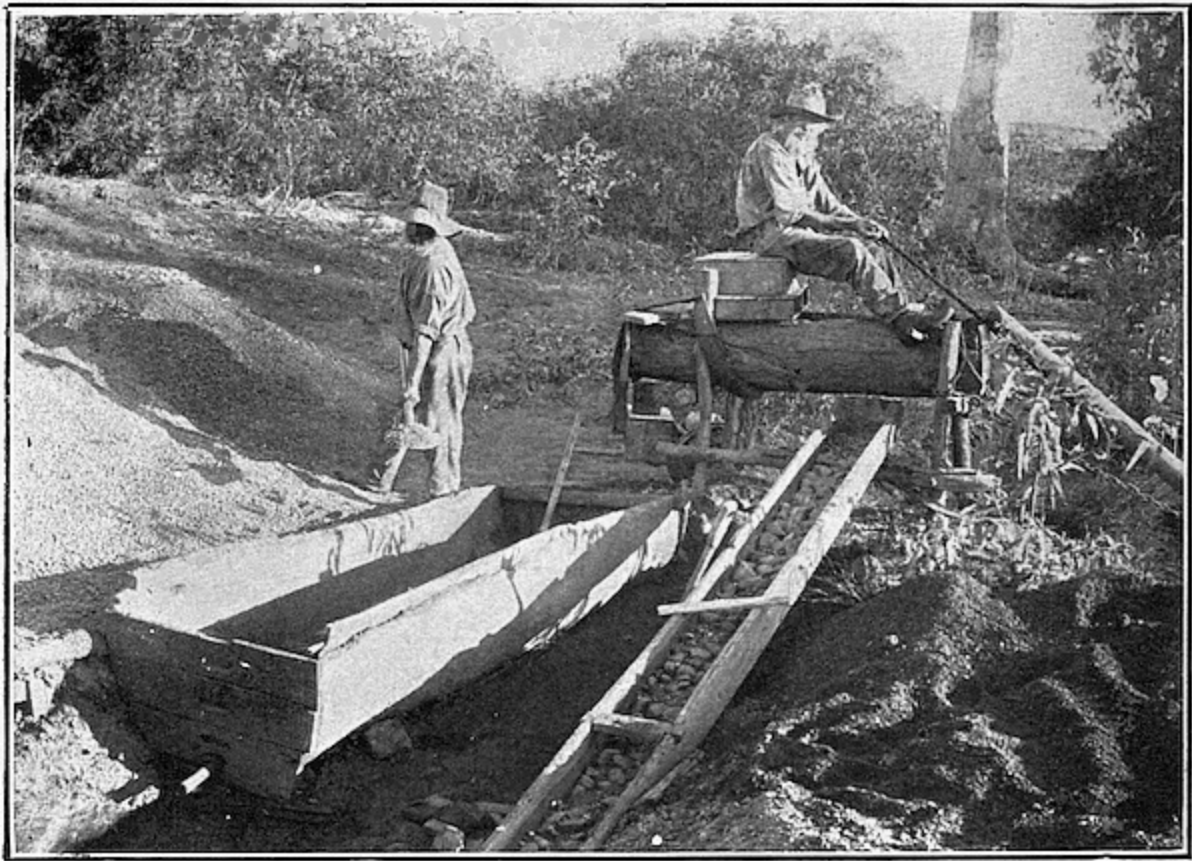
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MUNDARING WEIR.

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ALLUVIAL MINING.



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CLIFFS IN THE GREAT BIGHT.

The coach will take us through more orchards, 61 towards Franklin on the Huon River, where we touch the eastern side of the broken country from which we turned back at Macquarie Harbour. There are settlements all along the Huon Estuary, which runs into the channel of D'Entrecasteaux. Here we have come upon an important remnant of past history. The name is that of a French admiral who was sent out in 1791 to seek for a port in Southern Australia, so that France might gain a footing in these new islands. D'Entrecasteaux actually surveyed the Derwent River, and a later French expedition spent some time in the neighbourhood. Partly through fear of the designs of France we occupied Risdon in 1803, and in the following year, Colonel Collins, not content with the site at Port Phillip in Victoria, which he had occupied for a short time, came over and moved the settlement from Risdon to Hobart. The evidence of the activity of the French explorers in this region is still to be seen in the names of capes and bays all along the eastern coast of Tasmania.

There are saw-mills at Hobart, and all around the Huon is a fine timber country, easily reached from the sea. Here we see the beginning of the end: the cutters ⁶² have built a rough platform above the undergrowth and up to the point where the trunk of the tree rises straight and even. They are using the saw, though sometimes the whole work is done with the axe. They are fond of the axe in this part of the world, and it enters even into their sports, since the chopping match is a favourite form of athletic contest among them.

The trees in this district grow to a great size; here we have a forest scene, with the huge logs scattered on ⁶³ the ground, or loaded on to trucks which will carry them to the mill. Geeveston is noted for its saw-mills, one of which we have before us. The logs disappear ⁶⁴ inside the mill, and we meet them again as sawn timber on the little wharf, ready to be shipped to all ⁶⁵ parts of the world.

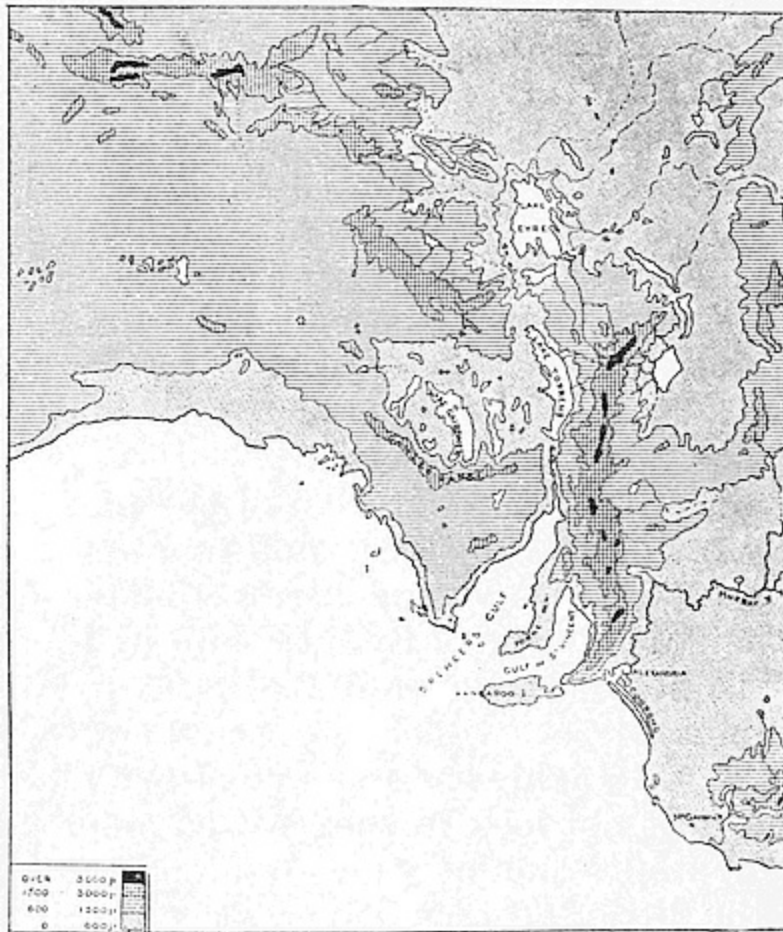
In the valley of the Huon, as well as in that of the Derwent, are orchards everywhere, proving that we are still in the sheltered lowland which we have traced from Launceston southwards. The cottages and gardens, with their masses of English flowers; the English trees, oaks and elms, and the hawthorn hedges all along the roadside, remind us strongly of England. Many even of the birds are English. In fact, this part of the island has been quite transformed by the colonists until it closely resembles the mother country. But we enter the forest and step at once from England to Australia. Here are the tall gums with their untidy bark and dead branches, and the swarms of honey-eating birds flitting among their blossoms. Here, too, are the wattle and banksia and many other plants peculiar to Australia, and the further we move from civilisation the less there is in the face of the country to remind us of England. But one difference may be noted between Tasmania and the rest of Australia: however far we penetrate into the wild interior we shall not meet the aborigines. They were few in number at the time of the first settlement, and the last survivor died many years ago.

LECTURE V

SOUTH AUSTRALIA AND WESTERN AUSTRALIA

On our visits to other great capitals we have found steamers unloading their cargoes in the very heart of the city; but Adelaide, founded in 1836 and named after the wife of King William IV., is neither on the sea nor on a navigable river. The original settlers were aiming at a purely agricultural colony, and so chose a position a few miles inland in the midst of fertile land and pleasant scenery. So we must land at Port Adelaide and take the train.

Partly owing to the separation of the port from the city, partly owing to the slower and more even growth of South Australia, Adelaide seems quieter and less crowded than Melbourne or Sydney, and its inhabitants consider it to be the model capital of Eastern Australia. King William Street, which we see here, with the statue of Colonel Light, the founder of the city, in the foreground, does not look in the least commercial. We get the same impression as we walk along the tree-clad banks of the little Torrens River, or cross it by the City Bridge. We miss the wharves and warehouses and steam cranes, and might almost imagine that we were on a backwater of the Upper Thames. The main streets, too, of the city remind us rather of the West End than of the City of London; while even the post-office stands in an open space with trees. In fact, the whole city, with its wide streets, its parks and gardens, gives the impression of spaciousness. If we make our way, however, to the railway station we shall see that Adelaide is not by any means without trade. Here are collected the products of all the country round; but as this is purely agricultural, and Adelaide is not the only outlet, there is not the rush of business which we saw at Melbourne.



SOUTH AUSTRALIA: OROGRAPHICAL.
By permission of the Diagram Co.

Here is a general view which will give us some idea of the position of the city. It lies on a plain; a few miles away to the east the view is shut in by a long, low ridge. If we climb this ridge and look back towards the city, we have in sight a large part of the original South Australia.

South Australia of to-day is a somewhat difficult country to analyse; but the ridge on which we are standing may give us the key to the whole. If we follow the heights northwards, we shall find that they disappear, hundreds of miles away, in the country south-east of Lake Eyre. South of Adelaide they curve round and end in Kangaroo Island, which stretches across the mouth of the Gulf of St. Vincent. These heights are really the edge of a plateau, and the plateau slopes gently away from the sea towards the basin of the Murray. The Murray, at the great bend, turns sharply

southwards and reaches its outlet in Lake Alexandrina just beyond the southern end of the highlands.

Between the plateau edge and the sea, Adelaide and the Gulf towns lie along a narrow strip of lowland. The Gulf of St. Vincent is merely part of a larger gulf which is interrupted by Yorke Peninsula, so that we have really one great inlet running up to Port Augusta at the head of Spencer Gulf. The west side of this gulf is formed by another plateau which slopes away from the sea towards the salt lakes and marshes of the interior. We thus have two plateaus and a lowland in between, partly flooded by the sea. A portion of the surface has dropped down between two lines of cracks or faults, and a rift valley has been formed. Lake Torrens occupies the northern end of this valley. It is the eastern side of the valley, together with a small part of the back of the plateau, which constitutes the real South Australia. We may notice that the local railways are almost confined to this area.

There is, however, one important piece of the State outside this area. South of Lake Alexandrina we see a long curving coast, bounded by sandbanks enclosing a string of lagoons, the Coorong; behind it is the scrub country which we have met already in the north-west district of Victoria. Beyond this, in the south-east corner, especially around Mount Gambier, we find the same conditions as on the neighbouring coast regions of Victoria. There is considerable rainfall and there are 10 even fresh-water lakes, as we see here. The soil, too, is fertile, since Mount Gambier belongs to the volcanic area of Victoria; so that there is agriculture and dairy farming, and oats are grown. But the district is cut off from Adelaide by land and has rather a detached existence, though the railways now being built or planned will alter this.

We will now explore the country round Adelaide. In the Mount Lofty ranges, east of the city, there are 11 streams and waterfalls; but the rivers of the plain are very small and do not suggest a very heavy rainfall. Everywhere are orchards and vineyards growing fruits which are not grown in the open air in England. Here 12 is an orchard quite near the city, and here is an orange tree laden with ripening fruit. The rainfall is very 13 light and comes mostly in the winter, while the summers are hotter than in most of the occupied regions of Australia. We have the sunny climate of the Mediterranean and a corresponding vegetation. Yet it is healthy for white

people, in spite of the heat, owing to the dry clear air; while the highlands only a few miles away offer a refuge in the summer.

On the eastward side of the plateau there are again no large rivers, and the rainfall is even less than at Adelaide; but there is enough to grow fine crops of wheat. We see much the same arrangement of zones as at the back of Sydney, but the wheat-belt is narrower and the rainfall rather less. Very soon we drop down into the rainless plains of the Murray basin, where the only cultivation is in the irrigated district round Renmark. Here we can see them drying the raisins ¹⁴ and loading them on to the little steamer which will take them down to near the mouth of the river. They ¹⁵ will then be sent on by rail to Adelaide, since the river has no good outlet to the sea.

Yorke Peninsula and the western side of Spencer Gulf have a few small towns on the coast; inland they have no rivers but only dry pastures, salt lakes, and marshes. Here is one of these lakes in the Peninsula. ¹⁶ The character of this district will change in the future, as much of it is adapted for the growing of wheat which has already been introduced. The only considerable population at present is on the coast strip from Adelaide to Port Augusta, and on the back of the plateau, never more than a hundred miles from the coast.

As we follow the railway northwards from Adelaide ¹⁷ we shall find that agriculture decreases with the decreasing rainfall; in place of crops we see cattle and sheep. Here is a typical station only a hundred miles ¹⁸ north of Adelaide, where the sheep seem to be in full possession. The further north we go the thinner is the settlement; and north of Port Augusta we shall only find it at a few favoured spots near the railway.

Beyond Lake Torrens the plateau edge to the east trends away and disappears, and we enter the Lake Eyre basin. At one time this may have been a vast inland sea, as the remains of extinct animals show that the climate must have been very different from the present. Now it is a great clay plain, broken by low plateaus and ridges of sandstone, and with much of its surface covered with stones or mulga scrub. The lakes are salt, while the long rivers, shown on the map as flowing into them, may be only a string of mudholes for years together. Much of this region is still unexplored, and nearly the whole of it is useless.

The railway ends at Oodnadatta, to which a train runs at rare intervals; and off the railway the camel, which 19 has been introduced into Australia for this purpose, is the only means of transport. To the east of this line something may be made of the country by boring through the clay to reach the artesian water, as we have seen already in Queensland and New South Wales; so that settlement may spread slowly towards these States. To the west is the arid plateau which covers so much of the central part of the continent—the Sahara of Australia.

Adelaide is a little south of Sydney in latitude, and 20 Oodnadatta a little south of Brisbane, yet what a difference between the two parallel journeys by rail! The explanation is to be found in the rainfall figures: north of Port Augusta we enter the zone where the annual amount is under ten inches. The railway reminds us of those starting from the east coast and ending at some remote point in the interior; but there seems very little country here for our line to exploit or develop. To understand fully its meaning we must look back at the past history of the region.

South Australia was founded by an Association formed in England with the object of building up a model agricultural colony. The plan was to sell the unoccupied land and use the proceeds to aid suitable emigrants in settling there. With this idea Adelaide was founded in 1836. For a few years the colony was poor, as it was intended to be self-supporting and the capital in private hands was insufficient to develop the country. But progress was helped by the various discoveries of copper, from 1842 onwards, at Kapunda, Burra and other places; and by 1855 the colony was able to export large quantities of agricultural produce to the other colonies, which had depended mainly on Tasmania up to this time.

It is curious that the State which is still mainly agricultural was the first to develop its minerals on a large scale; but as the mineral was copper and not gold, it led to no rush of settlers, but only to a steady growth of population. The older mines have been long worked out, but those in Yorke Peninsula and in the 21 Flinders Range still produce large quantities. Here are views of the Wallaroo and Moonta mines in the Peninsula. 22

With the exception of some iron ore, which is not much worked, South Australia has no other important minerals of its own: yet at Port Pirie, on the east side of Spencer Gulf, we find large smelting works. To explain this

we must look back again at the railway map. From the port a line runs north-east for two hundred miles or more, to the Silverton country, just inside the New South Wales border which we have already visited. It is the natural outlet for this district, as Sydney is more than twice as far away.

Here we see Port Pirie and some of the smelting ²³ works. The works also handle iron ore which is brought down to the opposite shore of the Gulf by a short railway ²⁴ from Iron Knob, near Lake Gilles, in the dry interior of the western plateau. In the case of this district we notice once again that the artificial boundary following a line of longitude has no correspondence with the natural features of the country.

We have already seen how, on the constitution of Queensland as a separate State in 1859, the country ²⁵ to the west was left as a detached portion of New South Wales. In 1855, Gregory had crossed what is now the Northern Territory, from the Victoria River in the west to the Flinders River, and so through North Queensland to Brisbane, following the line taken earlier by the explorer Leichhardt, but further inland. At the same time, various explorers had been following up Eyre's discoveries in South Australia, and miners and shepherds were pushing steadily northwards from Yorke Peninsula along the line of the present railway. In 1859, South Australia offered a prize for the first explorer to cross the continent from south to north, urged on by the proposal to connect Australia with England by a cable which must be landed somewhere on the north coast. A party from Victoria, under Burke, started first, and following roughly the western boundaries of New South Wales and Queensland, came out by way of the Flinders River to the Gulf of Carpentaria. But on their return Burke and Wills perished at Cooper's Creek of starvation, and only one member of the party was rescued.

In 1862, Stuart, for South Australia, succeeded in crossing to the north coast and returning to Adelaide by the route west of Lake Eyre which is now followed by the railway and telegraph. The immediate result was that the Northern Territory became politically part of South Australia, instead of being an annex to New South Wales; but it has recently been transferred to the Commonwealth Government by which it is now administered. By 1872, the telegraph line was completed to Palmerston, the northern capital, on Port Darwin, where the cable is landed. By 1889, the railway from the south had been pushed forward to Oodnadatta, nearly seven hundred miles from

Adelaide; while in the north one hundred and fifty miles of line was built southwards. Some day, probably in the near future, the rail will stretch from shore to shore, but there still remains a gap of over a thousand miles.

North of Lake Eyre, and on the Tropic, the Territory is crossed by the Macdonell Ranges, running east and west. Then comes a stretch of five hundred miles of sandy plains, with scrub and spinifex, and then the peninsula of Arnhem Land, a low plateau with a considerable rainfall. Right on the coast are mangrove swamps and tropical rainfall, as in North Queensland. Though some of this country is suited to cattle, and gold has also been discovered, progress has not been very rapid, in spite of the importance of the magnificent harbour of Port Darwin. The population consists of a few hundred Europeans, as many Asiatics, mainly Chinese, and some thousands of aborigines.

After Stuart's journey there remained only to complete the conquest of the desert from east to west. In 1840, Eyre had succeeded in travelling from Adelaide to West Australia along the shores of the Great Australian Bight. It was 1870 before Sir John Forrest traversed the same route in the opposite direction. The result was the occupation of Eucla, in the middle of the Bight, and the completion of the overland wire from Adelaide to Albany in 1877. Finally, in 1874, Forrest crossed the middle of the great plateau, from Geraldton to the north-south telegraph line. A few daring journeys and the two telegraph lines still represent the only land links between the detached areas of settlement which fringe the central plateau block; but a stronger link will soon be forged. Already the work is in hand. A survey of the route was completed in 1909 by the Federal Government; and at the end of 1911 a Bill was passed for the construction of a railway from Port Augusta to Kalgoorlie, a distance of over 1000 miles. This line will not only shorten the mail route to the eastern States, but may also lead to some pastoral settlement, as not all the country traversed is desert.

South and Western Australia, with the Northern Territory, together include nearly two-thirds of the continent. Western Australia alone includes nearly a third. It is rather more than five times the size of Spain, but its population is smaller than that of any of the States of the mainland. The reason for this contrast is partly a matter of history, as the State is comparatively young, and partly due to geographical causes, as we shall see when we have examined the country.

The map shows Western Australia stretching from 26 north to south, across the whole breadth of the continent, with the Tropic running through the middle. It corresponds in position to Queensland and New South Wales; for Cape Londonderry is in the same latitude as the middle of Cape York Peninsula in Queensland, while Albany is a little south of Sydney. The greater part of the State is occupied by a broad plateau, from a thousand to fifteen hundred feet high; between this and the sea is a narrow coastal plain, irregular and deeply indented in the north, narrower and more uniform towards the south, and disappearing in the south-west corner where the plateau edge approaches the sea. We may notice a certain likeness here to the structure of the eastern end of the continent. We shall find nearly the whole of the agricultural population collected along a comparatively narrow belt of country from Geraldton to Albany, with Perth, the capital, in the middle. On the plateau there are no great rivers, but many short streams run down from its western edge to the sea; and though the map shows large lakes in the interior we shall find that in this case the map is not entirely to be trusted.

The Swan River was discovered by Captain Stirling on his exploring mission from Sydney in 1827. Two years later Captain Fremantle took formal possession, and the Swan River Settlement was founded. The Home Government was at first doubtful about the project, but was urged on to the settlement through fear of French occupation. Perth itself lies twelve 27 miles up the river; its port is Fremantle, at the mouth of the river, on a deep and safe harbour, crowded with 28 wharves and shipping. It is the main outlet for the trade of all this part of Western Australia.

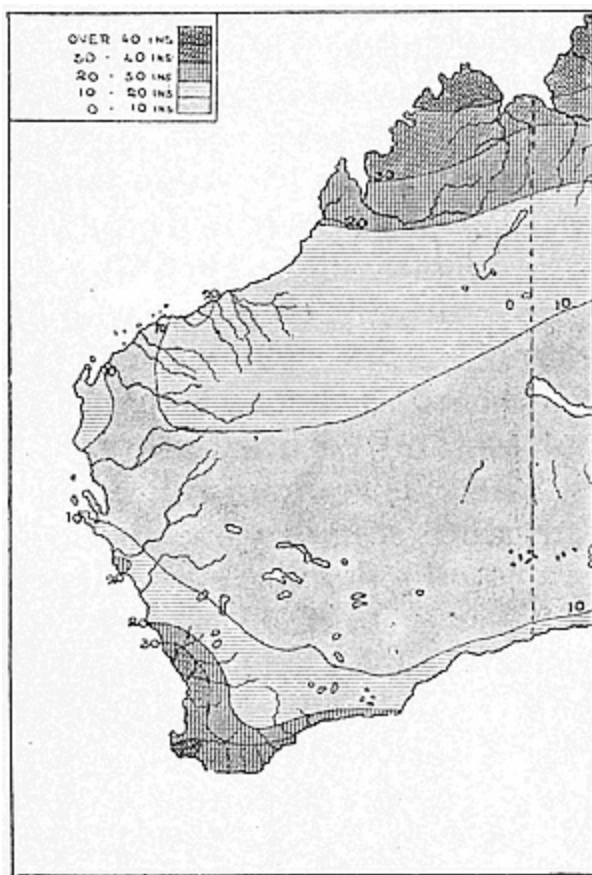
Instead of taking the train we will travel by launch up the Swan River; on our way we notice the large 29 flocks of black swans which are now collected here and preserved by the Government. The city of Perth is smaller and more irregular than Adelaide or Melbourne, and we see it at its best as we approach by the river. It lies rather in a hollow, and from the higher land in 30 the King's Park we get a fine view along the river front. The main streets differ little from those of other Australian cities; but in St. George's Terrace 31 they are working on the road, and we notice that the old tree stumps are not yet removed. Although Perth is making very rapid progress, everything is as yet somewhat quieter, more picturesque and on a smaller scale than in the great capitals of the eastern States. On the outskirts

of the city, especially overlooking the river, are the gardens and houses of the wealthier residents. Here is one of these; notice the 32 lemon trees laden with fruit, although it is winter. In the country round we shall feel quite at home. Here we have a picture with pasture and scattered trees and fat cattle; and we pass a poultry farm which might 33 well be in a corner of Surrey.

Let us follow the Swan, now become the Avon River, inland. First we come to Newcastle, where there are many orange orchards in the broad valley; then we pass the township of Northam, where we meet a native woman 34 on the road, and finally we reach York. From one of the surrounding hills we look down on a broad expanse of 35 plain, dotted with farmhouses, and with a background of hills in the distance. There is a flour mill which 36 might be in our own Yorkshire, and a very English-looking church on the sloping bank above the river. 37 All looks settled and civilised. On the other hand, the King's Head hotel is entirely primitive, and carries 38 us back to the days of the early settlers, as it is one of the oldest buildings in the State. As we cross the bridge we notice that the trees are standing in the water, for the river is in flood.

We have here a country full of English place-names, and with scenery which often reminds us of England; but at the same time we find wheat and oranges growing side by side, and trees in fruit and flower in the winter. This suggests something very different from our own climate. We must remember that Perth is in latitude 32° S. and that this district therefore corresponds to Egypt or Morocco. It is a land where frosts are unknown in the lowlands and valleys near the sea, and where the summers are hot and dry, though tempered by the sea breezes. Most of the rain falls in the winter, which is therefore the growing time; we have already noticed the Avon in flood. Even in the winter there is plenty of sun, as the rain falls largely in heavy showers at night.

If we travel eastward from York we shall soon find 39 a change in the face of the country, as the rainfall decreases quickly towards the interior. But for the present we will keep within the zone of moderate rainfall, not far from the coast, and continue our journey southwards. Much of the land round Perth has been cleared for agriculture, though it was formerly covered with forest. Beginning north of Perth, and stretching southwards at a distance of twenty to thirty miles from the coast, is a long belt of timber-country, marking the zone of heaviest rainfall. This belt broadens out and fills the



WESTERN AUSTRALIA: RAINFALL.

whole peninsula in the south-west corner, between Flinders Bay and Geographe Bay, and finally disappears a few miles to the north-east of Albany. The dense forest clothes the western face of the plateau, and thins out eastward to open scrub country with scattered gum trees of various kinds. It consists in the main of Jarrah, with some blocks of Karri in the south and south-west.

Jarrah is a very hard red-coloured wood which is useful for jetties, railway sleepers and all other work where there is great exposure to damp and weather. We may often walk over it in the streets of our cities, as it makes an excellent pavement. Here we see

the appearance of the original tree: it is tall, perhaps a 40 hundred feet high, with a trunk a yard or more thick and a dark-grey furrowed bark.

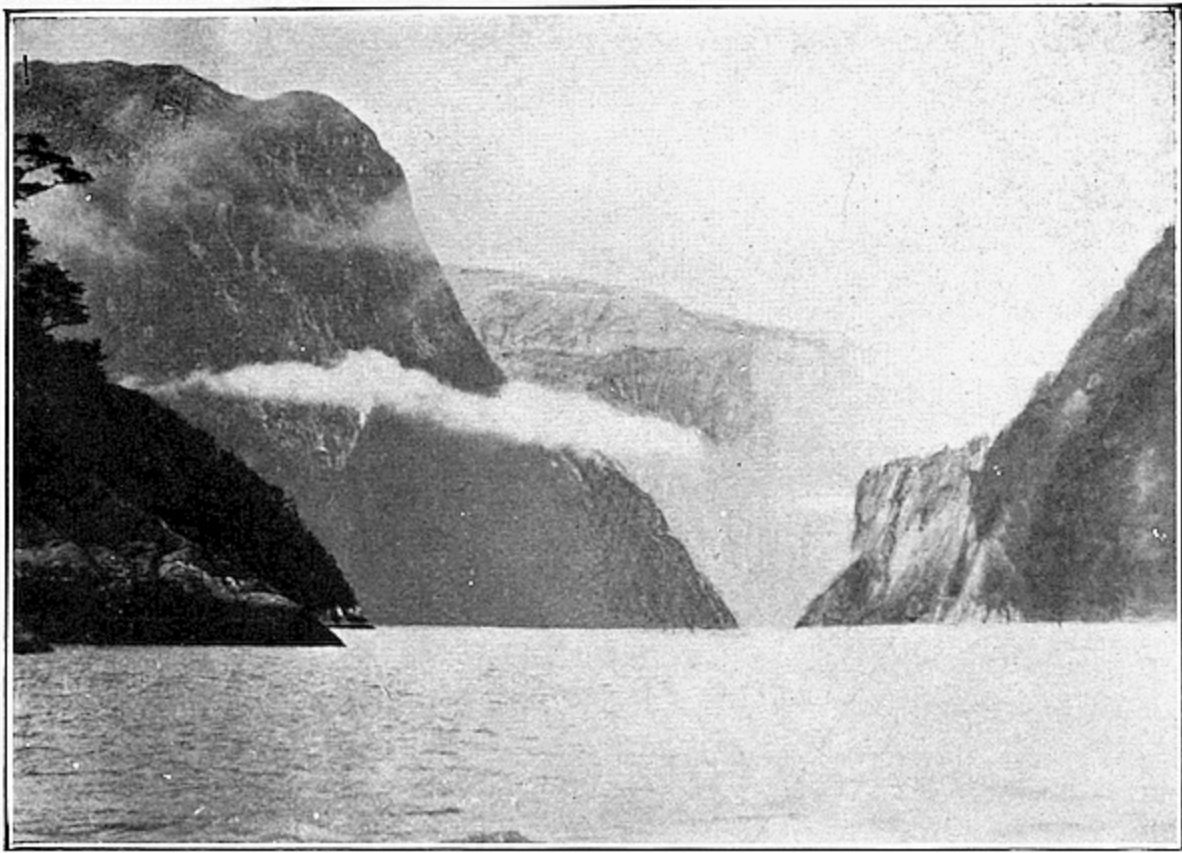
With the Jarrah is found the Karri, of much the same character but less durable. It is one of the finest and tallest trees of the Australian forest. Its bark is yellowish white, and peels off, leaving the tree clean, so that the Karri is sometimes called the white gum. The tree sometimes grows to a height of three hundred 41 feet, and is far too large to handle; while even the smaller specimens need a whole team to haul a single log. So that, although there is a very large area of forest land, we can understand that cutting only goes on within easy reach of one of the short branch railway lines running down to the sea. The forests of Western Australia have recently been estimated to contain two hundred and twenty million tons of valuable timber, worth about three hundred million pounds sterling.

Where the forest thins out eastward the red gum is one of the most useful trees, as it is spreading and 42 branched, and so gives more shade than the other gum trees. This is important for the stockowner in a dry, hot country.

We have already seen something of the agricultural district round York; and we shall find little difference as we follow the railway southwards to Albany, keeping always about a hundred miles from the coast. There are the same wheatfields and sheep and occasional orchards; it is not very different from the country round Perth, except that it is rather cooler as we travel southwards. In the summer, especially on the eastern edge of this belt, it is very dry and parched; but after the winter rains, though there is too little water for trees, the flowers and grasses spring up everywhere, especially everlasting flowers which grow in great profusion, as we see from the picture before us. 43

So we have a coast belt associated with timber, and a parallel belt inland where agriculture is developing along the track of the railway. The southern terminus 44 of the railway is Albany, on King George's Sound, one of the best harbours of the Australian continent. 45 Albany is a fortified coaling port, and is likely to grow in importance with the organisation of Australian defence in the future. This seems a small corner on the map, yet the area available for agriculture is considerably larger than the total area under cultivation in Great Britain; while beyond it, in the region of lessening rainfall, is a wide belt suited for sheep-raising. Further eastward still, beyond this belt, we enter another type of country. The railway from Northam Junction carries us across the dry country and through Southern Cross to the mining centre of Coolgardie. A little further on, at Kalgoorlie, it turns northward, to end at Laverton, about four hundred miles from Perth and the same distance from the sea to the south. This is the limit of that part of West Australian settlement which is based on gold.

The gold-bearing country is in two bands, running north and south, through Southern Cross and Kalgoorlie; and we find quite a large population in the midst of waterless country. Here is a view of Kalgoorlie, a prosperous-looking town, in spite of its arid 46 surroundings. This is the region of Australia where gold-mining has developed most rapidly in recent years. It is very different from Ballarat or the Queensland mines, where the gold lies in the midst of a fertile agricultural country, easily reached from the sea.



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MILFORD SOUND.

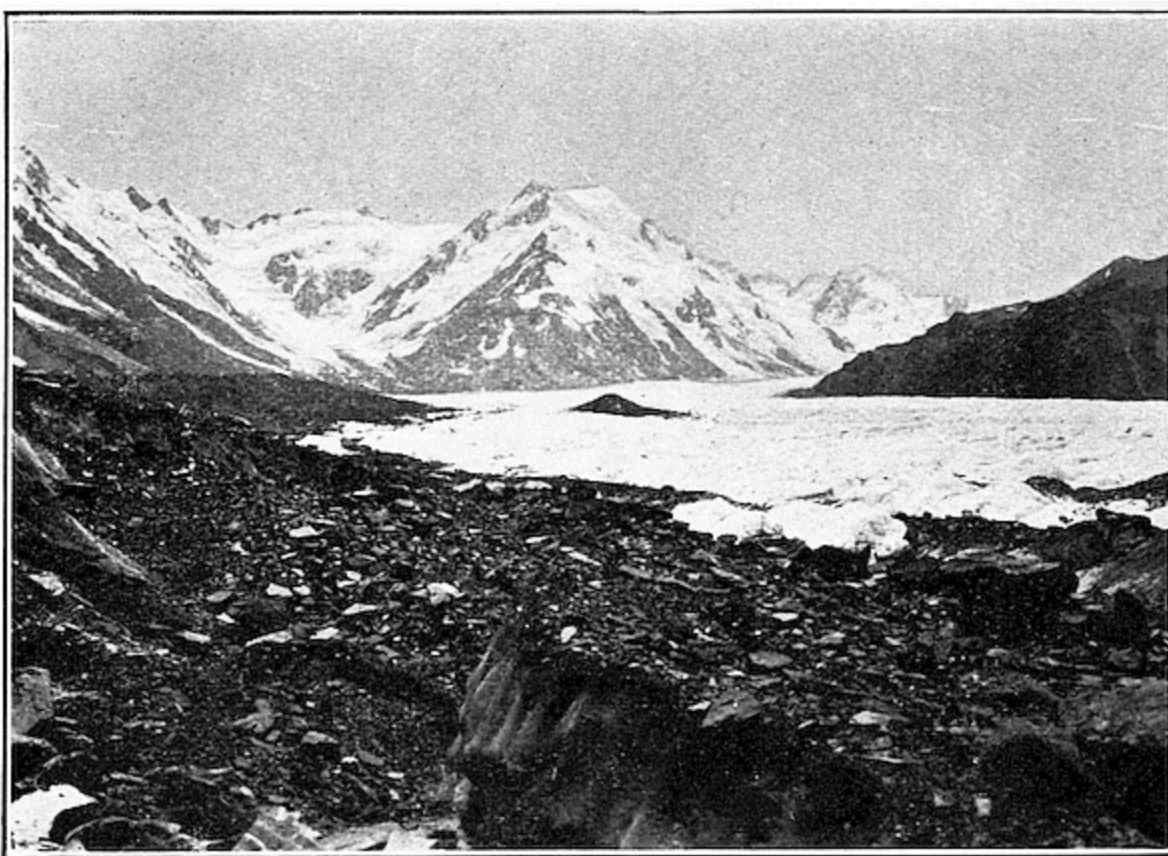
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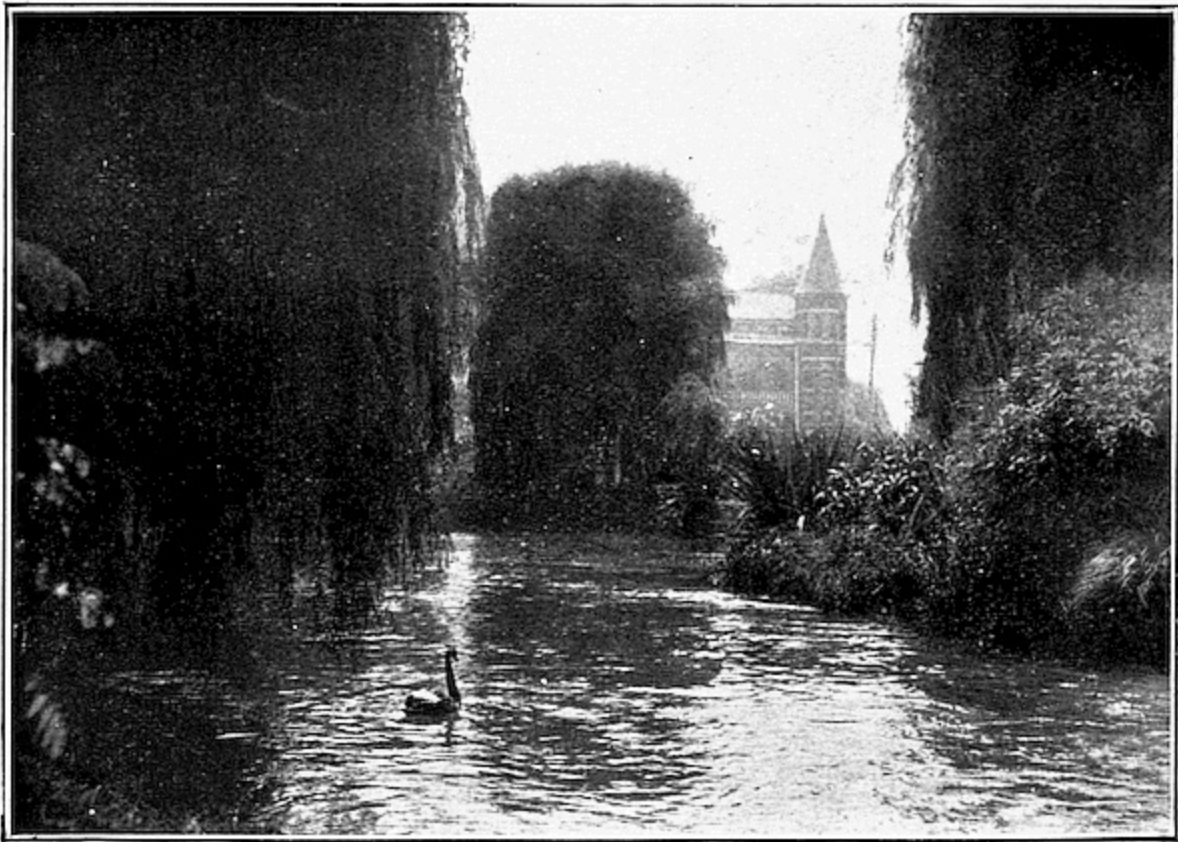
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LOOKING UP TASMAN GLACIER.

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THE AVON: CHRISTCHURCH.

After leaving the agricultural belt we pass through a stony country with low hills covered with deadwood and scrub. Yet the map shows us many lakes and pools, and as it is winter we shall find them according to our expectations. Here is a pool of fresh water ⁴⁷ just after the rains; notice that it even has trees growing round it. Here again is one of the larger ⁴⁸ lakes; though there seems to be plenty of water here it is really only a flat clay marsh, flooded to the depth of a few inches. Notice the gold mine in the distance. In the hot weather the water in these shallow lakes soon evaporates, often leaving a deposit of salt, so that in place of a sheet of water there is a dreary clay flat covered with stones. On these flats near the shrunk ⁴⁹ lakes the only vegetation is the salt-bush, with occasional patches of very dry scrub.

There is nothing to attract the settler to country such as this, except gold. There is little water, and that mostly salt or brackish, so that in the early

days of the mines it was distilled and sold by the gallon at a high price. There was none available for washing the gold, so that it had to be separated from the earth and rock by the method known as dry-blowing. Now all this has been changed. The mines are supplied with water from far away in the Perth district, where the rainfall is heavy. The whole river is held up at Mundaring, twenty-one miles from Perth, by a 50 gigantic weir, and the water is raised at one pumping station after another, and so carried up and over the plateau to the mining districts. The pipes are laid along the railway and the water is carried for about three hundred and fifty miles, or rather further than from London to Berwick. Without this great work, the Coolgardie region, with its large mining population, could not exist.

Let us now examine the mining. Here we see its primitive methods in the bush. There is probably 51 alluvial gold in the bed of a stream, buried under the surface sand and clay. The stuff is merely hauled up 52 in buckets and thrown out on to the “dump” to await further treatment. Here again, instead of sinking a 53 shaft, the miners have burrowed into the bare hillside. In some places, where water is to be had for a short time after the rains, the miners contrive a primitive 54 washing machine such as we see here. The large trough is merely a hollow log, where the gold-bearing dirt is washed to free it from clay, while the larger stones are picked out. Then it is put into the long trough and sluiced, so that the gold settles among the small stones while the lighter stuff is carried away. The troughs are of wood and the pump is distinctly home made; the water is used again and again until it becomes too thick or evaporates altogether.

Now let us turn to another picture. Here is an up-to-date mine, with machinery and tanks for the 55 chemical extraction of the last atom of gold, after the mass has already been treated in great batteries. 56 There is a deep shaft here with mechanical winding as in a coal mine; but often the gold-bearing quartz is near the surface, and we have an “open cut,” or 57 practically a quarry. Here is one showing the rock at close quarters: the white is the rock which contains 58 the gold, while the dark is the “mullock,” or “country” rock, as the miners style it. There are other goldfields nearer the coast, in the district north-east of Perth. Although near the sea this is a very dry country with much mulga scrub; for, as we go northwards, the scanty rainfall is more and more limited to a very narrow

strip along the coast. We have noticed already that the forest zone ends not far north of Perth.

The rest of the interior in this part of the State is a stony desert whose chief product is the spinifex, or porcupine grass, which we have seen before. This often forms an impenetrable scrub, as many explorers have found to their cost. Over very large areas the surface is covered with great boulders; these may be troublesome to the traveller, but they are not without their uses. The storm water runs off them and does not contain salt, so that it can be caught in a tank for the cattle on the stock routes, as we see in this picture.

In the south-east, towards the great Australian Bight, there is a change. There is much limestone country here, with pasture, but the water sinks away through the porous rock. Access, too, is difficult, owing to the cliff barrier stretching for fifteen hundred miles unbroken by any stream or gap. Here is a view of these cliffs; sometimes they come down to the sea, sometimes they lie some distance inland, but they are always present. Here, too, we find the water, which sank into the plateau above, breaking out at the foot of the cliffs and giving us unexpected vegetation.

One part of West Australia we have not visited, that beyond the Tropic to the north. Here is a moderate rainfall similar to that in the Northern Territory and in Queensland; while in the extreme north we meet the heavy downpour of the Tropics. Much of it is well suited for cattle, but at present it is hardly touched, and it is separated from the populous district of the south by a wide stretch of dry and inhospitable country.

LECTURE VI

NEW ZEALAND—SOUTH ISLAND

We now return to the east and leave Hobart, with a westerly wind astern, and following the course of the great Dutch navigator Tasman, steam south-eastward across the deep sea which still bears his name. We are bound for the southern end of New Zealand.

The islands of New Zealand are often compared to the British Islands, but we must be careful not to press this comparison too far. We must notice that their position on the globe corresponds not to that of Britain, but to that of Italy; while the long toe to the north is even nearer to the equator than the toe of Italy. In shape, too, the main islands remind us somewhat of Italy; for Cook Strait is shallow, and we may think of them as forming one continuous land mass. We may notice also that Stewart Island lies on a bank which extends from the mainland southwards as far as the Auckland Islands. There are similar shallows to the northward, with island points upon them projecting above the surface of the sea; so that we may think of the whole area as one great bank below the water, with the higher parts showing as land, while east and west of it lie the deep abysses of the Tasman Sea and the Pacific. So though we have at first sight a group of scattered islands, they have a certain unity if we look at the under-sea platform on which they stand. There are also certain scattered islands, more remote from the New Zealand group, but dependent politically on the Dominion. Of these the chief are the Chatham Islands, 500 miles east of Lyttelton, the Kermadecs, 600 miles north-east of Auckland, and beyond them the Cook Islands, far out in the Pacific.

Though in shape and position New Zealand may resemble Italy, in one important respect it differs alike from Italy and the British Islands. It is surrounded by the open ocean, and there is no great continent near enough

to affect its climate or the life of its people. Its plants and animals are largely peculiar to itself; its native races are widely different from the aborigines of Australia; its past development has been on independent lines, and we may remember that it still remains outside the circle of Australian Federation. We may contrast the position of Italy, in a land-locked sea and with a continental frontier; or the close connexion with the neighbouring mainland which has affected all the past history of our own islands.

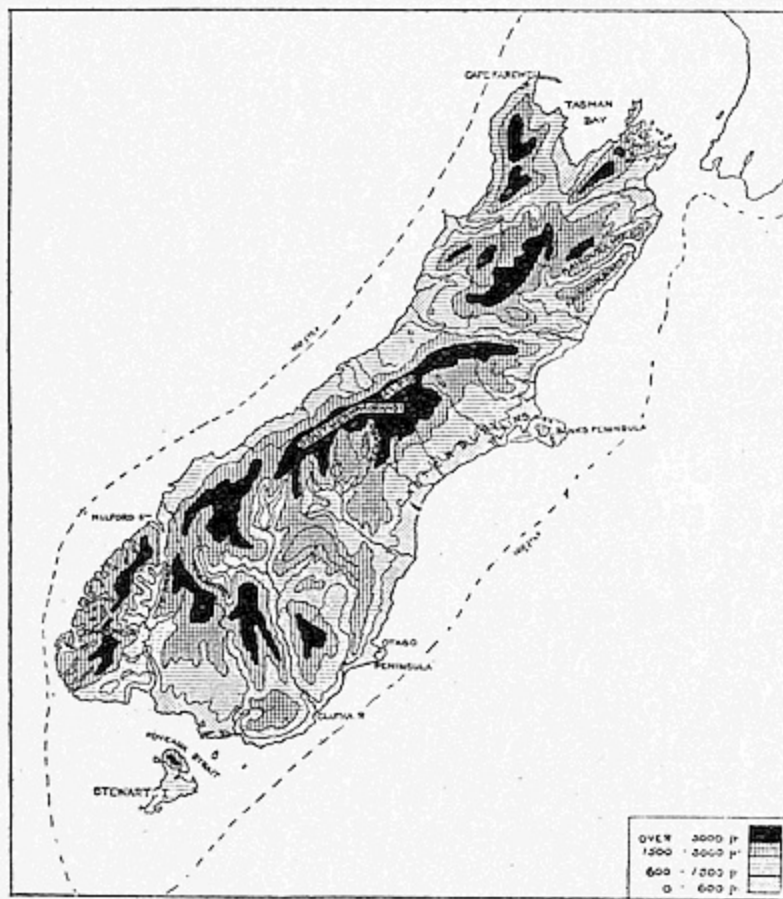
We are visiting South Island first, since its geography is more simple and easier to grasp than that of the North Island; we shall find, too, that some of the conditions are not unlike those which we have lately seen in Tasmania. Our steamer will land us 1 at Bluff Harbour, in the extreme south of the island. Fifteen miles away, across Foveaux Strait, we can distinguish the hills of Stewart Island, famous for its oysters. Bluff is the port of Invercargill which lies some distance up the estuary of the New River, and is not visited by large vessels. It is a busy little town, and here we see the steamers loading wool, frozen meat, and perhaps oats. The tall chimney in the background belongs to one of the largest meat-freezing factories in the colony, while that smooth green mass is the Bluff itself. The wharves of a seaport are an excellent guide to the district which it serves; and wool, meat, and oats may suggest to us the nature of the country which we are about to visit.

A journey of seventeen miles by rail brings us to Invercargill. We have seen the products of the country at the port; here we shall learn something more of the people who own them. Invercargill is a grey, solid-looking town, with very wide and clean streets and a general air of comfort and prosperity. The town is full of people who have come in for the great summer wool sales. Among the crowd we come across some highland pipers—a rather surprising meeting in the southern hemisphere. Our surprise will not last long. Let us look around us: here are Dee Street and 2 Don Street, whose very names, like that of the town itself, carry us back to Scotland. Or let us look at the people and listen to their talk: we shall find Scots everywhere, not in this town only, but all over the south of the island. In fact, the original settlement at Otago, which we shall visit later, was a Scottish Free Kirk colony.

From Invercargill we can reach the mountains of the south-west and some of the most magnificent scenery in the southern hemisphere; but

before we start we must take a brief glance at the island as a whole. The country which is to be surveyed is about the same size as England and Wales. We remember that we approached it from the west but landed in the extreme south; the map may suggest a reason for this. Not far from the west coast, from end to end of the island, runs a backbone of mountains, with a steep face towards the Tasman Sea but a long and more gentle slope to the Pacific. Thus the rivers flowing to the west are short and rapid, for the most part mere mountain torrents, of no use for ships; while their narrow, deep-cut valleys give little space for settlement or roads into the interior. Their mouths also are blocked by sand-bars, owing to the strong winds and currents which sweep along the coast.

In the south-west corner, instead of rivers, we find a whole group of fiords, deep inlets running up between mountain walls and leading nowhere. Here is some of the most beautiful but least known country in New Zealand; there are no railways and practically no settlement, as we shall find on our visit. In short, the island looks to the Pacific and turns its back on the Tasman Sea. On the east side there is more room. The province of Otago is hilly, but further north is the great expanse of the Canterbury Plains. North of the plains are more mountains, the Kaikoura Ranges, lying parallel to the east coast; so that even on this side of the island there is no through communication from end to end. We shall see later that this peculiar grouping of the physical features has had a great deal of influence on the nature and history of settlement in South Island, or Middle Island as some prefer to call it.



SOUTH ISLAND: OROGRAPHICAL.

We will now start on our trip to the mountains, lakes, and fiords of the south-west. The first part of our journey is by train, past fields of oats and rough grazing land with sheep scattered about, and here and there a clump of firs. Soon we are running through a valley between the hills, and following the line of a grey-looking stream, to the town of Lumsden. We 4 are coming to the end of the railway and must change into a motor car to complete our journey. Our destination is the beautiful Lake Manapouri, one of a group of lakes which wind in and out among the valleys of the western mountains just as the fiords which run in from the sea. Here is a piece of our road, with the mountains in the distance; notice the team of horses 5 carting wool.

We sleep at an accommodation house, and in the early 6 morning go down to the little pier and board the launch which is to take us up the lake. Here we see it lying 7 below one of the tree-clad slopes. In the distance,

veiled in cloud, are the Hunter Mountains; here is a 8 nearer view from which we can form some idea of the wonderful colours in which these mountain lakes abound. Further up the lake on the north side we see 9 the Cathedral Peaks, with patches of snow still lying, although it is past midsummer; while all the lower slopes, right down to the water's edge, are covered with green beech, called birch by the natives, with here and there the scarlet blossoms of the *rata* tree making a patch of bright colour. Finally we see the end of the northern arm, with Spire Peak shooting up in the 10 background.

From Manapouri we drive across to another lake, Te Anau, where we again find a launch. From the northern end of this long, narrow and somewhat uninteresting lake we must walk the thirty miles over the Milford track, if we wish to see Milford Sound, one of the grandest of all the sea fiords. We are now beginning to realise how very much out of the world we are in this corner of the island.

Our path lies for many miles through a dense forest of beech. From the roots, and far up the trunks, the trees are covered with mosses, lichens, and small ferns. On the ground are more thick mosses and myriads of ferns great and small. Everything is moist and green in this part of the island, where the perpetual westerly winds bring heavy rain all the year round to the mountain slopes. In fact, we shall be lucky if we finish our walk without meeting with a downpour. Here we see part of our track on the way up to the Mackinnon 11 Pass, with the dense bush below and the snow-capped mountains above. All the way through the trees we 12 have glimpses of cataracts, seaming the rugged slopes above the limit of the forest. Many of the torrent beds are dry now, like the one in front of us, but we 13 can easily imagine the force of the flood water from the appearance of the stones and debris strewn about it.

It is a long walk, and we stop for the night, wet and tired, at the Government huts at Pompolona, where we sleep, if we can, in rough wooden bunks fixed round the inside of the walls. In the early morning we start again, making our way by the side of the Clinton River which flows down the pass between dark woods not yet touched by the sunshine. So we 14 go on over the Divide and down the valley of the Arthur River, through more dense jungle of tree ferns, to Sandfly Point, on Milford Sound, where Donald Sutherland is to meet us with his launch. It is growing late, and

there are sandflies in plenty, but no sign of the launch or its independent owner. Sutherland was the original discoverer of the Sound and of the famous falls named after him, and for a long time he lived here in sole possession. He is now a kind of guide and guardian of its magnificent scenery, and has the air of a landowner showing visitors over his private estate. Here is his portrait, that of a typical blue-eyed Scot. He arrives at last and takes us by moonlight 15 to the little landing near his house.

In the morning we make a voyage down the Sound. All around us are steep mountain slopes with wisps of cloud hanging upon them not yet dissipated by the sun. At the head of the Sound we can see the Bowen 16 Falls, and our boat will take us right up to their foot for a closer view. We pass Mitre Peak, with its remarkable 17 shape, and then are shown Two Man Beach, where Sutherland landed first with one companion, prospecting 18 for gold, thirty years ago. The rock above is dark purple-grey, with splashes of green and scarlet from the small trees clinging to it and the blossom of the *rata*. Finally we reach the Heads and the open sea, and 19 turning look back. Notice how the mountain walls close in on the blue-black water. Sheer above the surface they tower for half to three-quarters of a mile, carved and polished by the great glacier which filled the Sound long ages ago. We may see something like this in the West of Scotland, though on a far smaller scale; but to find a close parallel we shall have to search among the fiords of Norway. Here is a map showing the district of the Sounds. When we have 20 seen Milford, we have seen the essence of all the thirteen Sounds; though each has its own special beauty; so we start on our long tramp back, turning aside a mile or two from the Quinton Huts to visit the Sutherland Falls, 21 which claim to be the highest in the world; and so to our motor car for Lumsden and the railway to Dunedin.

Before reaching Dunedin we cross the Clutha, one of the largest of the New Zealand rivers, as it drains three lakes among the mountains, and these in their turn are fed from the snows of the southern Alps. The river flows for the most part through a narrow rock-bound channel, in hilly country, and so has no broad valley for settlement; but its sands are rich in alluvial gold, and we may perhaps catch sight of an 22 ugly floating dredger, moored in the stream. Much of the gold in New Zealand is alluvial, and dredging for it in the river-bed is one of the most common and profitable methods of mining. This part of Otago owes its population and prosperity almost

entirely to its gold and the sheep which are scattered everywhere over its rolling downs; in the west, as we have already seen, it is occupied by mountain, fiord, and forest, and is scarcely inhabited.

Dunedin, the Edinburgh of New Zealand, is the most important commercial city in the South Island. It is much larger than Invercargill and has many fine buildings of the Oamaru stone, the best building stone to be found in the island. Here was the original Scottish settlement in 1848, and the natural outlet of the rich gold and pastoral district behind. The city lies at the head of a long inlet, Otago Harbour, which is formed by a narrow peninsula connected only by a small neck with the mainland. The peninsula and the surrounding hills together make a kind of funnel through which sweep the winds from the north-east or south-west. It is always cool and breezy here, a fit place for a strong, healthy, and energetic race; but people coming from the warmer regions of the North Island do not always appreciate the change. Here we have a wide view of the town and harbour, looking northwards, from the Town Belt, a fine stretch of public land on the hillside encircling the city. Here is a nearer view from which we can gain some notion of the size and character of the city itself. Notice the church spires, the tall factory chimneys, the warehouses and offices. Dunedin has its flour mills, its woollen mills, its manufactures of boots and clothing. Here is the interior of one of the large woollen mills, showing the combing and spinning machines; we might easily imagine ourselves in Yorkshire. Another picture gives us a glimpse of one of the many beauty spots on the coast within a short distance of the city.

Just as the ocean trade of Invercargill was carried on at Bluff, so the outlet for Dunedin and the usual landing-place for large steamers is at Port Chalmers, some miles further down the inlet; though a recently dredged channel now enables them to reach the wharves of Dunedin itself. Here we see the port from the west, looking towards the sea. Dunedin is the headquarters of the largest steamship company in the southern hemisphere; the company trades especially with the coast ports of New Zealand and with Australia. It is a curious accident which has given us a fine harbour in this position. The east coast, like the west, takes the form of long, smooth bights, with very few natural harbours. The peninsula which forms Otago Inlet is volcanic in origin, as is the larger Banks Peninsula further north. There are no live volcanoes in South Island; but these interesting relics of

past activity have proved of the greatest value to settlement. They even attracted here the Maoris, who confined themselves for the most part to North Island.

We must now resume our journey, by the trunk 29 line of railway, which follows the coast all the way, with occasional branches running inland between the south-eastern spurs of the central mountain chain. We pass Oamaru, whence comes the stone for many of the public buildings of New Zealand, and stop at Timaru, where we change trains. We are going up one of the branch lines to visit another district of magnificent scenery, though quite different from the southern lakes and fiords. This is the district of the New Zealand Alps, the Switzerland of the southern hemisphere. At Fairlie the railway ends and we must once again trust to a motor car. In the distance, as we start in the early morning, we see a great mass of snow-capped mountains: this is our goal.

Our road at first lies across a flat plain, with sheep and cattle, and here and there a field of corn or a clump of trees and an isolated homestead. Then we reach Burke's Pass, with its firs and poplars and little group 30 of houses. Mount Cook, the highest peak in the island, is just visible in the distance. We are crossing over one of the long spurs which spread eastward from the main chain. Soon we find the mountains, with snow peaks, closing in on either side, as well as in front, and so we come down to Lake Tekapo.

31

Here we cross the end of the lake by a suspension bridge; notice how the light is reflected from the snow on the summit of the mighty range in front of us. Next we come on a sight peculiar to the country; this is a boundary dog, a lonely sentry chained up at a 32 point where the boundary of a sheep run crosses the road and there is no gate to prevent the sheep from straying. Around him is a heap of bones, the remnants of many meals. Then we reach another lake, Pukaki, with its chalky-looking water; and in the distance up 33 the lake we have a view of the great mass of Mount Cook. We drive along the lake side right into the heart of the mountains, and in the evening reach the Hermitage at the foot of Mount Sefton. Here is a view 34 of Mount Sefton and here are the Hermitage and the mountain as we should see them in winter. 35

We sleep at the Hermitage, and in the morning start with a guide and the usual equipment of the climber in Switzerland. We cross the Hooker River by a 36 suspended cage; the water looks quiet enough now, but here it is in flood; a very different picture. All 37 along the west coast there are mountain streams of this kind; and we can easily imagine what a serious barrier they offer to movement from north to south across the line of their channels.

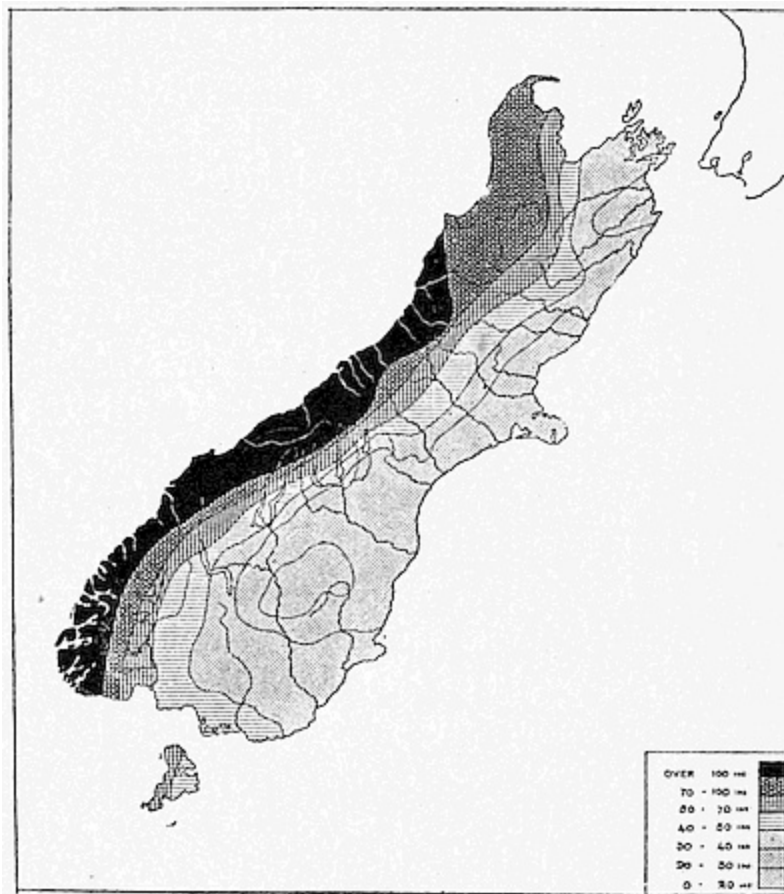
It would be impossible, without a large scale map and plenty of space, to describe the wonders which we can see on a two days' walk, sleeping at Ball Hut, a long way up the mountain slopes. But we may gain some notion of the different kinds of scenery to be found. All around us are snow-clad peaks, with great glaciers flowing down the valleys between and ending in long moraines—masses of rock and debris with chalk-white streams flowing out from beneath them as in the picture before us. This is the Mueller Glacier with 38 Mount Sefton; and the river coming out from the terminal face is the Hooker, which we crossed earlier 39 at a lower point in its course. Here again we are looking up the famous Tasman Glacier, towards the Minarets, and here is the wonderful scene at the head of the 40 glacier. On the same glacier we find a beautiful specimen of the ice river; and not far away, on the 41 great Hochstetter, we see some magnificent ice falls. In the background is Mount Silberhorn. The picture 42 suggests the origin of the name. Finally we have a view of Mount Cook, the Aorangi of the Maoris, with 43 its topmost peaks rising into the clouds. It is the highest mountain in Australasia, and we have already seen it in various aspects from a distance. Here is 44 the summit, twelve thousand feet above the level of the sea, or nearly three times the height of Ben Nevis. Mount Cook is worthy to stand by the side of the giant peaks of the Alps, and even in Switzerland it would be difficult to find scenery grander than that of these southern mountains.

We must not spend too much time here, so we make our way back to Timaru. Timaru is a small but busy town and port, with flour mills, meat-freezing works, and steamers loading at its wharves. We must pause here for a moment as we are entering on a new region in South Island, very different from those which we have visited. Timaru is the chief outlet for the centre and south of the province of Canterbury. The eastern part of the province consists of a strip of level or gently undulating country, a hundred

and fifty miles from north to south and from thirty to forty from east to west. This plain slopes gently from the foothills of the western mountains, and has been built up largely of alluvium brought down by the rivers which now flow across it in shifting channels through wide belts of shingle. It is a sharp contrast to the forest regions of the west: a land of comparatively small rainfall, with pastures, hedgerows, and open cornlands; treeless except for scattered clumps, planted by the settlers. Dropped suddenly in this district we might easily imagine that we were somewhere in the Midlands of England.

On these treeless plains and the eastern slopes of ⁴⁵ the hills, the early settlers found pasture ready-made for their flocks; as there was no clearing of forest to be done. So, to-day, this part of South Island has still the most important sheep-rearing industry in New Zealand.

Here is a typical farm near the foothills, with ⁴⁶ cattle feeding on the rich pasture. Again, we have a scene nearer Canterbury, showing a great expanse of ⁴⁷ grain, with the reapers at work. The rainfall is so small that in some parts of the plain there is not enough water for the farmer, so that artificial irrigation is ⁴⁸ necessary, and there are now many miles of water races strung across the country. It seems strange to talk of drought and irrigation after our experiences of the west coast, but we may understand it better if we look at the figures of annual rainfall. At Hokitika, and all along the west coast as far north as Greymouth, the fall is a hundred inches or more; while in the extreme south-west corner it is over two hundred. On the other hand, all the low-lying parts of the Canterbury Plains have less than thirty inches; in fact, we may compare them to the east and south-east districts of England. In England, as in South Island, we find heavy rains in the west, and the explanation is similar in both cases. The prevailing moist winds are from the west, so that the eastern plains lie under the lee of the mountains.



SOUTH ISLAND: RAINFALL.

If we remember the latitude of South Island we might expect to find it rather warm; but we must take account of other things than latitude. It is true that there are occasional hot winds blowing down from the mountains over the Canterbury Plains, but the island as a whole is cool. The great mass of highland, with its winter covering of snow, the perpetual moist winds and the cool currents of the ocean all help to modify the heat of the sun's rays, and to give us a climate not in the least enervating for English settlers.

As we approach Christchurch the farms are closer together and the whole country is enclosed and cultivated. For mile after mile we run past green hedgerows as in England; and in the spring these are ablaze with the yellow of the gorse blossom, while the scent hangs heavy in the air. Then, through suburbs with villas and gardens, we reach the city itself. Christchurch is as typically English as Dunedin is Scotch. It was founded by an association of High Churchmen who bought land from the New Zealand Land Company in 1850, and it still boasts the only cathedral in the Dominion. Here is a

general view of the city. We 49 shall find much to remind us of England in the social life, the clubs and sports of the city, and its beautiful park planted with English trees. The Avon flows through it, with its riverside houses whose lawns slope down to the banks; there are willows 50 overhanging the water and swans swimming about, and the whole scene might be taken for a quiet 51 backwater on the Thames. It is often said that emigrants from Britain carry a piece of their homeland with them; there can be no better illustration of this than Christchurch and Dunedin.

The cathedral, the parks and gardens and beautiful residences show us one side of English life; but Christchurch has another side. Its people are not idle. We see here warehouses, offices, and busy factories, as befits the chief centre of the whole Canterbury district. In one respect, however, it differs from the other large cities of New Zealand; it is not on the sea. We shall see its trade and shipping at Lyttelton, 52 seven miles away by rail. Lyttelton lies on an arm of the sea to the north of the Banks Peninsula, a volcanic 53 mass breaking the line of the long smooth coast, just as the Otago Peninsula at Dunedin. But the activities of the two cities are not the same: Dunedin, the outlet of the coal and gold-mining area of Otago, has its School of Mines, as well as of medicine and other subjects; while Christchurch, the city of the fertile plain, in addition to a great engineering department, has its Lincoln Agricultural College. Here are the college buildings, and here we see some of the students 54 at practical work, threshing wheat in the fields. Thus we see a contrast which is purely geographical reflected 55 in the world of education.

If we follow the great plain northwards from Christchurch, we shall find that it narrows gradually and comes to an end at last, against the seaward range of the Kaikoura Mountains. Our road north is stopped, but we can follow the railway north-westward to Springfield; thence we go by coach, over the Alps through Arthur's Pass and down the Otira Gorge, until we strike the short railway which will take us to Greymouth and the west coast. This is the only coach road over the Alps, and we can learn much in the course of our exciting forty mile run.

We climb up, by a twisting road, over open slopes covered with brown tussock grass. Here and there we pass a mountain tarn, and all the time above us are purple mountains capped with dazzling fields of snow. 56 We cross the saddle at the summit of the pass, 3000 feet above the sea, and then

plunge down through a deep cañon gorge, following the narrow bed of a mountain 57 stream. Instead of the open mountain side, all around us are high fern-clad cliffs, steep slopes and dense forest, all ablaze in summer with the scarlet rata blossom. We have seen, on our short journey, the east and the west countries, in the sharpest possible contrast. Soon, the rail-heads on either side will be linked, and a rush through miles of dark tunnel will take the place of the splendid drive on the coach, though doubtless many will still prefer the road to the rail.

From Greymouth another line of rail and coach will take us through another famous gorge on a zigzag course to Nelson on Tasman Bay. We realise that communication in this part of the island is very difficult, and we shall find that the isolation of the different areas of settlement has been an important factor in the political development of New Zealand.

Greymouth, on a small artificial harbour at the mouth of the river, carries on a trade in gold, coal, and timber; Westport, further north, owes its existence mainly to coal, though there are also gold mines in the country behind it. Hokitika, which we can reach by a short railway journey, southward from Greymouth, is also the outlet of a gold district. The coal in this district of Westland is of good quality, and some day may give rise to a large industrial population; at present it is mined chiefly for export. Here we see the little township of Brunner, with its coal mines; notice the 58 forest and mountain in the background. Round Nelson there is a strip of sheltered lowland noted for its agriculture and fruit growing; as it was easily reached, and near the line of traffic through Cook Strait, the district with its fertile soil and mild climate was one of the earliest to attract settlers. Here is a view of Nelson 59 on its little plain ringed in by bare hills.

The mountainous north-east corner of the island, now the district of Marlborough, was difficult of access; so for a long time it had no separate existence, but was 60 merely part of the larger province of Nelson; but now we may reach it easily by steamer and might spend many days exploring the inner coast and bays of the Sounds which penetrate deeply into the land. These Sounds are different from those which we visited in the south-west of the island. They are broad, lake-like expanses of calm water, backed by forested hill slopes, with here and there clearings and farms near the water's edge. The early whalers established themselves here, and to-day it is a

favourite haunt of the yachtsmen. Here is a view of part of Pelorus Sound.

61

If we look at the structure of the island, we now see that there were three detached areas favourable for early settlement; Nelson in the north, Canterbury in the middle, and Otago in the south. All these were occupied separately, between the years 1840 and 1850, and the only means of communication between them was by sea. We can understand why at first these districts formed independent Provinces, each with its own local Parliament. Indeed, for a few years, even the southern part of Otago enjoyed a separate existence as Southland. The provinces still remain as provincial districts for certain purposes of local government, though Westland, between the Alps and the sea, has been split off from Canterbury, and Marlborough, as we have noted, from Nelson.

Among the cities Christchurch and Dunedin still stand alone with over sixty thousand inhabitants and two of the colleges of the University of New Zealand; while the greater part of the trade of the island passes through their ports. The districts behind them also contain the great majority of the sheep reared in the island. Next in importance to these cities is Invercargill, with less than twenty thousand inhabitants; while no other town has more than ten thousand. These numbers may seem small to us, but we must remember that they are large in proportion to the total population of New Zealand, which, including the Maoris, was less than a million in 1906. In Christchurch, in addition to the Cathedral and University College, we find a Supreme Court of the Dominion; and here we may see also the old Provincial Council Buildings, a remnant of the former system of government.

Though these capital cities have attracted relatively large populations, yet there is not that enormous concentration of life and wealth in a single city which we noticed in some of the Australian States. The sea is partly responsible for this. The population tends to scatter in a fringe along the coasts, since the natural means of access to the agricultural and mining districts of the interior is from the nearest point on the sea-board. There is no great concentration of routes at any one point, and though Christchurch and Dunedin have a long start, many of the smaller towns are growing rapidly, through the development of the areas behind them. As we have seen, the whole structure of the country is against concentration, and the

scattering of the population may become more rather than less marked as the newer districts are opened up from the coast inwards. The geographical conditions which controlled early settlement are still operative to-day.

We have treated of the settlement of South Island as though the first settlers found the country uninhabited and were free to develop its resources as seemed best to them, with no difficulties in the way except those due to climate and structure. We have made no mention of the aborigines for the good reason that less than five per cent. of them are to be found in South Island; so that the native problem has never been of a serious character. The chief attraction for the Maoris lay in the greenstone for their weapons which the island provided; and though they had a few settlements on the coast, their number was too small to form any real obstacle to white occupation.

The character of the climate alone would give sufficient ground for their avoidance of the island, as they were a race of immigrants from the warmer regions to the north; also, as we shall see later, the structure of the country was not well suited to their habits of life. The case is very different when we come to deal with North Island; we shall find the Maori there a very important element in the history of its development. On our way from Nelson we shall pass through the narrow channel between Rangitoto Island and the mainland, where we shall probably meet "Pelorus Jack," the huge pilot fish, or rather dolphin, who regularly escorts vessels through the strait. He is protected by Act of Parliament, and is so well known that he has become a subject of Maori legends. A short 62 steam from Pelorus Sound across Cook Strait brings us to North Island and within sight of Wellington.

LECTURE VII

NEW ZEALAND—NORTH ISLAND

We may remember noticing, in our first survey, that Cook Strait was much shallower than the surrounding seas, and that there was an evident connexion between ¹ the North and South Islands of New Zealand. The strait at one point is narrower than the Strait of Dover, and the connexion between the two sides is still more clear when we look at the structure of the land. The parallel ranges of the Kaikoura Mountains, which we saw in the north-east corner of South Island, reappear on the other side of the strait. We can trace the continuation of the seaward Kaikouras in detached fragments of highland along the east coast of North Island; while inland there runs an unbroken range from the coast at Wellington right up to Cape Runaway at the eastern corner of the Bay of Plenty. So we have a backbone in North Island as well as in South Island, but on the eastern instead of the western side. The result is that the plains on the east are smaller than those in South Island. There is one of these plains behind Wellington and another south of Napier; the railway runs through them and connects up a whole string of small agricultural towns. North of Napier there is no railway, and the country is hilly or mountainous with few settlements. The whole arrangement is similar to that which we have seen in South Island, and is equally dependent on the position and structure of the main mountain backbone. Napier lies ² at the mouth of an estuary, while Wellington, the only good port in the south of the island, is in a drowned river valley, running up between the seaward spurs of the hills and sheltered by them from the gusty winds of Cook Strait.

As we steam in from the sea we might well imagine ^{3,4} that we were entering one of the fiords of South Island. The hills come down to the water's edge so ⁵ that the city which they shelter is somewhat crowded and cramped for space. The old settlement was on a terrace up the hill; but a

modern port must be close to the water-level, so the wharves and warehouses 6 are built largely on land reclaimed from the foreshore. In fact, the foundations are piles driven into the mud, and the buildings in the pictures before us seem to be growing out of the water. Here is the passenger wharf 7 with our steamer, and beyond it the town in the early morning haze. From the tower of the Customs House we can obtain a good idea of the business part of the city. On one side of us are tall buildings, offices, and 8 warehouses; on the other are factories, sheds, and railway sidings, with the harbour and shipping beyond. 9

It would be a rather ugly city except for the terraced hill in the background, where we find some of the finest buildings. If we look closely at some of these, we may be struck by a remarkable feature in their construction; many of them are made entirely of wood. The chief reason for this was the fear of the earthquakes to which the whole of this district is subject. Here is another curious result of the same thing. We see a fine cricket-ground, on which a match is in progress between Australian and New Zealand teams: the ground is in the Basin Reserve, which was originally 10 intended for a dock but was raised in level during the great earthquake of 1848. The Government offices are said to be the largest wooden buildings in the world; 11 notice the curious style of the architecture and contrast it with some of the many public buildings of stone which we saw in Australia. Even the Houses of Parliament are of wood. Parliament meets here, since Wellington is now the capital and seat of Government for the whole Dominion. The present meeting place 12 is the old Government House, since the former Parliamentary Buildings were burnt down in the summer of 1907. Only a fragment of them remains, chiefly the 13 library which we see in this picture; the marble statue in front is that of John Ballance, a former Premier of New Zealand.

Wellington, by its position, is a natural centre for politics and for trade, whether across the ocean or round the coasts; it is the chief port of the island and the starting point of a whole system of railways which has grown outward from it. As a trading centre, Auckland comes a good second, and Lyttelton third.

We have noticed already the railway to Napier, 14 which reminded us somewhat of that in the Canterbury district. The lines running to the north and north-west carry us into an entirely new type of country, and before

following them up we must look carefully at the build of the island on the other side of the mountain backbone. Here our experience of South Island will help us little. Instead of a gentle uniform slope to the ocean, we find a confused and bewildering structure. The mountain ranges in the long northern neck and in the Coromandel Peninsula, with its continuing islands, run from north-west to south-east, almost at right angles to our main dividing range. In the broader part of the island these ranges disappear, and we have a broken plateau in the region round Lake Taupo with short ranges lying on it in irregular fashion. We may notice that rivers radiate from this centre in every direction except the east. Finally, in the south-west corner, we see an isolated and symmetrical mountain block. The forces at work here seem to have been very different from those which shaped the long ridges of South Island. The Taupo district will give us the key to the whole: it is volcanic and contains active cones, geysers, and hot springs in great number. There is another contrast: in the south and west of the island we have no fiords or inlets, but long, smooth coasts hollowed out into bights; whereas the north is made up of islands, inlets, and peninsulas. We shall see later that these differences in structure have more than a purely geographical interest, since they have had a most marked influence on the settlement and history of the island.

charged with sulphur. Further on is the Waikiti, at the bottom of a deep chasm.

As we draw near to Lake Taupo we notice here and there puffs of steam rising from the scrub at the side of the road, and on the roadway we meet Maoris, tattered and dirty looking. Here we see two of them meeting and 17 saluting one another by touching noses. At last we drive on to the jetty at Tokaanu at the southern end of the lake. There are more Maoris here and on the steam launch which is to take us to the other end. Taupo lies right up on the central volcanic plateau, and we are about to visit part of the great geyser region a short distance from its northern end. It is the largest of the lakes of New Zealand, and is interesting not so much for its beauty, as it has little, but for its close connexion with Maori history. Here is a view 18 of the lake looking back towards Tokaanu and the volcanic range of mountains.

A short drive from the landing place at the northern end brings us to Wairakei, where we walk down a pleasant green valley to view the geysers. First we visit the Champagne Pool, a little lake with steep red 19 walls and deposits of white silica on the lower rocks. Here the water is always bubbling and spouting at a temperature above boiling point. The deposits of silica often form beautiful terraces: the most famous in the world were at Rotomahana, not far from Lake Rotorua which lies at the north end of our district. Here are two pictures of the Rotomahana terraces; notice 20 the people bathing in the hot pools. These great terraces were destroyed by the eruption of Mount Tarawera 21 in 1886, but there are many smaller formations of the same kind to be seen in the district. Here, for 22 instance, is one gradually being built up round the Twin Geysers.

Further down the valley we come to the great Wairakei Geyser which plays for three minutes and then is quiet for eleven. Here is a distant view from 23 the opposite bank of the river; and here we see the geyser at work with great vigour. Notice the curious 24 projection in the foreground of the last picture; it is the trunk of a tree, petrified by the deposit of silica, which is not white here but pale coffee-colour. There is a rainbow, too, in the vapour above, which is unfortunately beyond the capacity of our camera. Other smaller geysers are all round us with deposits of claret-colour, black and yellow, due to the different salts dissolved in the hot water. The ground shakes with the explosions of steam beneath us, and everywhere is heat and vapour; yet all the time the cold waters of the Wairakei River are flowing by

within a few feet of us. Scattered over the valley are beautiful lakes and pools of different colours. Here is one of them, the Blue Lake, lying in a corner of an old crater; the perpendicular walls which form the bank in front of us are white striped with orange.

Everywhere in this district volcanic agencies have changed the face of nature. In one part we find long stretches of grey pumice plains, with no vegetation but dusty brown fern. In another part, great deposits of sulphur have been laid down over the earth. It is a country with a beauty of its own, but more attractive to the tourist than to the colonist and settler. It still remains a barrier separating the centres of white settlement and only recently crossed by the trunk line of railway.

It was in this wild country, to which the rail and coach now bring curious visitors from all parts of the world, that not many years ago the Maoris made their last stand against British power. East and west of the lake is the King country of which we hear much in history; except for the tourist it is still in the main left to the Maoris.

In Australia, as we have seen, the real obstacles to settlement were the climate and the character of the country; the aborigines were pushed back into the wilder districts or simply crushed out of existence by the civilisation of the white man. Not so the Maori: he has been a very important element in the life of New Zealand, whether as friend or enemy of the white invader, from the time of the earliest settlers—the whalers of Cook Strait, and the traders, boat-builders, and missionaries of the Bay of Islands which lies on the east coast of the long northern peninsula. The British have been settled in New Zealand barely three-quarters of a century, as the proclamation of annexation dates only from 1840; while the Maori has been in possession for five or six centuries at least and shows no disposition to be crowded out. He was, when first we met him, on a far higher level than the aborigines of Australia.

Who, then, are the Maoris? They are a brown race, a race of seamen who came in their long double canoes from the islands to the north and settled along the neck of North Island—the Fish of Maui as they called it—and at Otago and some few other points in South Island. They brought with them their island customs, especially the division into families and clans which were always fighting among themselves. In New Zealand they found all the

materials for their usual mode of existence: timber for their houses and sea-going canoes, native flax for the weaving of clothes, stones for their weapons, as they had no knowledge of metals; and fish everywhere in the rivers, lakes, and seas. So here they built their houses, with little patches of garden for their few vegetables, while the surrounding country was the common possession of the clan, whence they added roots, berries, and wild birds to the larder. Everywhere, in the best positions for defence, were dotted about their fortified enclosures, 27 or pahi, in which they took refuge in the numerous clan feuds. Their religion consisted of a kind of fetichism, and their whole life was subject to the principle of *tapu*, or taboo as we commonly spell it, a principle common over all this part of the Pacific, and among savage races elsewhere. The Maoris were chivalrous fighters, eloquent orators, and careful preservers of the traditions of their past history. Such was the race which, in the nineteenth century, contested the settlement of the island with us.

Here is a portrait of a chief showing the method of 28 tattooing the face. Here again we see him with his *taiaha*, or staff of office: he looks civilised enough 29 now; in fact, dressed as he is, he might be mistaken for an Englishman, but doubtless he was very different in the old days. He is nearly ninety years old, and so must have seen the settlement and fighting almost from the very beginning. The building behind the group is a meeting house of the usual Maori shape, though the galvanised iron roof seems scarcely to agree with the elaborate ornamentation of the woodwork. Notice the curious scroll carving on the fetich pillars: it is a favourite design among the Maoris and appears also in the tattooing of their bodies.

We see the Maori here in European dress and we find him entering Parliament and the learned professions, or becoming a successful farmer and grazier in every way as civilised as ourselves. But he is not anxious to be a mere imitator of the white man, as he is intensely proud of his own race and past history. In the remoter districts he still clings to his ancient fashion of life. Here is a typical group, in native dress, and here is the 30 simple private house of a chief. Again we have an interesting portrait of the daughter of a chief, holding 31 her father's heavy staff; her curious dress is largely made up of the feathers of the wingless Kiwi, which we 32 have already seen.

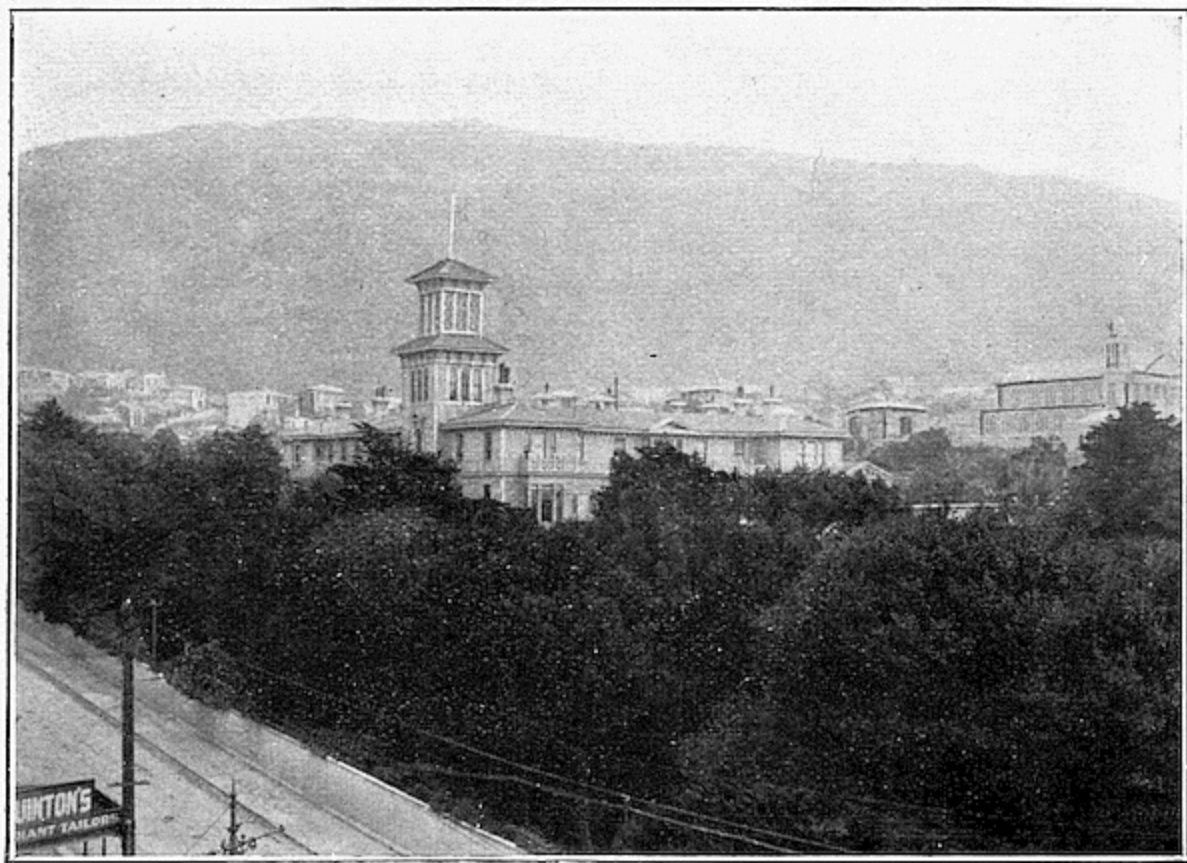
The Maori dances are among the most interesting of their social customs. We may perhaps see the *haka*, a survival of the old war-dance, performed by 33 half-clad warriors. It is now a ceremonial dance of welcome. Very different is the *poi*, the pretty action-dance 34 and song of women and girls, which we may see in the dancing-house of any native village. We notice that the costumes of the dancers are made up of native flax and feathers. The chief native sport is the canoe race, as the water is everything to the Maori. Here is one of the huge dug-out war canoes, hewn from a single 35 tree-trunk, and now becoming somewhat rare; and here again we see his canoe used to mark the grave of a chief. 36

We are now in the district south of the Bay of Plenty and near Lake Rotorua, which we reach by a railway running up the valley of the Thames. Here is a typical scene. In the group before us a woman 37 is smoking a pipe; we may often see the same thing among the peasants of Ireland. Down below, a crowd 38 of Maori children are bathing in the stream, while the tourists watch them from the bridge. The place is called Whakarewarewa, a name rather difficult to pronounce. This is the northern end of the geyser district, which we have already visited. Here we see 39 the tourists again, with their cameras, waiting for the great Wairoa Geyser to start; and here is the geyser 40 playing. Not far away are some Maori women cooking by the steam heat provided by nature. 41

The centre of this region is Lake Rotorua, calm and cold, while all around on its banks are hot springs, geysers and boiling mud-holes. Here, where a few years ago was only bush, or primitive village, a fine modern town 42 has sprung up for the benefit of the tourist in search of beautiful scenery, or the invalid coming to the health-giving waters. Not far away is another lake, Rotomahana, 43 in a huge basin formed by the eruption which destroyed the famous terraces. Here we may enjoy the strange experience of boating on boiling water, with the geysers working all around us. Volcanic activity is everywhere in this wonderful country, and it is only natural that round it have gathered countless Maori legends.

We can easily understand the fondness of the Maoris for the water. They were a race of islanders and fishermen, never really happy unless their houses were built close to the water's edge. This determined their mode of settlement; they only took to the dry land of the interior as a refuge from their enemies. So we found them, when first we entered the island, scattered

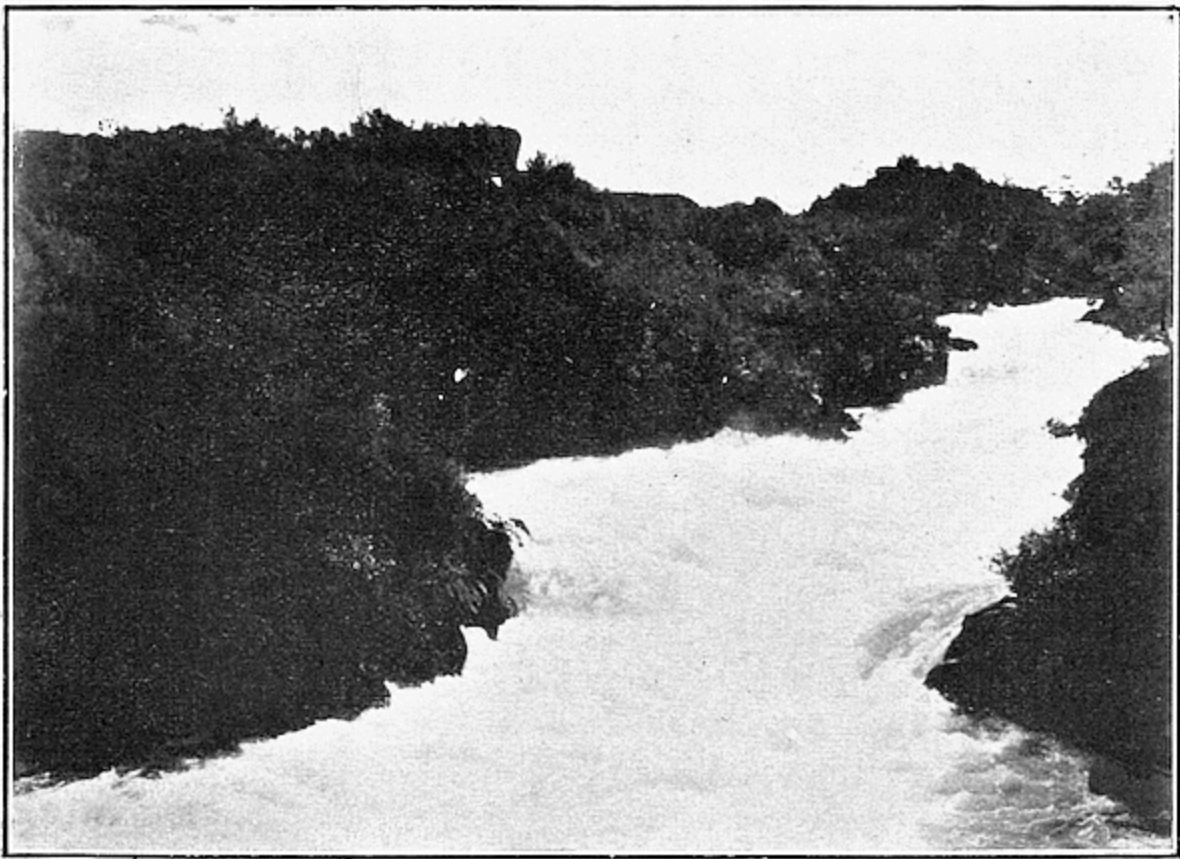
along the narrow part of the Fish, where are bays and inlets in abundance, and along the coasts of the Bay of Plenty and the Taranaki Bight. Lake, river, and marsh were their dwelling places; and when we find them established of their own will in the interior, it is on the banks of Lake Taupo, a small sea in their eyes. The white settler wanted, in the main, the drier districts for his farms and cattle, that is, the parts of least value to the Maori; but none the less, settlement was not effected without a generation of trouble and a long period of petty warfare.



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[See [page 105](#).

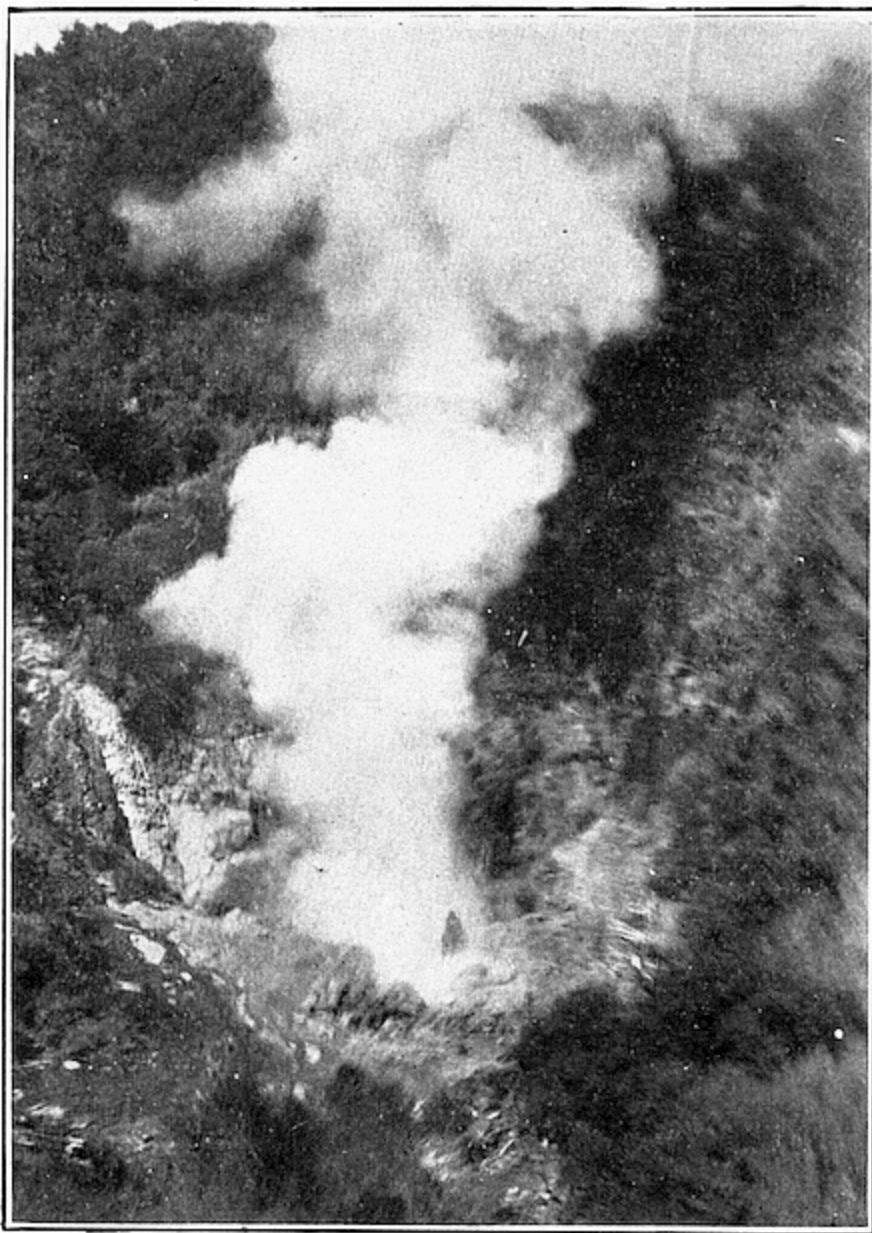
OLD GOVERNMENT HOUSE: WELLINGTON.



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RAPIDS ON THE WAIKATO.

[See [page 116](#).



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GREAT WAIRAKEI GEYSER.

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[See [page 111](#).

A CHIEF'S DAUGHTER.

Down to 1830, except for the missionaries, the Maoris knew us chiefly through the trading ships which visited the islands, and the few white men, not real settlers, who lived the life of the natives and sometimes took part in their clan feuds. The main occupation of the Maoris was fighting among themselves; and it was the petition of some chiefs in the Bay of Islands for protection from another clan which first led to the dispatch of a British official from New South Wales. This official had no real power and

depended for his information as to the natives entirely on the missionary interpreters. Then a Frenchman intervened, and in 1835 proclaimed himself sovereign of New Zealand, apparently on the strength of some land which he had bought years before at Hokianga. This led to a league of the northern tribes, headed by the missionaries, which demanded British protection.

Next we hear of a French bishop setting up a mission at Hokianga, and a French syndicate buying land in Banks Peninsula, in 1838; while the English Land Company, started by Gibbon Wakefield, dispatched its first batch of colonists in the next year. As the result of these movements, the British Government was forced to take action, and the Governor of New South Wales promptly proclaimed our authority over the whole of New Zealand. His agent, Captain Hobson, concluded the Treaty of Waitangi, at the Bay of Islands, in 1840; by this agreement the native chiefs recognised British suzerainty. South Island was occupied just in time, as Hobson's officers reached the Banks Peninsula only four days in advance of a French warship and party of French colonists. So that it was really France which hurried us on to the effective occupation of New Zealand.

The later history of our relations with the natives all hinges on the land question. The New Zealand Company, and many private speculators as well, bargaining through interpreters, believed that they had bought enormous areas of land for a mere nothing: the Maoris imagined that they had sold very little. Moreover, it was very doubtful who really had the power to sell the land, since the waste lands of the Maoris were all held in common. To settle the disputes the Home Government sent out a special Commissioner who decided that most of the bargains could not be allowed to stand. Even so the Maoris found it hard to appreciate our ideas as to sale and possession of land. There was endless trouble until Sir George Grey was sent over from South Australia in 1845, learnt the native language, and reduced both sides to order. For comparatively small sums he succeeded in buying gradually those parts of the country which were most essential to the white settlers, while leaving plenty for the support of the original inhabitants; for we must remember that the Maoris were very few in proportion to the area of the country. Grey was popular with the Maoris, and when he left, in 1855, the country was fairly well organised.

But the land question was not yet finally settled. In 1857, the Maoris tried to federate under a native king, the main object of the association being to

oppose the further sale of land to the white settlers. The whole of Central Maoriland, east and west of Lake Taupo, was unfriendly to us. The trouble came to a head over a land claim in the Taranaki district, and the result was over ten years of raids and casual fighting which could hardly be called organised war, since it moved from district to district, and the different tribes had no common policy or unity. As the sections were subdued from time to time, the usual penalty inflicted was confiscation of land; so that by 1870 the whole of the coast strip, from the north of Hawkes Bay round by Wellington to the Mokau River on the west, was in our possession. So, too, was all the land on both sides of the Lower Waikato, and a narrow fringe along the Bay of Plenty. Thus there was practically a ring fence put round Central Maoriland, and until quite recently there was no connexion between Wellington and Auckland across this isolated block. Sir George Grey, who had returned as Governor, in 1861, was largely responsible for laying down the lines on which the Maori problem was finally and successfully solved.

One interesting result of the war was that Wellington became the capital instead of Auckland. Auckland was the original settlement, moved from the Bay of Islands in 1842, and marked out by Hobson as the future capital. By its position it divided the tribes of the north from those of the south, and this was a great advantage in the early days of settlement. Port Nicholson, later Wellington, was founded in the same year by colonists of the New Zealand Company; while New Plymouth, in Taranaki, was settled in 1841. So we have three original and contemporary settlements, Hawkes Bay being split off from Wellington later. We have already seen a similar process in South Island. After 1865, the population of South Island increased very rapidly, until it exceeded that of North Island, while the Maoris in the latter were thought to be as numerous as the white people. It was inevitable that the capital should be brought nearer the centre of power of the white population, and so, in 1865, Wellington was chosen as the capital by a committee of Australians to which the question was referred.

We have seen the importance of the Taupo district in past history, and how throughout, as in South Island, the structure of the country has profoundly influenced its development and the relations of the settlers with the Maoris. If instead of travelling up the trunk line of railway we had continued along the coast to New Plymouth, we should have realised the meaning of the purchase and confiscation of the narrow strip all round. We

noticed before how the rivers radiate from the Taupo district in three directions to the sea; but in their upper courses they are of no use, since they flow between forest-clad hills, in deep gorges, where the stream is often broken by rapids, ⁴⁴ such as we see here, on the Upper Waikato. The river up here does not seem to be of much value for navigation, and though the scenery is beautiful the river valley is not attractive to the settler. Lower down it is different.

We may reach the coast by following the line of the Wanganui, the most beautiful river in New Zealand. For scores of miles we rush down, in the little launch. In one reach the river is swift but smooth, running between sloping, densely forested banks; in another ⁴⁵ steep fern-clad cliffs close in to form a gorge through which the water foams and swirls in rapids. Above, ⁴⁶ we may catch sight of a Maori village; all the way down we meet canoes, slowly poling against the swift current; while lively parties of Maoris board us at every landing. Below Pipiriki the scenery changes, and near the sea we pass through open level country dotted with sheep and dairy farms, and these are steadily pushing the forest rim further back. Along the coast strip there is a whole string of small agricultural towns which suggest to us that the country round is of great value to the farmer. It is the same all the way to New Plymouth which lies under the shadow of Mount Egmont, a huge detached volcanic cone. Everywhere, in the clearings by the railway, are prosperous-looking dairy farms, the Taranaki district being specially famous for the quantity and quality of its dairy produce. Why is this so? The district is warm, it is also moist, as it lies on the west coast; in fact Mount Egmont enjoys the heaviest rainfall in the whole island. Here is a farm in Taranaki with the ⁴⁷ mountain in the distance, and here we have a near view of the great cone. ⁴⁸

The economic progress of North Island, so far as we have seen it, has consisted largely in the occupation ⁴⁹ of the lowlands near the river mouths, and the gradual clearing of the forest further inland—a rather different forest from that of South Island, since the climate is warmer and drier. To open up the country for farming we had to reduce the hunting grounds of the Maori, and even to drain, in spite of his opposition, some of his favourite marshes. As a result of the settlement we find in the south-east, south and south-west, butter, wool, meat and timber produced in large

quantities; while in the centre we have nothing but the scenery and some forest as yet untouched.

If we think of the products which are most valuable to New Zealand at the present day, we shall find that with the exception of timber they have been introduced from abroad by the white settlers and were unknown to the Maoris. But the marshland, dear to the Maori heart, produces one plant which was necessary to the natives in the past and is still very useful to us. This is the New Zealand flax, though it is more like hemp than the flax from which linen is made in the northern hemisphere. The Maoris wove the fibre of this plant into clothes. Here we see the long blade-like leaves 50 being cut in the swamp; we follow it to the mills where it is unloaded and sorted into lengths. Inside the mill it is put through various processes until the fibre finally appears as a rope or bale of fine white twine 51 ready for the market. In this form it is sent from the mill all over the world and commands prices as high as Manila hemp with which perhaps we are more familiar.

We will now leave the south and pay a short visit to the northern districts. Here the conditions are somewhat different. There is gold to be found at the base of the Coromandel Peninsula, and it is mined largely from the quartz rock, not merely washed from the alluvium as in South Island. This has led to the growth of a group of small mining towns. There is coal, too, in the basin of the Waikato and in the districts north of Auckland. The presence of these minerals in itself would lead us to expect a difference between the north and south of the island; but the difference extends further, to the vegetable products of the soil. We are moving out of the west wind region into the zone where, as in Victoria, the greater part of the rain falls in the mild winter, and the summer is dry and hot though tempered by the nearness of the sea. It is a land where the orange and the vine grow well, in great contrast to the cool and moist region in the south of South Island. Many people find the change from Auckland to Dunedin or Invercargill very trying to the constitution.

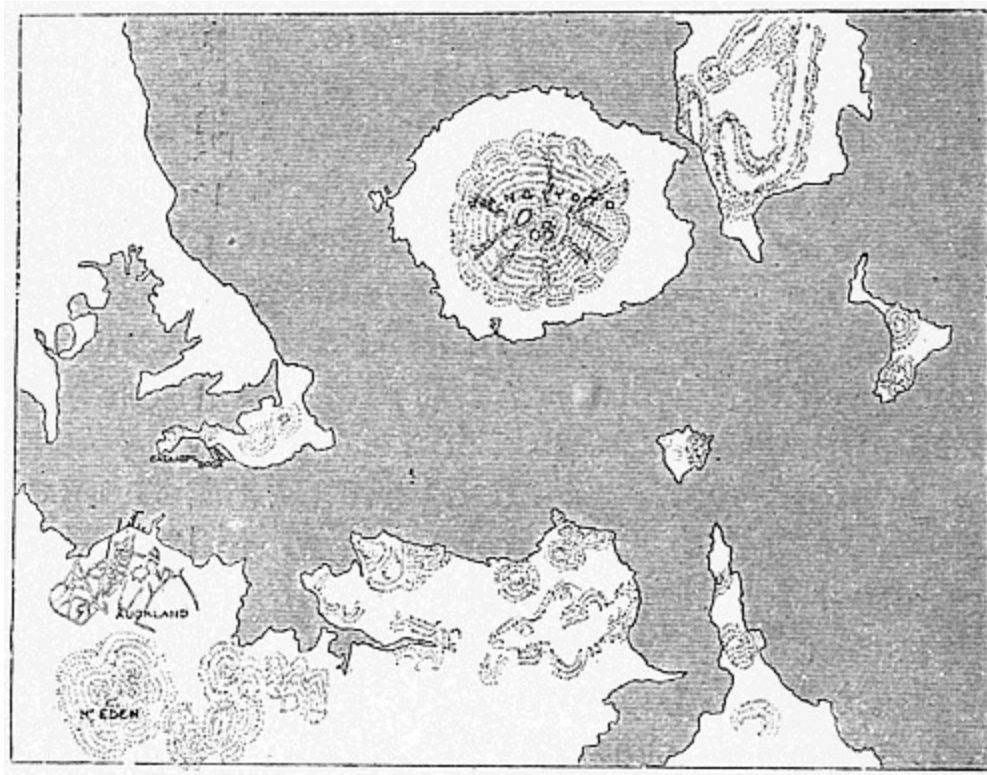
North Auckland has its forests, but they are different from those of the south, for here and here alone grows 52 the Kauri pine. Here is pictured the life of the tree. First we see it growing proudly in the forest; next 53 the lumbermen are at work on the huge trunk; finally the logs are hauled away, and at any of the little 54 ports of the peninsula we may see the timber ships

loading; here, for instance, is a Norwegian sailing ship ⁵⁵ taking in a cargo for Liverpool. The Kauri pine is responsible too for a curious local industry. The forests of the past have left masses of gum buried in the swamps, and many labourers are employed in spearing or digging for it. They become very expert in finding likely spots in which to probe. Here we have a group of Croatian diggers ready to set out for the day's ⁵⁶ work; notice the long slender spears which they carry and the great swamp saw for digging out the gum. We may see them scraping the gum and then follow them home in the evening over the desolate ⁵⁷ plain to their primitive camp. It is a strange industry carried on by these foreigners, and the life seems hard, but the resin is one of the most useful for making our varnishes.

We have already noticed the great number of bays and inlets along the long northern neck of the island. In the far north is the Bay of Islands, the site of the ⁵⁸ earliest settlement, represented now only by the small town of Russell. Fifty miles further south we have the deep inlet of Whangarei Harbour, where a port and a short railway line mark the presence of coal in the country behind. Finally, on Waitemata Harbour, a western arm of the wide Hauraki Gulf, stands Auckland, ⁵⁹ the former capital and still the largest town in New Zealand. The population of the city with its suburbs is now over a hundred thousand. Only eight miles away by rail is Onehunga, on Manukau Harbour, ⁶⁰ a gulf almost landlocked but opening into the waters of the Tasman Sea.

The best way to appreciate Auckland is to approach it from the sea. As we steam along the coast from the north, on our left is the curiously shaped Rangitoto Island, an extinct volcano; on our right is a long peninsula, with two low rounded hills, also volcanic, joined together by a low neck of land on which stands the suburb of Devonport. Across this neck we get a brief glimpse of the city standing on two groups of small hills with narrow valleys running down to the harbour between them. Then we turn round the North Head and suddenly the whole bay, two miles broad, opens out before us. On our left are the residential suburbs of Remuera and Parnell, with houses and gardens running down to the water's edge; further up the bay is the main city with its long quays busy with traffic, and large steamers anchored in the deep channel. Beyond are more suburbs, and we can follow the winding inlet for fifteen miles through beautiful scenery which reminds us somewhat of Port Jackson. On the north side again another city is

growing up, so that Auckland seems to be built round a great lake, open at the two ends. The air is clear and free from smoke and everywhere the sparkling blue water reflects the bright sunshine. Here we see Auckland, looking across the 61 water from Devonport, and here we look down on it from Mount Victoria. 62



AUCKLAND.

Auckland, for its beauty, has been styled the Naples of the South, and, like Naples, it has its volcanoes. We have noticed some of these already, and in the neighbouring country over sixty small cones, such as the one before us, can be counted within 63 a radius of ten miles. Even the hills on which the city stands are partly built up of volcanic debris. Fortunately for Auckland the volcanoes are no longer active, and it lacks also the picturesque dirt and squalor of Naples. It is modern, clean and prosperous. It has also something more than beautiful scenery to recommend it. Though it suffered in some ways by the removal of the capital to Wellington, it is still, and is likely to remain the chief port-of-call in the Dominion. It is near to Sydney and is the natural centre for the trade of the neighbouring Pacific islands. It is the last port of departure for vessels sailing between Australia

and America; and when the Panama Canal is finished, it will stand on the shortest route from the eastern United States to Sydney and Melbourne, and an alternative route to the Suez Canal for the United Kingdom and the whole of Western Europe. It is the chief link connecting New Zealand with the outer world, and a fit point of departure for our final cruise among the islands to the north.

LECTURE VIII

FIJI AND THE WESTERN PACIFIC

We started our tour with the great continental land-mass of Australia; we shall end it with a visit to some of the many hundred scattered islands of the Pacific which are under British control and protection. 1 North-east from Auckland, for a thousand miles, we steam through the open ocean, sighting no land except perhaps the lonely Kermadec Islands, which are, as we have seen, attached to the Government of New Zealand, until we reach the fringe of the many groups of coral islands and reefs which fill so much of the Western Pacific. We shall touch first at Tonga Tabu, the southernmost and largest of the Friendly Islands, as Captain Cook called them, which lie just north of the Tropic of Capricorn and rather more than halfway round the globe eastward from London.

At Auckland we left behind us hills and fiords; on our right as we enter this group is the small but high volcanic island of Eua, the only one in the whole group which contains a river; the main island, Tonga Tabu, or Holy Tonga, lies on our left, and has a very different aspect. A low mass, green with what prove to be coconut palms, rises out of white surf and spray breaking over reefs of coral rock. We round a low headland and thread our way through winding channels among the reefs; on one side is the low mainland, on the other a string of green islets capping the reefs. After anchoring in the roadstead for a visit from the 2 doctor we are hauled alongside the little ferro-concrete wharf. Beyond is a picturesque white town, Nukualofa, 3 the capital of the island.

At the wharf half the population seems to be gathered to meet us. Some are clothed, native fashion, in the famous Polynesian mats or in the bark-cloth called *tapa*, or even in cotton; others are in European dress. The town with its white houses and verandas looks quite civilised. The inhabitants

remind us of the New Zealand Maoris, though their colour is somewhat lighter; in fact, both peoples belong to the same great family, the Polynesians, or light-brown people of the Pacific, which is spread over many of the island groups which lie to the north-east and east of Tonga. It is this race which roused the admiration of early voyagers in these regions, both for its physical appearance and its character.

The Friendly Islands, though a British Protectorate, and, as regards all matters in which European interests are involved, under the jurisdiction of the High Commissioner of the Pacific, still have a native king and Parliament. The inhabitants are Christians, and are in a sense the most school-taught people in the Pacific. They seem to delight especially in mathematics and music, while shorthand is their usual method of writing. Tonga was the last of the independent kingdoms to come under European control. It was left to us as the result of our negotiations with Germany, at the end of the nineteenth century, and is now practically an integral part of our Empire in the Pacific. During the fifty years before we assumed control, it had made rather remarkable progress, under its native ruler, King George Tubou I., in the adoption, or perhaps the imitation, of Western ideas. This was largely due to the Wesleyan missionaries, and especially to the work of one missionary who occupied the post of Prime Minister.

We will now land and pay a call on the father of the present king. He receives us in civilised fashion 4 on the veranda of his house, but we notice that chairs are provided only for the guests; the rest sit on the ground in native fashion. A bowl of *kava* is brewed for us by pounding, squeezing and straining out the juice of a root of a plant belonging to the family of the pepper-worts. It is the native substitute for alcohol, the use of which by any but white people is strictly forbidden in Tonga. Without *kava* drinking, no social ceremony is considered complete, even to-day; and it has played an important part in the political and religious gatherings of the past.

Let us now look round the island. It is only about (2) thirty miles long and ten wide, shaped rather like the human foot, with two bays running far inland in the broadest part. The streets and the sea front of Nukualofa are mostly covered with short grass, and the town is partly hidden in the groves of coconut palms. It is the same all over the island. In the drier season we can follow broad grass tracks, for the most part running round the island; on both sides of us is a dense growth of palms, bananas, and tropical trees,

such as we see in the picture before us. Everything is moist ⁵ and green and flat; yet, in Tonga Tabu, as in all the islands of this particular group, except Eua, we find no rivers, since the island is merely a block of coral rock, with a thin covering of rich soil, raised a few feet above the level of the sea. It is a form of coral island very common in this part of the Pacific.

We now leave Tonga Tabu for the northern islands of the group. Our decks are crowded with passengers. The natives are fond of travelling, and camp out on our vessel with a miscellaneous collection of luggage and food, including a number of pigs. We are bound for the Haapai Group, through a mass of small islets and foaming reefs; but away to the west rise several lofty volcanoes, some of which are still active. We are touching here one of the great volcanic lines of the world, a line which we have already seen continued in New Zealand. Off one of these islands, Tofua, was the scene of the famous mutiny of the *Bounty*.

We come to anchor on the leeward side of Haapai, at the one town, called Lifuka. There is no wharf here, and we lie a long way from the shore, as we see in this ⁶ picture. We may land in the launch or in one of the native boats which come out to meet us; and we can ⁷ survey the whole island in a short walk, as although it is five miles long it is less than two miles wide at the point where Lifuka stands. There is the same rich foliage and tropical fruit as in Tonga Tabu, but if we cross the island to the windward side we shall see a difference from our calm anchorage on the west. Here the south-east Trade Wind is blowing on-shore and the rollers are pounding incessantly and breaking ⁸ into surf against the coral rocks. It is a contrast which we find in most of the islands in this region.

We must hurry on to Vavau, the most northerly of the three main groups of the Friendly Islands. Here the main island is hilly, with high limestone cliffs and ridges. It is the top of a vast mass, heaved up by volcanic agency from the depths of the Pacific. So we have Vavau Sound, studded with islands and protected from the sea by cliffs and headlands running out on either hand. Here is a view of the Sound; ⁹ it may remind us perhaps of a Scottish loch, until we notice the coconut trees covering the hills and coming right down to the water's edge. Gradually the Sound narrows, and after a rather abrupt turn we enter the landlocked harbour of Niafu, with its little wharf and ¹⁰ its group of houses buried in the trees. There are palms and bananas here as in the other islands, but Vavau is especially the home of

the orange. The whole country round is a mass of orange trees, and the ripe fruit strews the grassy roads on which we walk. The picture is spoilt somewhat, especially near the town, by the style of the buildings. There is timber from New Zealand on the jetty, and the native grass and reed hut is giving place to the wooden house with galvanised iron roof, which is ugly and not well suited to the climate. The importance of Vavau lies in its deep and safe harbour, in a part of the world where such are somewhat rare.

Many of the islands of this part of the Pacific are often inaccessible, even for small vessels. Here we have an interesting method of landing on them; the figure 11 in the water is a native postman, who is swimming from our steamer to the island and carrying the mails sealed up in a water-tight can. The natives are fine swimmers and as much at home in the water as on the land.

From Vavau, we turn north-west, and after crossing about two hundred miles of open ocean reach the Lau Islands. These are the easternmost of the Fiji Islands and are, in Fiji, often spoken of as the Windward Islands. Here is one of the islands of this group: notice the white line of the fringing coral reef. 12

In this part of our voyage we follow the course of the Trade Wind, as the Tongans have done for many generations, and the missionaries after them. As a consequence of the easy voyage down wind, the Tongans have had great influence on Fijian affairs in the past; they even established for themselves a kingdom, in the Lau Group, shortly before Fiji was taken over by Britain. Great navigators though they were, they had less skill than the Fijians in boat-building and carpentry; so they came to Fiji for their canoes. In Fiji canoe-building is a hereditary occupation. In former times they used great twin canoes, with a deck between; but now the usual form is a single canoe with raised sides and 13 a solid outrigger, and carrying mat sails. This canoe is capable of great speed, though it does not look very 14 safe. It is giving place now to boats of a European type.

The Fiji Islands, the most important group in this 15 part of the Pacific, are really the higher parts of a great bank in the ocean; the bank is fringed by reefs and coral islands, and in the middle is the Koro Sea, like a vast lagoon, with a wide opening to the south. On the western side of the bank are the islands of Viti Levu and Vanua Levu; these names, in the Fijian

language, mean respectively the “Great Viti” and the “Great Land.” These two islands are by far the largest in the group. Both are volcanic, with high mountains and long rivers, and are quite different from the ordinary coral island.

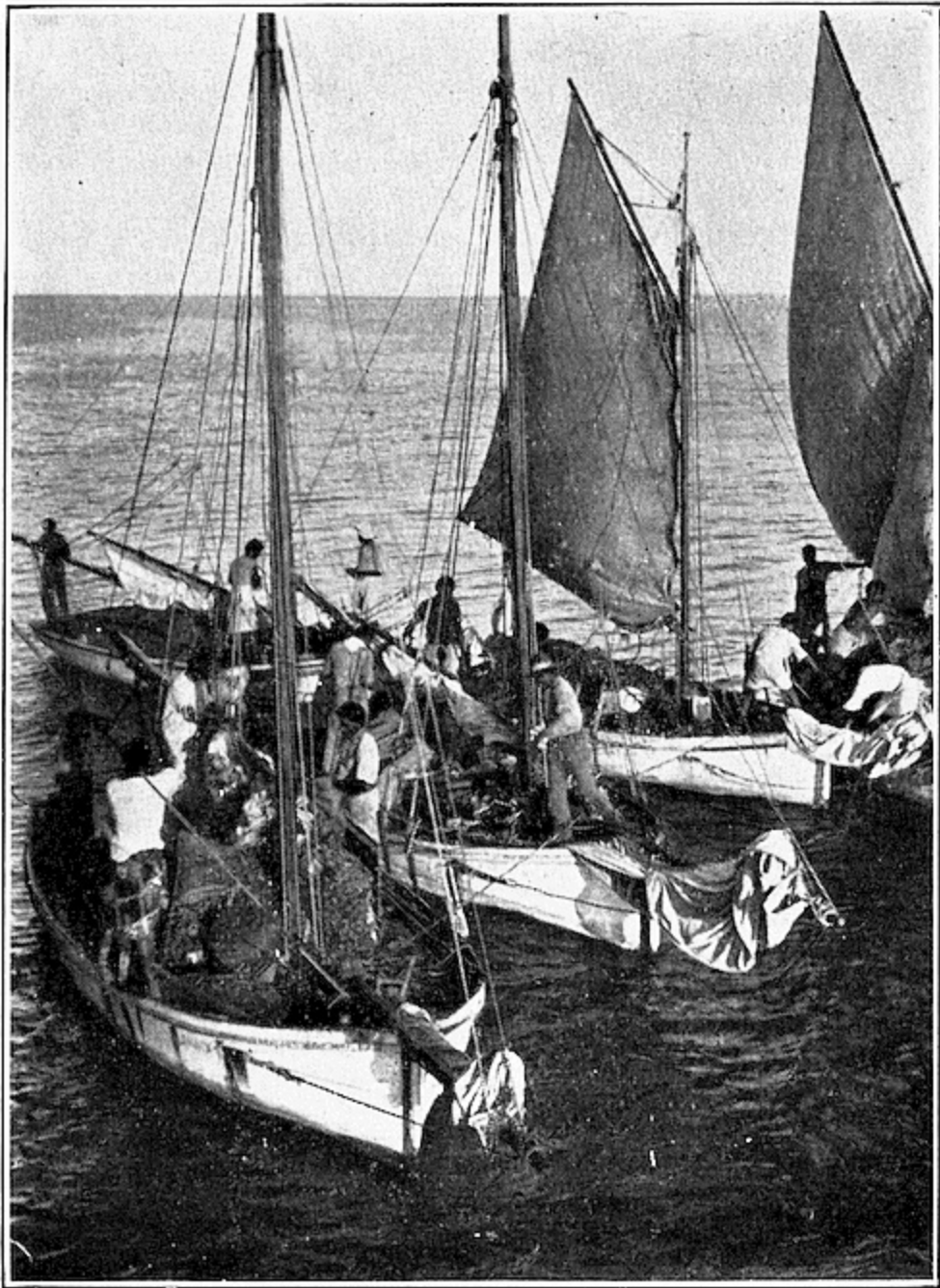
We touch first at Ovalau, a small volcanic island eight miles by six. It lies some fifteen miles off the nearest point of Viti Levu, from which it is separated by shallow water much interrupted by reefs. Here is a view from our steamer of one end of Ovalau. We are approaching Levuka town, which stands on a narrow strip of lowland hemmed in by a wall of mountains. Levuka was one of the earliest settlements of the white traders, and at a later period was the capital of the group; but we can see that there is little room for expansion. So the capital has been transferred to Suva, on the southern side of Viti Levu, where the conditions are very different. Before leaving the island let us have a glimpse of the western side. Here we see a typical coast village; across the narrow strait is another island, and beyond it in the distance rise the heights of the mainland.

On our way from Ovalau to Suva we pass near the tiny islet of Bau, which is so near the mainland of Viti Levu that we can walk almost dry-shod from one to the other at low tide. Bau, small as it is, was once the stronghold of the most powerful of the Fijian chiefs from which they long effectually resisted both Tongans and Europeans. The importance of Bau has departed, but its chiefs are still looked up to as the real aristocrats of the group, and their way of talking is the standard for classical Fijian.

After passing Bau and the wide mouth of the Rewa River, we reach Suva, which lies on the margin of a wide bay almost enclosed by the protecting reef which we see in the foreground of the picture. Inside the reef is a spacious and safe anchorage for small or large vessels. Here is a panoramic view of the harbour; and here are some native vessels; notice that their cargo consists of bananas. The houses are partly hidden in the trees, so that the streets are very different from our own. Here is a street scene: notice that the women in the foreground are Hindus. The Fijians are darker in colour than the Tongans, and many still retain their strange national habit of wearing their hair frizzed out in a huge mop. They differ from both of the two great races of this part of the Pacific, from the Friendly Islanders whom we saw to the east and from the Solomon Islanders whom we shall presently see to the west. In fact, the Fijians are

almost certainly a mixture of these two races, with the addition of yet other strains.

We should expect that the mode of life and the history of the Fijians would differ from that of the inhabitants of the small coral islands, since they have for their home a comparatively large area of land with marked geographical peculiarities. Viti Levu is over 23 eighty miles from east to west and sixty from north to south. It is a land of mountain and river; whereas the ordinary coral island has no rivers worth the name. A range of rugged mountains runs along the northern coast, at no great distance from the sea, the highest point being Mount Victoria, which rises to 4500 feet. It is from this part of the island that the long rivers Rewa and Singatoka flow to the south-east, and a smaller stream, the Ba, to the north-west. The third largest river, the Navua, rises in other heights towards the south of the island, which we shall visit later. Here is a scene on the 24 Navua. The Rewa is long and winding, but it is navigable for shallow draught steamers for about forty miles from its mouth. The whole of the south-eastern part of the island is wet and was originally covered with forest; but the coast-lands to the north-west, under the lee of the mountain ridges, are drier and more open, as we may judge from this picture. We 25 can look across the forest, with its dense undergrowth of fern and creeper, and narrow trails along which we walk, and see in the distance the outlines of the jagged 26 mountains of old volcanic rock. There are no active volcanoes in the island, though hot springs in many parts show that volcanic activity is not yet entirely exhausted. Here are some of these springs. 27



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BANANA BOATS: SUVA.

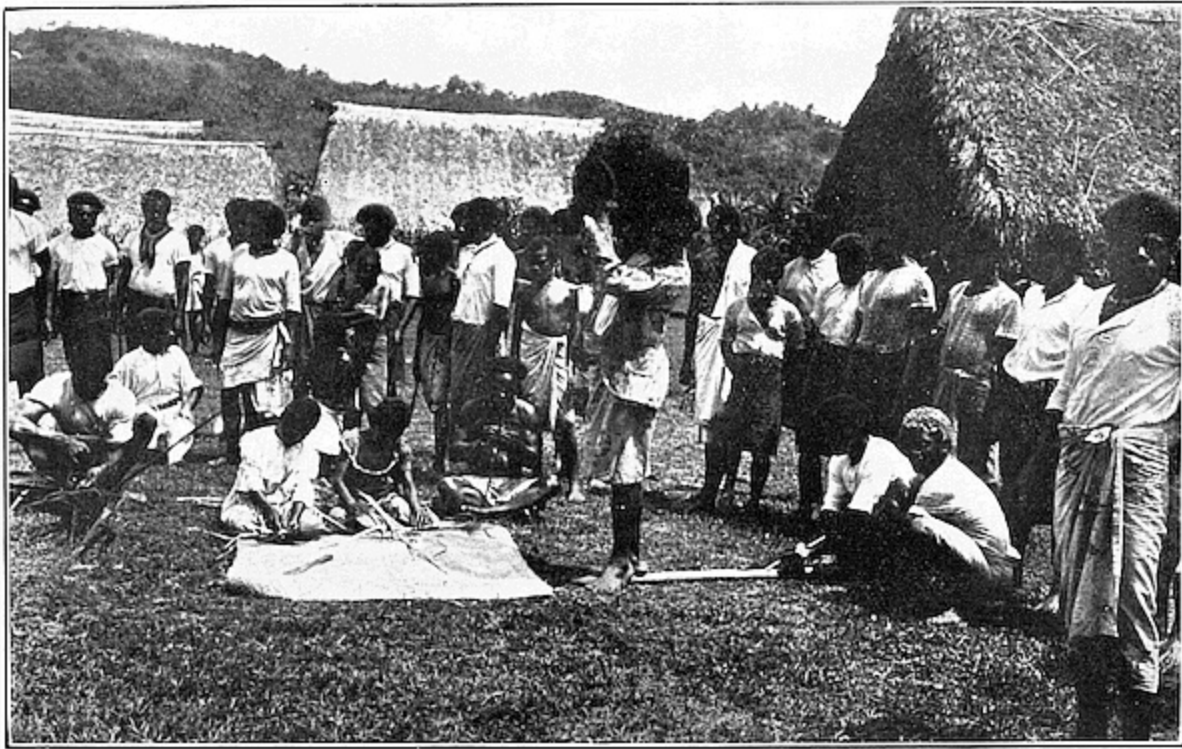
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COCONUT TREE.

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MAKING MATS: FIJI.

Rivers have been an important element in the making of Fijian history, since it is in the fertile soil of the river deltas that the crops of the natives flourish best. Breadfruit and coconuts, together with the roots of the taro and the spindle-shaped yam form their principal food. The taro grows best in the wet districts, or where there is running water; so that the natives have long been familiar with simple methods of irrigation. These staple foods are supplemented by many kinds of wild tropical fruits and roots. The natives have pigs and fowls, but these are kept for state occasions; fish is the only usual non-vegetable food. The Fijians are clever fishermen, whether using the spear or arrow or net, or the fish-fences which they build in the estuaries of the rivers. In fact, the island provides them amply with all the food which they need, though sometimes there is a shortage as they lack means to preserve it. The taro and yams are stored in earth-covered heaps, as we store potatoes; and some of the vegetable food supply, especially breadfruit, is buried in pits and used in a partially fermented state. Here we have a formal presentation 28 of food, yams and turtle and the *yangona* root for making

the native drink called here *yangona* but elsewhere *kava*. It is a typical Fijian scene.

The native is content to cultivate his patch of land with simple implements, such as the digging stick, and is not anxious to work harder than is necessary for his own needs. He is not very ready to work for a European employer, while all his traditions, and the communal system under which he lives, make it impossible for the industrious individual to accumulate any private property.

The cultivation of produce for export is due to the initiative of Europeans, and is largely carried on by the East Indian immigrants who now form a very considerable element in the population. We have already noticed the Hindu women in the streets of Suva. For a few years, during the American Civil War, Fiji exported excellent cotton, but at the present day its export trade consists practically of sugar, copra or dried coconut, and bananas. The sugar-cane is a native of the wet and fertile lowlands of the deltas; but the cultivated kinds are mostly introduced from elsewhere. We find the cane fields covering much of the land along certain parts of the coast, and on the river deltas of Viti Levu and Vanua Levu; and these are connected by nearly two hundred miles of steam tramways, which carry the cane to the centrally placed mills. Here is a view across the Rewa, showing a 29 sugar mill; inside the ugly buildings we find the elaborate European machinery for extracting the juice 30 from the cane. Here again we are in the cane fields with Indian women at work. It is all unlike the native 31 agriculture, though it is very profitable to Fiji, since sugar is the most valuable of all its exports.

The coconut industry is far more picturesque; it is chiefly confined to the south side of Vanua Levu and to Taveuni and the other islands. The nut is grown for its kernel, which, when dried, is called copra, and yields the coconut oil which we use for various purposes. The tree grows everywhere in this region, but does not flourish in Viti Levu owing to an insect pest which seems peculiar to that island. The nut has always been much used by the natives for food, but it is now carefully cultivated for the production of copra. Here 32 is a plantation with the bungalow of the planter; notice the hills in the background; here again is a 33 tree carrying its fruit in curious clusters just below the crown of leaves. The natives, who are excellent climbers, swarm up the taller trees to gather 34 any fruit which is wanted in

a not quite ripe state, as for eating or drinking; but the bulk of the nuts, intended for copra, are allowed to ripen on the tree until they drop off. The ripe nuts are then cut open, 35 the kernels extracted and dried in great trays and put into bags for shipment.

We will now leave the coast for a short trip inland, to see something of the country and the people in their more primitive condition and less mixed type. We travel up the Rewa river for about twenty miles, and there turn westward across country. We have left behind the steamer and the European trader and planter and plunge suddenly into a strange and wild world. At our first stopping place we are entertained with a display which reminds us that a very short time ago the Fijians were fierce fighters and cannibals. This is the *meke* or native war-dance and song, now only an interesting survival. First we see the dancers in the distance, entering the village in two lines. Their 36 faces are hideous with lampblack and vermilion, and they wear strange-looking dresses made of leaves. They go through many complicated evolutions. They rush towards us, stabbing with their long spears and swinging their formidable clubs, and as suddenly rush away. They stamp and charge and shout and imitate all the movements of a battle. Finally they subside quietly. 37 The women also have their special dance, of which we have here a picture. The war-dance is now only a 38 game, but it was far different before our occupation of the islands. Though we still utilise the old tribal organisation, and govern them through their native chiefs and councils, we have forced the Fijians to understand that fighting, raids and massacres are an amusement no longer permitted. The constabulary, 39 which we see here with their rifles and maxim guns, serve generally only in the coast regions, where they represent a form of law and order which the Fijians readily understand; but even in the interior certain Fijians, less formally organised into a sort of rural police, keep effective order. Our little trip might not have been so safe or pleasant forty years ago.

We now leave the Rewa and strike across country south-westward. Our road is a mere track, and often we find streams to be crossed but no bridges to help us. To our carriers this does not matter, as they are not overburdened with clothes. They plunge in and 40 wade through the shallows, and they will carry us if necessary. On the wider streams canoes must be used, or a bamboo raft lashed roughly together. We may notice that our carriers have slung our baggage on poles carried on the shoulders of one or two men; this

is the regular native means of transport, for carts are only used near the plantations, and by Europeans, where roads are available. In the old time the women would have carried the burdens, as it was thought beneath the dignity of a man to carry anything but his weapons.

Presently we reach a village where we discover an interesting native industry, the making of mats from 41 a kind of reed. Here we see girls at work weaving the mats, and in one of the native huts they are making baskets from the same material. The Fijians are clever at this work, and both mats and baskets are important articles in their daily life. They also used to make a peculiar but very artistic pottery, such as we do not find in the islands of Polynesia further east; but this 42 art is no longer practised except in a few places, and for the production of pots for domestic use.

Perhaps the most noteworthy native manufacture is that of *tapa* cloth for their dress. *Tapa* is made by beating out the fibrous bark of the paper mulberry, and sometimes of certain other trees. The art is known in Polynesia generally, and the Fijians, like many of the other islanders, also print the cloth in various patterns and colours.

We continue our march, passing many villages. Here is a corner in one of them; notice the native huts 43 of grass and reeds, very different from the wood and corrugated iron of Suva. Notice, too, the coconut palms growing all around. Here again we have a 44 more elaborately finished house, in the old style; and here is the interior of the home of a chief; it looks 45 somewhat unfurnished to our eyes, though it is cool and airy. At the next village the natives receive us with a solemn presentation of food and *yangona* in a rather dark hut. Here they are making the *yangona*. 46 Finally we reach Namosi, a picturesque native town perched up in the mountains. The town lies in a kind of pass between steep rocks, the "Gate of Namosi," as 47 we see in the picture. The inhabitants are summoned together by the beating of the town drum, a hollow log of wood, and receive us sitting on the ground beneath 48 an ancient tree. We have reached the limit of our journey; but short though it is, it has given us some idea of the real Fijian as he is to-day, away from direct European influence.

The conditions in the other large island, Vanua Levu, are much the same as in Viti Levu. The island is long and narrow, but with the same irregular mountain structure, the same vegetation and the same contrast of wet and

dry on the opposite coasts. We may remember that Ovalau, off the coast of Viti Levu, was much concerned in the past with the politics of the neighbouring mainland; the same was true of Taveuni off Vanua Levu. The dwellers on the rich coastlands of the larger islands led a very uncertain life between the raiders from the sea and the wild tribes of the interior. The sea was always the more important factor in the life of the whole group of islands; it united, while the land more often divided. We have already noted its importance in the relations of Fiji and the Friendly Islands.

Before we leave Fiji it is interesting to note that the line which divides East from West passes through this group, cutting the island of Taveuni into two; so that in one part of Taveuni we can stand with one foot in the East and the other in the West, or in other words, one foot may be in a place nearly twenty-four hours ahead of that occupied by the other according to Greenwich Time.

The Fiji Islands are governed as a Crown Colony—the only Colony so governed in the Pacific; and hitherto the Governor of that Colony has also been High Commissioner for the Western Pacific, that is to say, he has been in charge of most of the scattered islands in the Pacific which are more or less under British protection. We have already visited one of his charges, the Friendly or Tongan Islands. The number of these islands under the High Commissioner is very great, but unfortunately there is no regular means of communication between Fiji and most of them. Every month a steamer starts from Auckland, calls at Tonga Tabu, Haapai and Vavau, and then goes on to Levuka and Suva, exactly along the route which we have followed. But from Suva this steamer goes on to Sydney in Australia. Once a month, too, a steamer starts from Sydney, along the same route but in the reverse direction, to Auckland. With none of the other islands is there any direct means of communication; and when the High Commissioner wishes to visit them he must go round in a warship. We will accompany him on one of these tours, in order to see something of the other British islands.

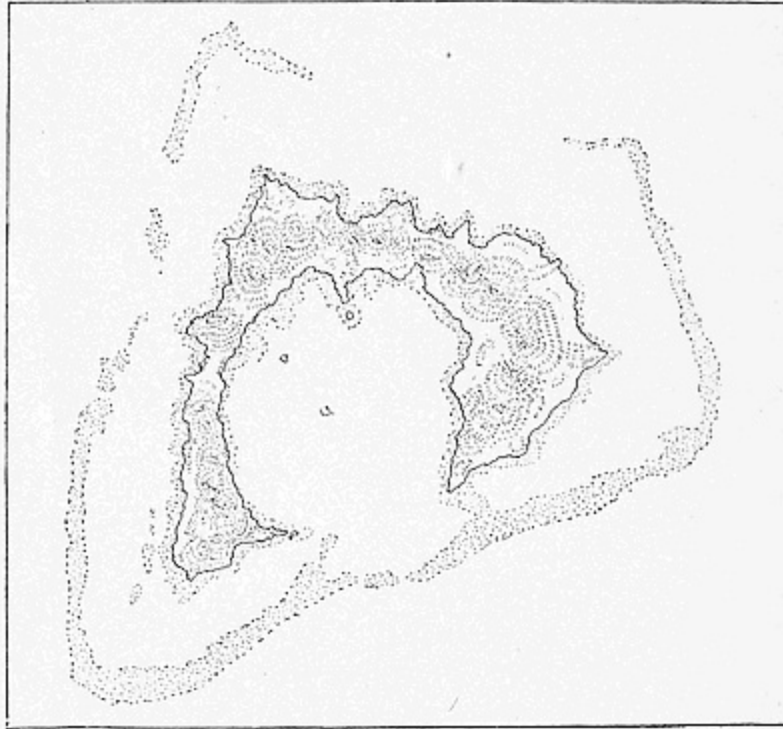


FUNAFUTI.

The dotted portion represents the coral reefs, the black portion land, and the arrows mark the deep-water channels.

Starting from Suva we steam a little west of north, along the chain of the Ellice and Gilbert Islands, which reaches quite up to the equator. These islands are all coral atolls; Funafuti, in the Ellice Group, serves as a type of all the rest. It consists of a lagoon, 49 about twelve miles across, roughly circular and surrounded by a reef; the top of this reef appears here and there above water in the form of coral islets, the largest of which is Funafuti proper. The whole atoll is merely the coral-rimmed summit of a huge mountain rising from the great depths of the Pacific. Funafuti is a complicated atoll; many of the smaller islands of the Fijian group are volcanic, and show a different kind of structure. In these we sometimes find the old crater filled with water, while the remnants of the slopes of the volcano form the shores of the lagoon. A good instance of this structure is Totoya, towards 50 the south-east of the Fijian group. Here the enclosing rim rises to over a thousand feet, but the sea has broken through on the

south and formed a lagoon. The entrance is almost closed by the coral reef which encircles the whole of the outside of the island. Coral reefs grow everywhere if the water is shallow enough, so that we find reefs not only around the coasts of the islands but growing on banks which are isolated and entirely covered by the sea.



TOTOYA.

The dotted portions represent coral reefs.

North of the Ellice Islands are the Gilberts, arranged also in rows from south-east to north-west. They, too, are coral atolls, with a thin soil which will grow nothing but coconuts and screw pines; none the less, on coconuts and fish a fairly dense population supports itself. They are, as we might expect, less civilised here than in the larger islands which we have visited further south.

From the Gilberts we turn south-west to visit another great row of islands, the Solomons and New Hebrides. On our way we pass Ocean Island, a small dot isolated from the rest of the Gilbert Group. Here we find, in addition to the natives, a considerable number of Europeans. The island is being exploited commercially, but instead of making copra they are digging out valuable deposits of phosphates. Here are some of the natives,

51 and here is a picture showing the excavation of the phosphatic coral rock.
52

Ocean Island has for us a further interest. North-west of it lies Nauru, where also phosphates are obtained. Nauru belongs to Germany, and the dividing line between the British and German spheres runs between the two islands. If we follow the line south-west, we see that it cuts through the Solomons and then turns sharply west to New Guinea which it divides roughly into halves. Germany shares with the United States the Samoa Islands, north-east of Tonga, but her main sphere of influence is in this western area.

On our voyage to the Solomons we pass another curious island, or rather group of islands. This is Ongtong Java, where the people differ from those of the islands which we have visited up to this point. We are approaching a region where we know little of the character and origin of the natives, a region more backward and savage than any other part of our dominions in the Pacific.

The Solomons consist of a double row of long and narrow islands, with high mountain ridges and many volcanoes, some extinct, others still active. The largest island, Bougainville, at the north end, is German; the rest are British. They have long been known to explorers, but until recently their history tells chiefly of resistance by the natives against Europeans who have attempted to open up intercourse with them. Lately, thanks to the good influence of the missionaries, and perhaps still more to the better regulations made by the agents of the British Government for intercourse between the wild men and the Europeans, considerable advance has been made, and plantations of coconut and other valuable products have been established in many of the larger islands. Many of the tribes, however, are still head-hunters and cannibals. The islands are covered with great forests, and the plants and animals, as well as the natives, resemble to some degree those found in New Guinea. The people go naked for the most part, except for necklaces and bracelets of shells and teeth. Their houses are often built on piles, like those of the Papuans. Here we see 53 a group engaged in a war-dance, and here is one of their curiously ornamented canoes. 54

South-east of the Solomons we pass the Santa Cruz Group; the name, as do many other names in this part of the world, reminds us again of the early

voyages of de Quiros the Spaniard, who at the end of the sixteenth century first discovered these distant islands. Spain has disappeared from this part of the Pacific, and the region of islands is divided between Germany to the north, Britain in the Solomons, and Britain and France in the New Hebrides.

Southward from Santa Cruz we pass almost at once the northernmost of the long chain of the New Hebrides, discovered towards the end of the eighteenth century, almost simultaneously, by our own Captain Cook and by the great French admiral Bougainville. Northernmost of all, in the sphere of joint Anglo-French influence, we pass the Torres and Banks Islands, great centres of the famous Melanesian mission. Next we come to Espiritu Santo, or Santo for short, at the northern end of which is the great Bay of St. Philip and St. James, within which, four centuries ago, Quiros built a town—a town which lasted but a few months. He rejoiced greatly because he thought that in that newly discovered land he had reached the beginning of the great Southern Continent which was supposed to extend thence to the Antarctic regions.

Here and there round Santo there are a few mission stations, and a few fairly prosperous plantations, some English and some French; but the natives in the interior are still very wild and occasionally raid the European settlements. Here we see a vast crowd of natives celebrating a feast. Now it is a peaceful ceremony: it was far different in the past.

Next, still steaming southward, we pass between two rows of islands, until, about where the two rows join, we come to the Island of Efate, with its growing town of Vila, the centre of Anglo-French administration. Here is the seat of the Joint Court, and many buildings of almost European type have recently been erected for its use.

Still further south the New Hebrides reach, now in a single line, almost down to the Loyalty Islands and New Caledonia. These last two places are, however, purely French possessions, and we may pass them by.

From Vila the Governor in his warship would probably return eastward to Fiji. We may leave him at Vila and take either the French or the British steamer, which calls there once a month, and so make our way direct to Sydney. But before leaving the subject of the New Hebrides we may note that the islands of this group are largely volcanic, and the people are not unlike those of the Solomons, though perhaps rather less savage. Here we

have a picture showing their ⁵⁶ former method of receiving visitors. It was from these two groups that the brown labour for the Queensland plantations was largely recruited in times past. The New Hebrides are fertile, though not healthy for Europeans; and when they have been reduced to some order by white administration they may develop a trade with Australia in various vegetable products; since they are not limited to the coconut as are so many of the coral atolls. The picture before us with ⁵⁷ its coral reefs, its forest and its background of volcanic hills, gives a very good idea of the scenery of these islands.

We have now completed the circuit of our chief possessions in the Western Pacific. There still remain a few detached islands and groups on its eastern borders which are under the jurisdiction of the High Commissioner. Of these only two are of much interest.

Fanning Island lies almost in mid-Pacific, and halfway along the usual route from Sydney to Vancouver. A steamer from Sydney travels 1700 miles to its first stop at Suva; then 1900 on to Fanning Island; thence 3400 to Vancouver, the starting place of the Canadian-Pacific route to Europe. Fanning Island is in consequence the mid-Pacific station of the Pacific cable. The island and its near neighbour, Washington Island, are small coral islands with lagoons, on which coconuts have long been profitably cultivated. Both also have valuable deposits of phosphates, due to the age-long deposit of the droppings of countless seabirds on the decomposing coral. These phosphates, as well as the very abundant coconuts, are already exported, and their value is likely to increase considerably before long. Moreover, the probability that Fanning Island may be made into a shelter and repairing station for vessels crossing the Pacific adds largely to its value as an asset of the Empire. Here are two pictures of ⁵⁸ the island, typical of the kind of coral atoll which is found isolated in the Pacific, instead of being, as are ⁵⁹ most of the others which we have seen, a member of a group.

Far away to the south-east, almost on the Tropic, ⁶⁰ and halfway to South America, lies the lonely Pitcairn Island, to which a few other scattered islets, British possessions, are attached. Pitcairn has a population of about a hundred and fifty souls, descendants of the mutineers of the *Bounty*, who settled here with their native wives in 1789. The present occupants represent those who returned after the experiment of removing to Norfolk

Island in 1856. They had found themselves overcrowded, since Pitcairn is a tiny island only two miles long by three-quarters wide, rocky and volcanic, though fertile. The island is of great interest from the point of view of the history of the Pacific, since there are remains on it of stone monuments, weapons, and images which prove that even in this distant corner of the ocean some early people must have settled long before the Polynesians. In fact, there are traces of 61 such settlements all over the Pacific, which suggest that the Polynesians themselves are merely modern colonists, occupying the homes of an earlier and perhaps more civilised race.

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